

RPI



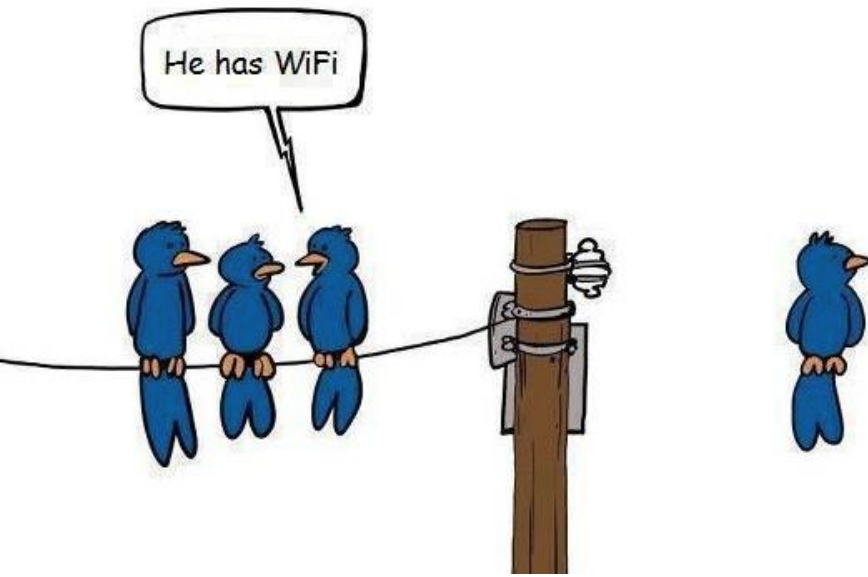
ECSE 4560/ 6560

Modern Communication Systems

Lecture 3: Wireless Systems Overview

Part 2

Fall 2025



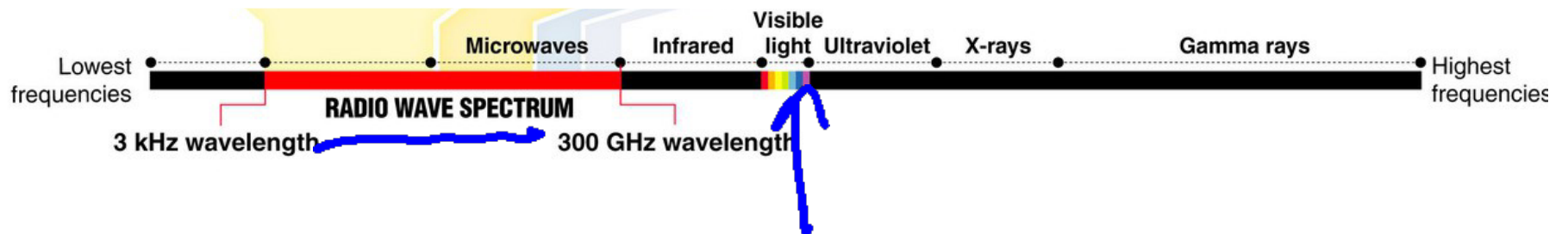
Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- GradeScope for assignment submission (You will be added)
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
 - Single attempt; Don't forget to submit, o/w not graded.
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Capacity**
 - 80% attendance mandatory, In-class participation 5%
- Projects
 - A list of papers will be posted
 - Group formation due 9/12
- HW1/Lab1 to be posted, due Friday 9/19

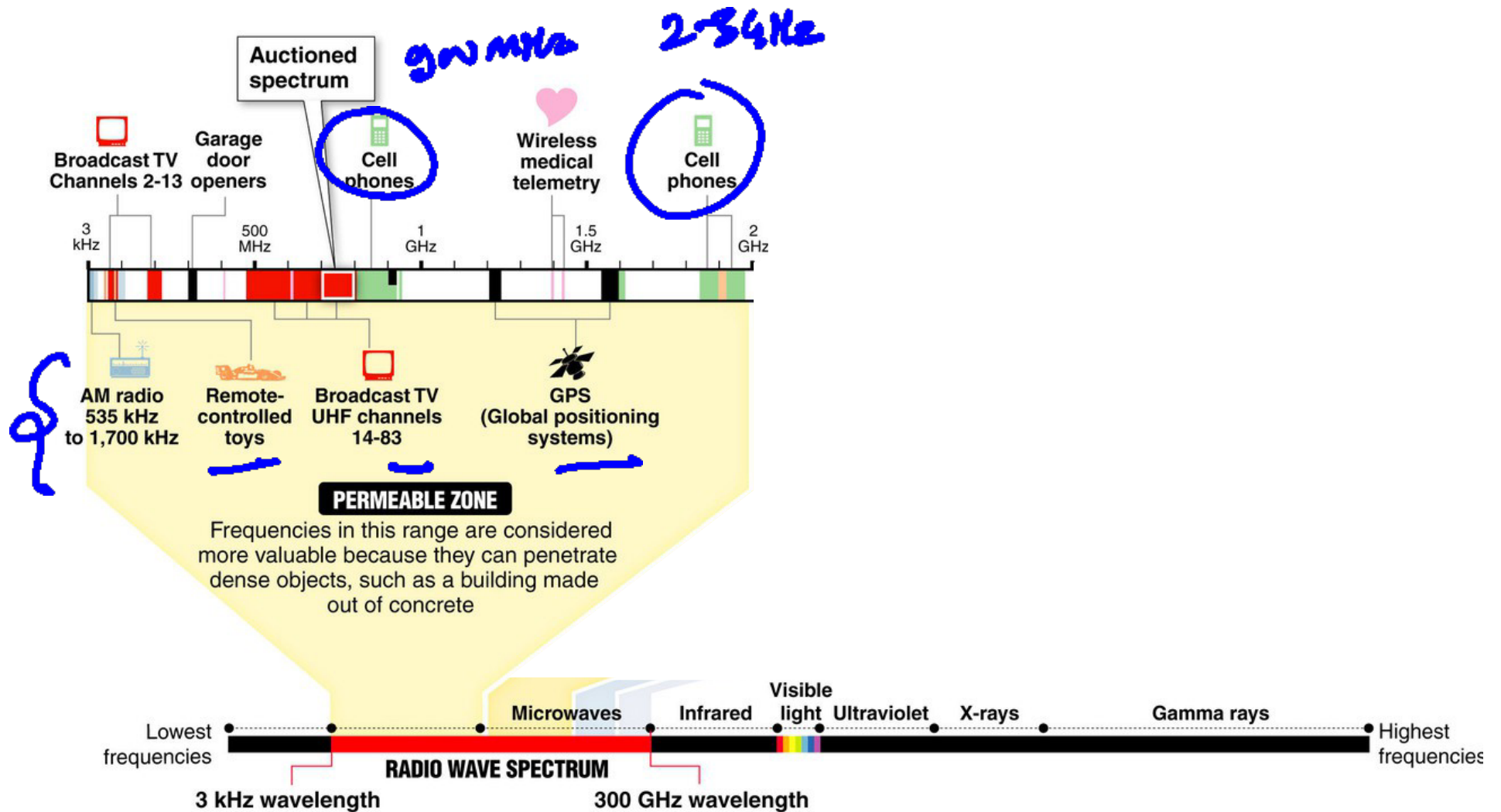
Today's Class: Wireless Overview

- Baseband/Passband Communication
- Up-Conversion and Down conversion
- Modulation
- What is SNR?
- Higher order modulation and its relation to SNR
- BER-SNR curves
- Capacity

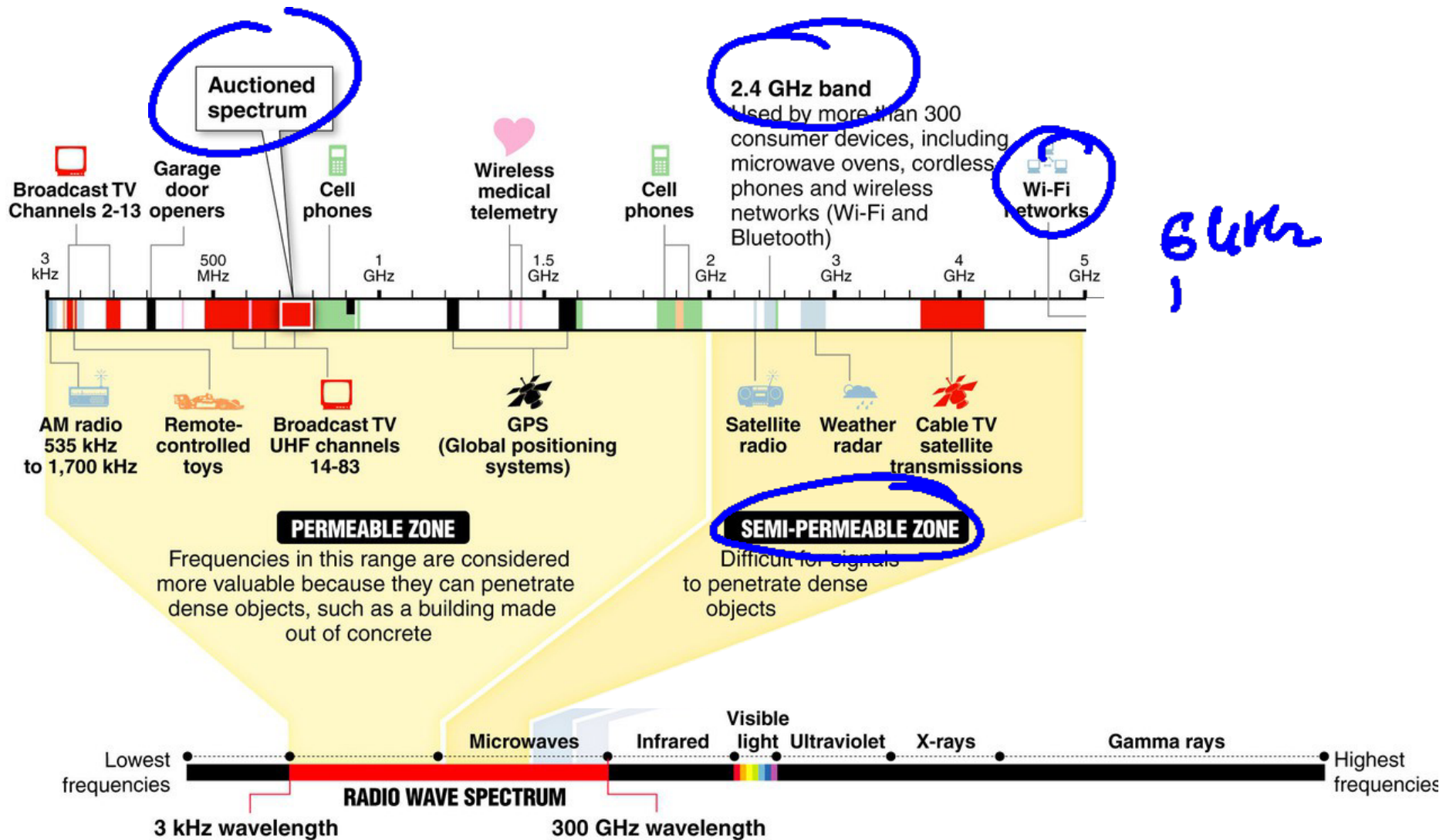
The Wireless Spectrum



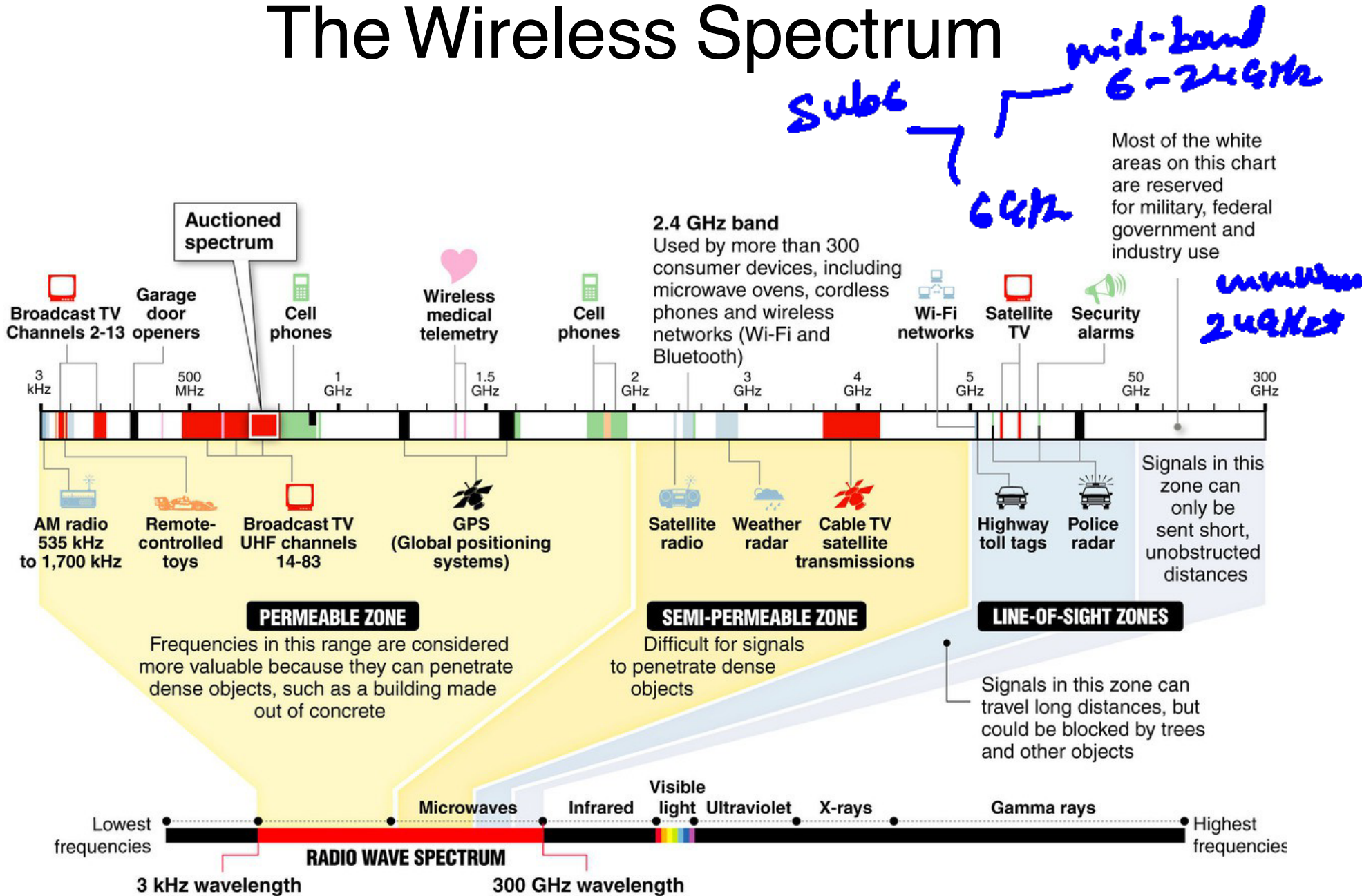
The Wireless Spectrum



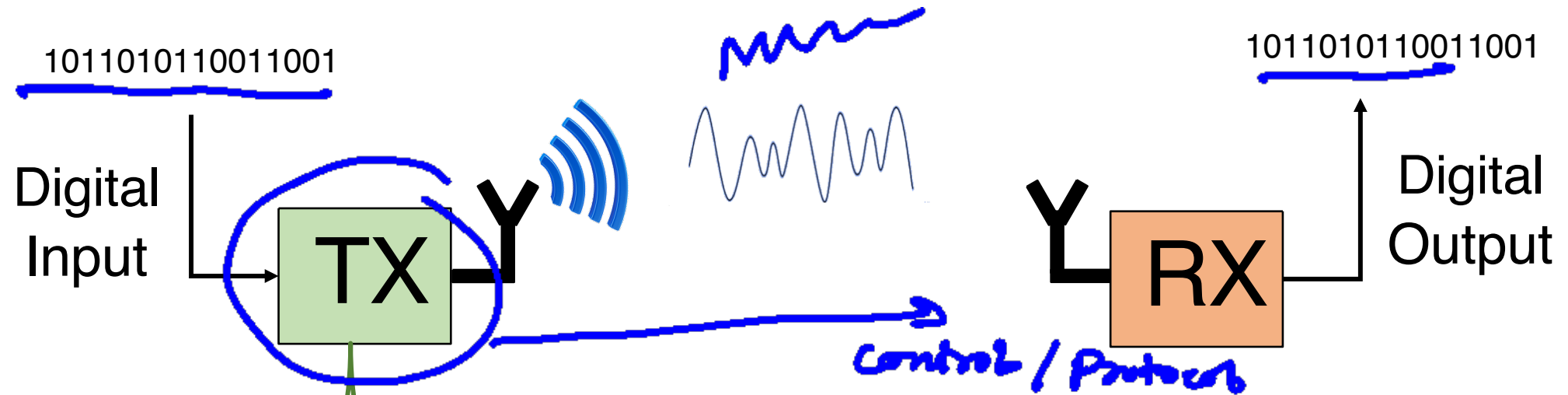
The Wireless Spectrum



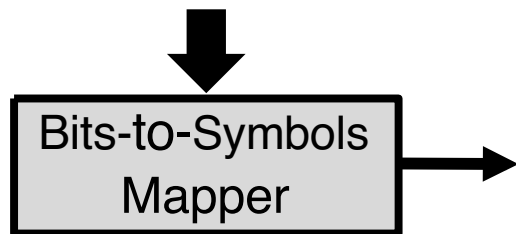
The Wireless Spectrum



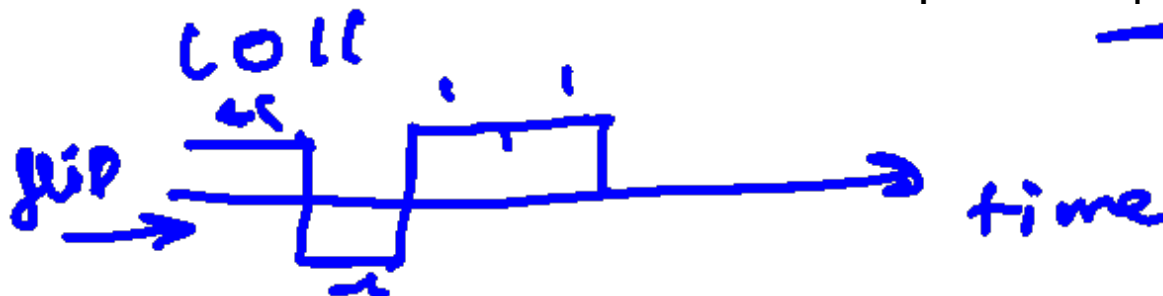
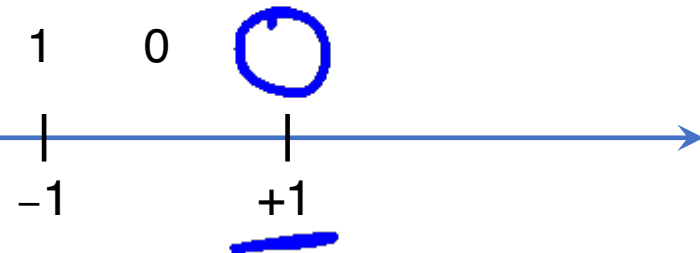
Wireless Communication System



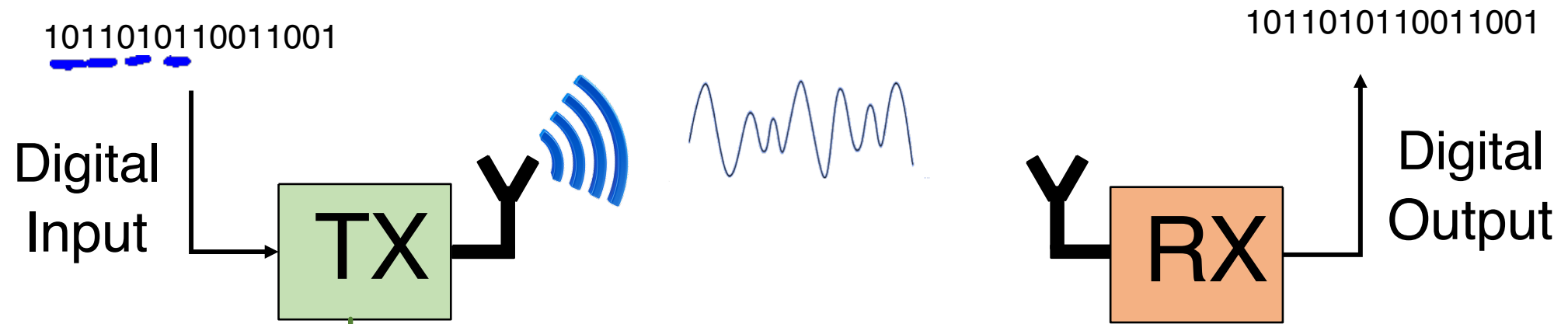
Modulation: Mapping of Bits to Symbols



BPSK: 1 bits/symbol

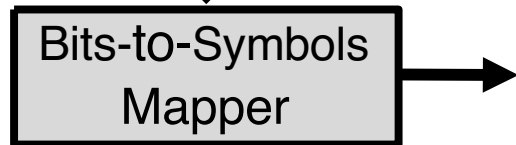


Wireless Communication System

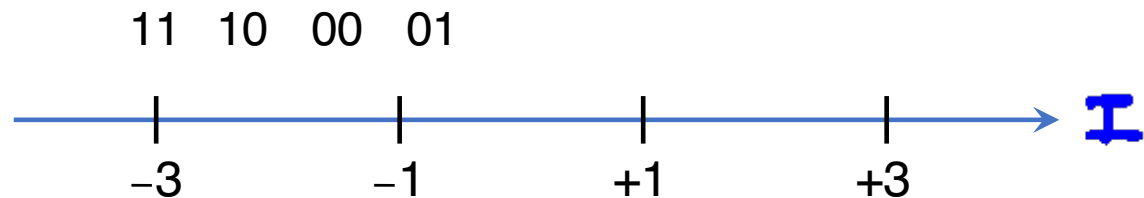
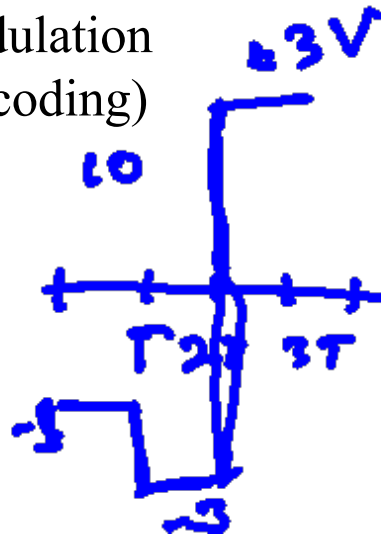


Modulation: Mapping of Bits to Symbols

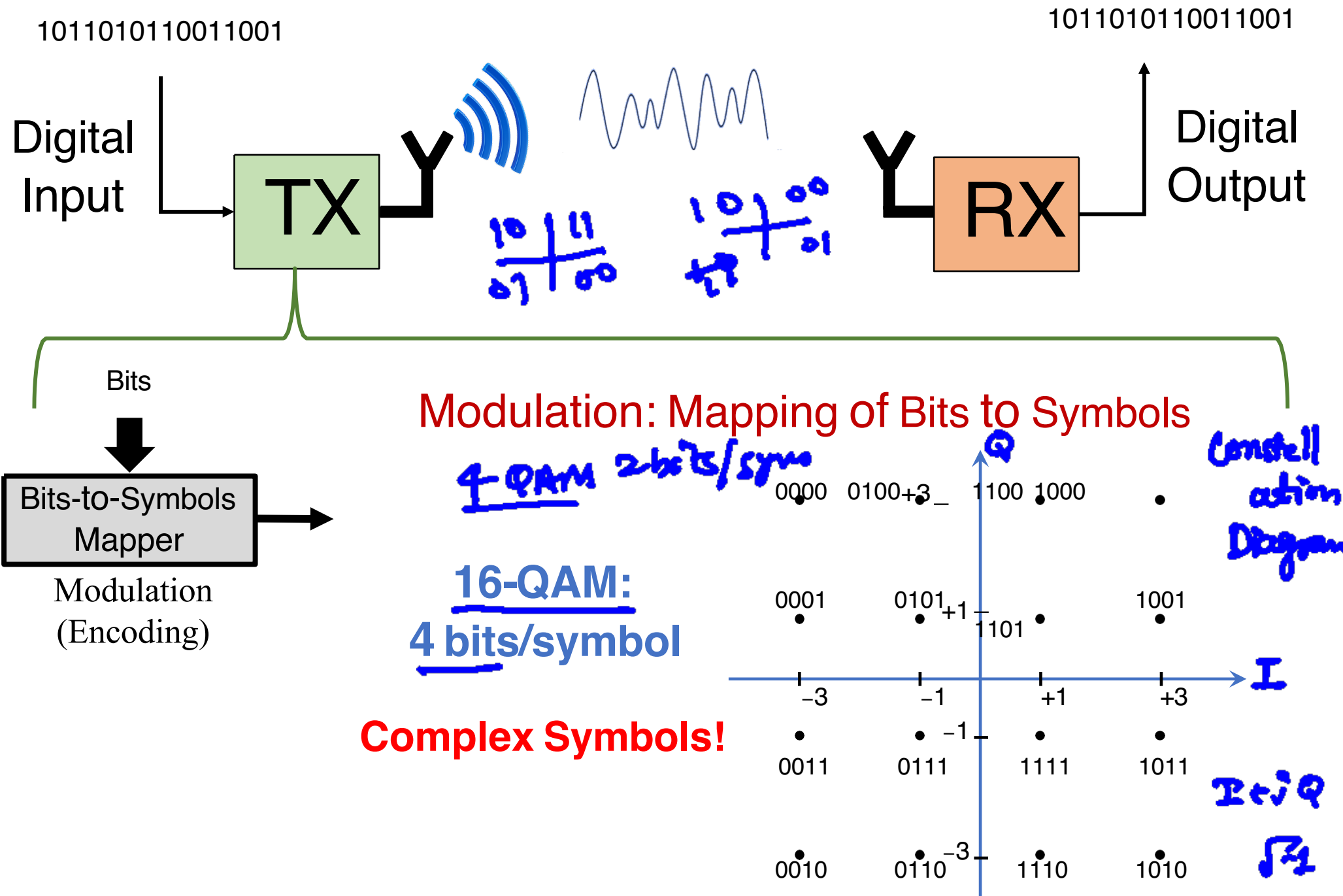
PAM: 2 bits/symbol



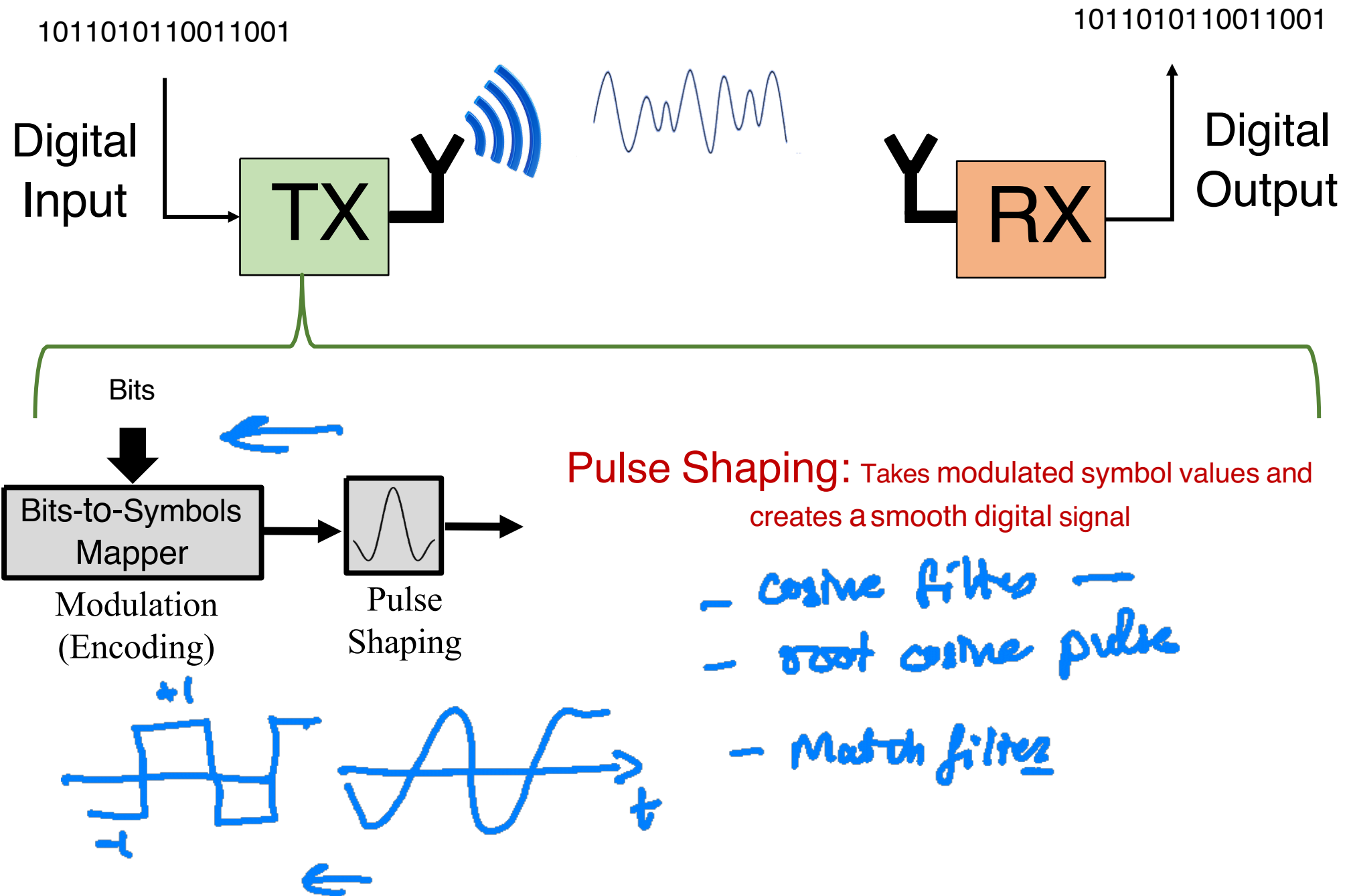
Modulation
(Encoding)



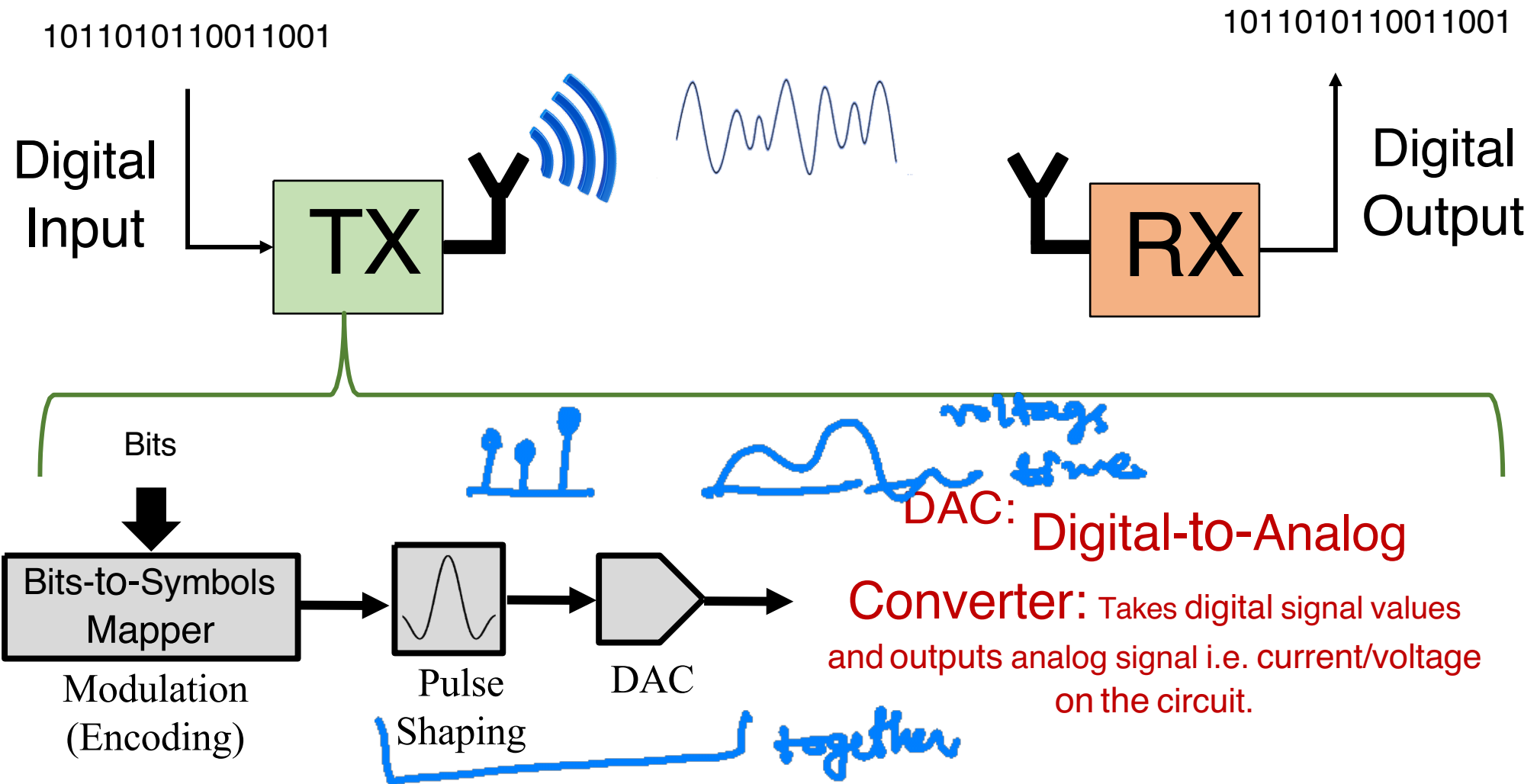
Wireless Communication System



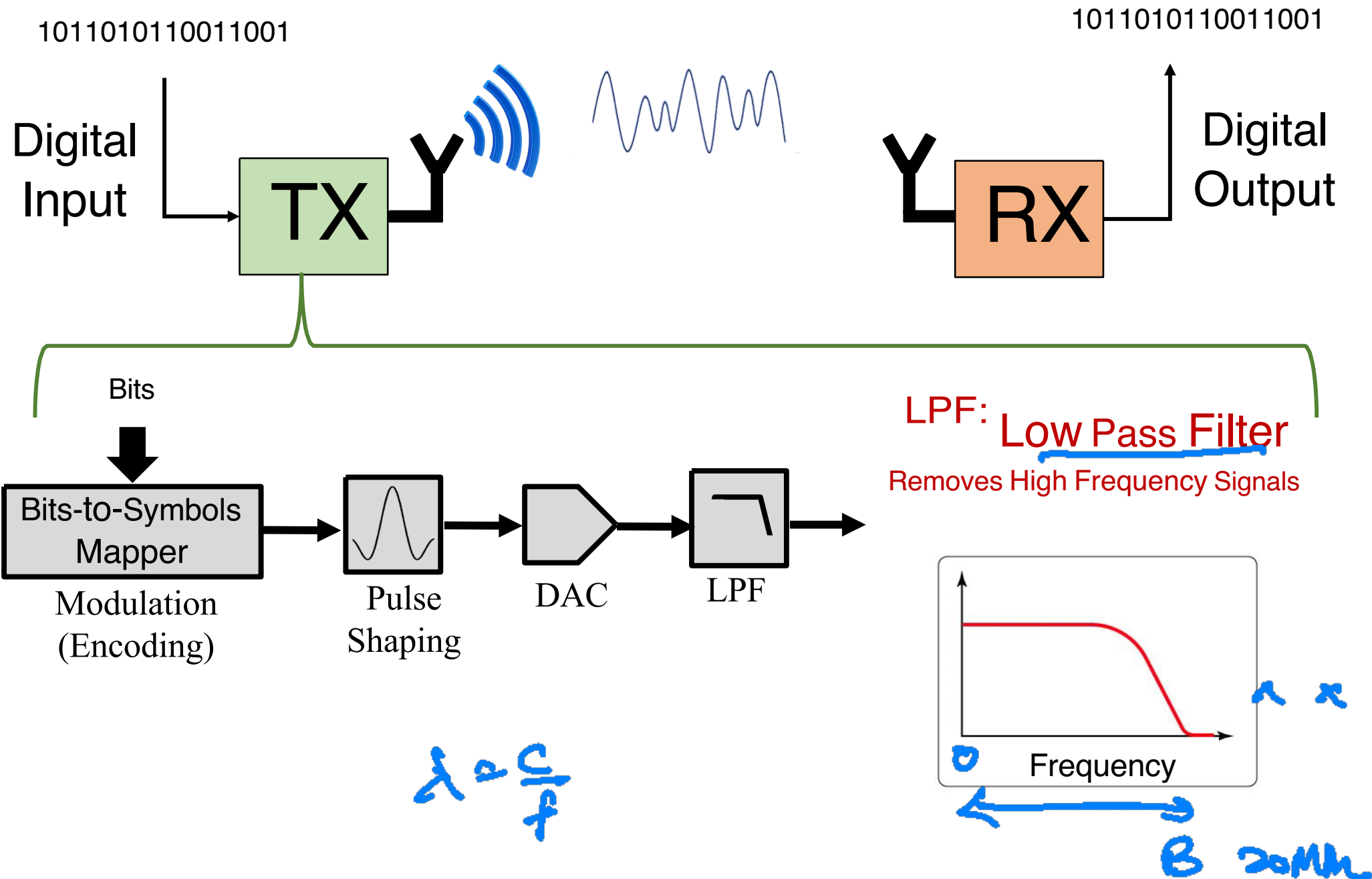
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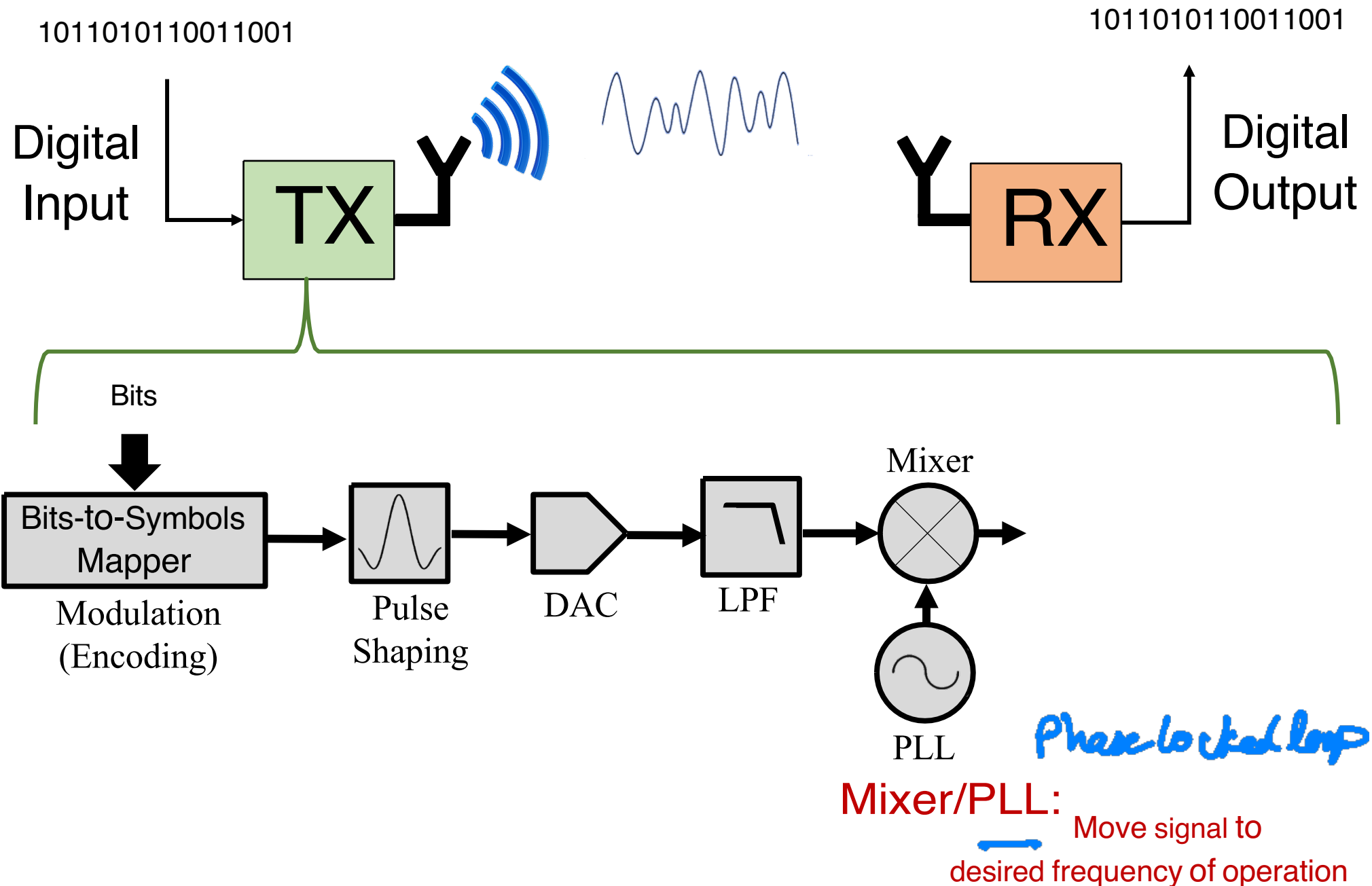
Wireless Communication System



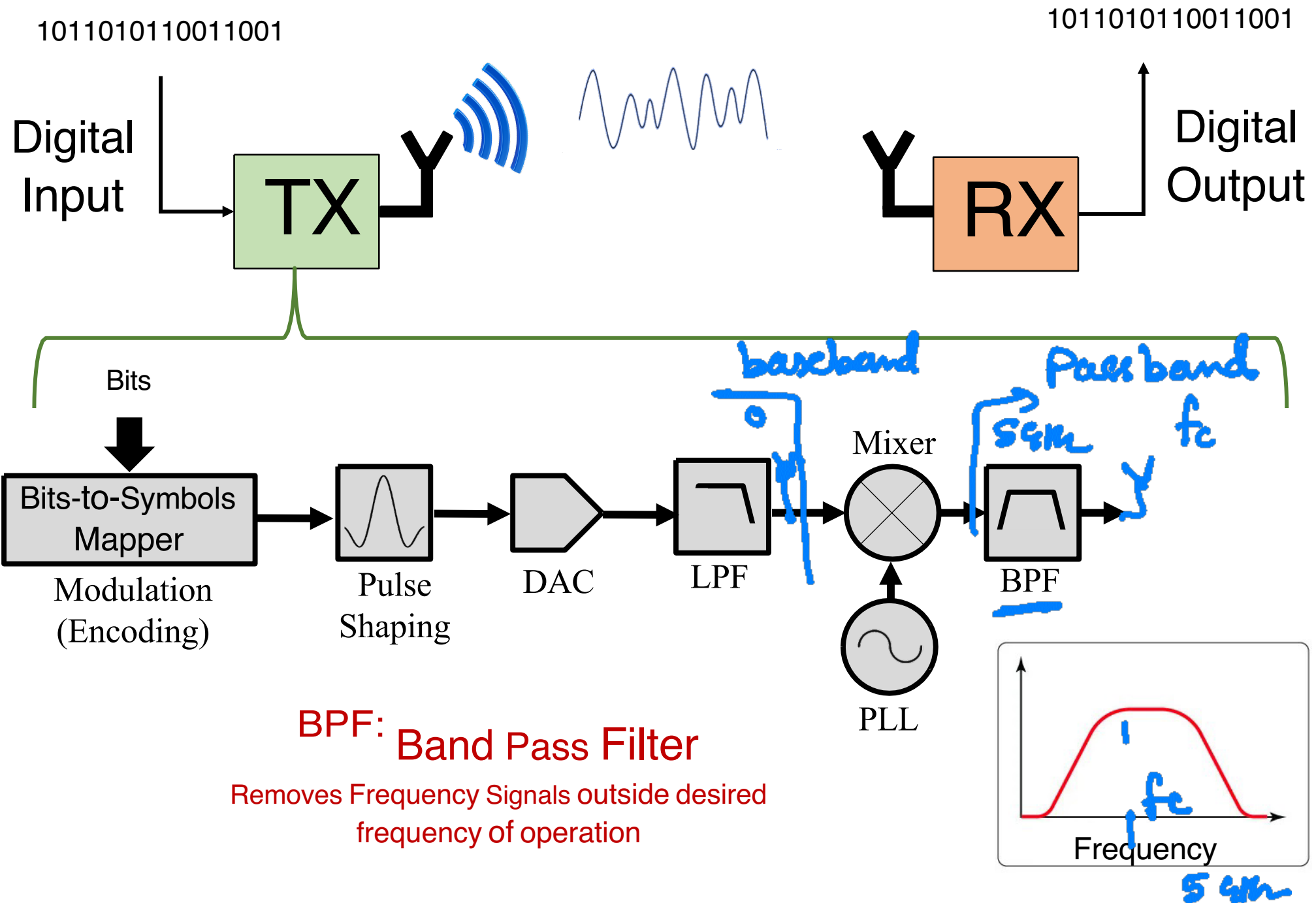
Wireless Communication System



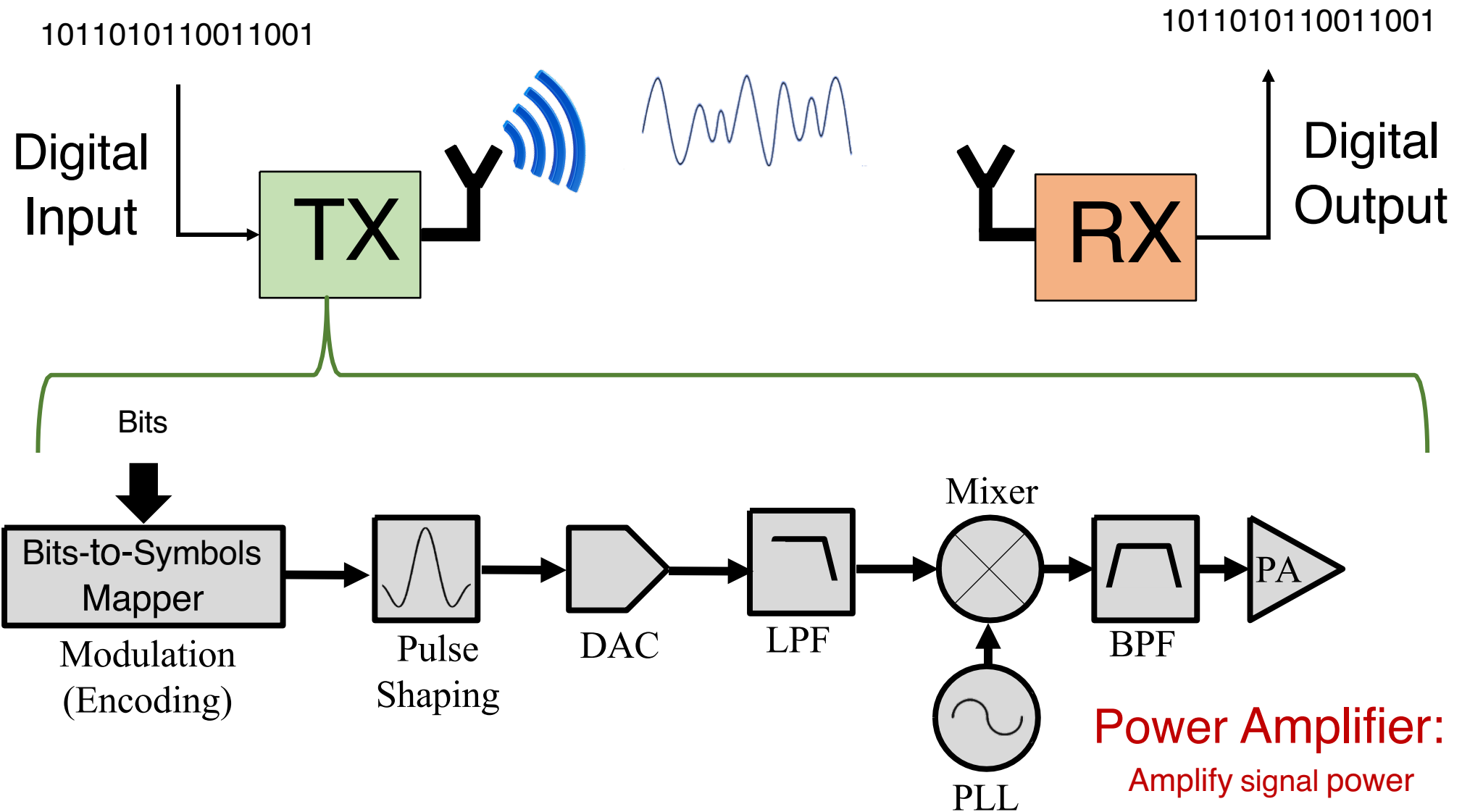
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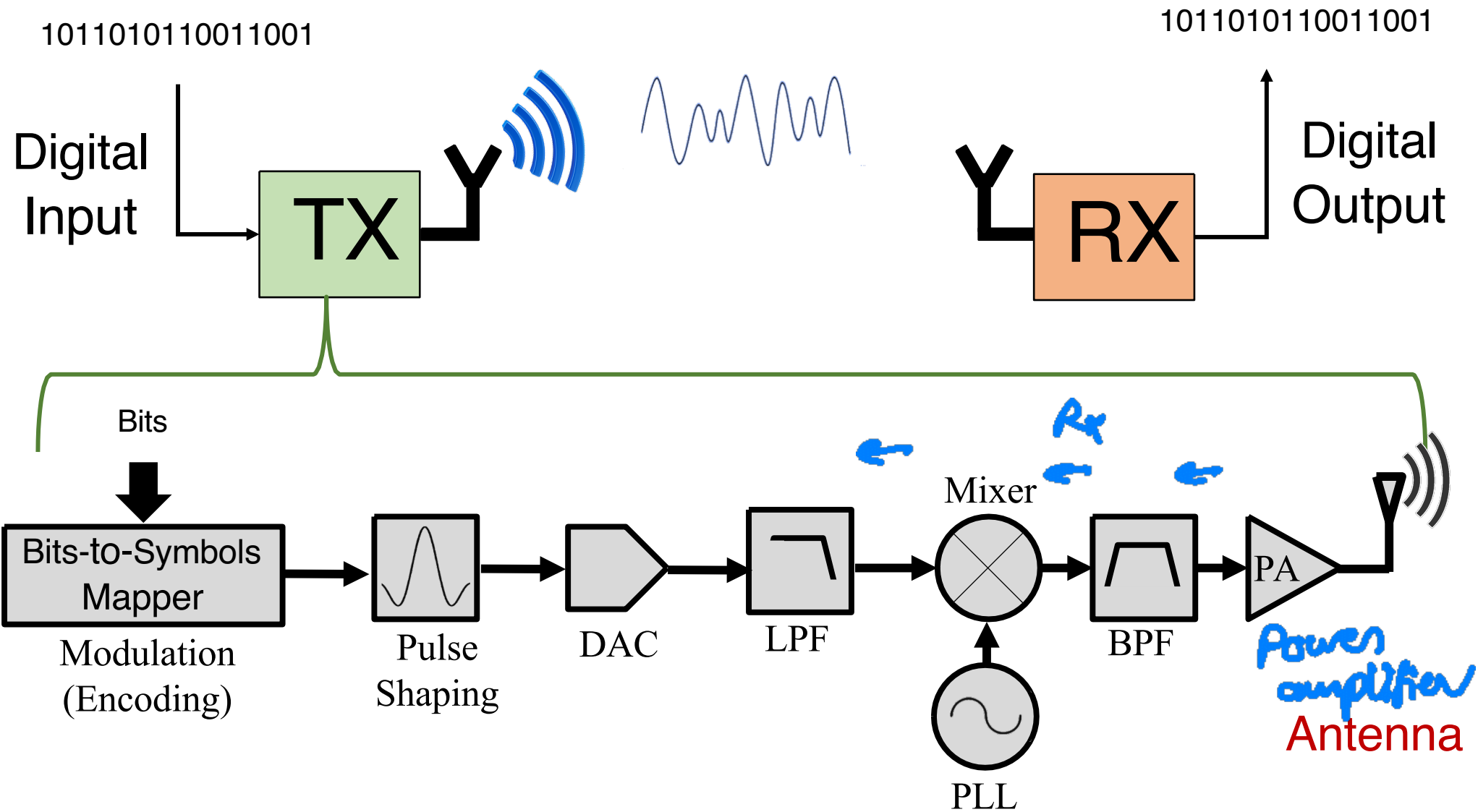
Wireless Communication System



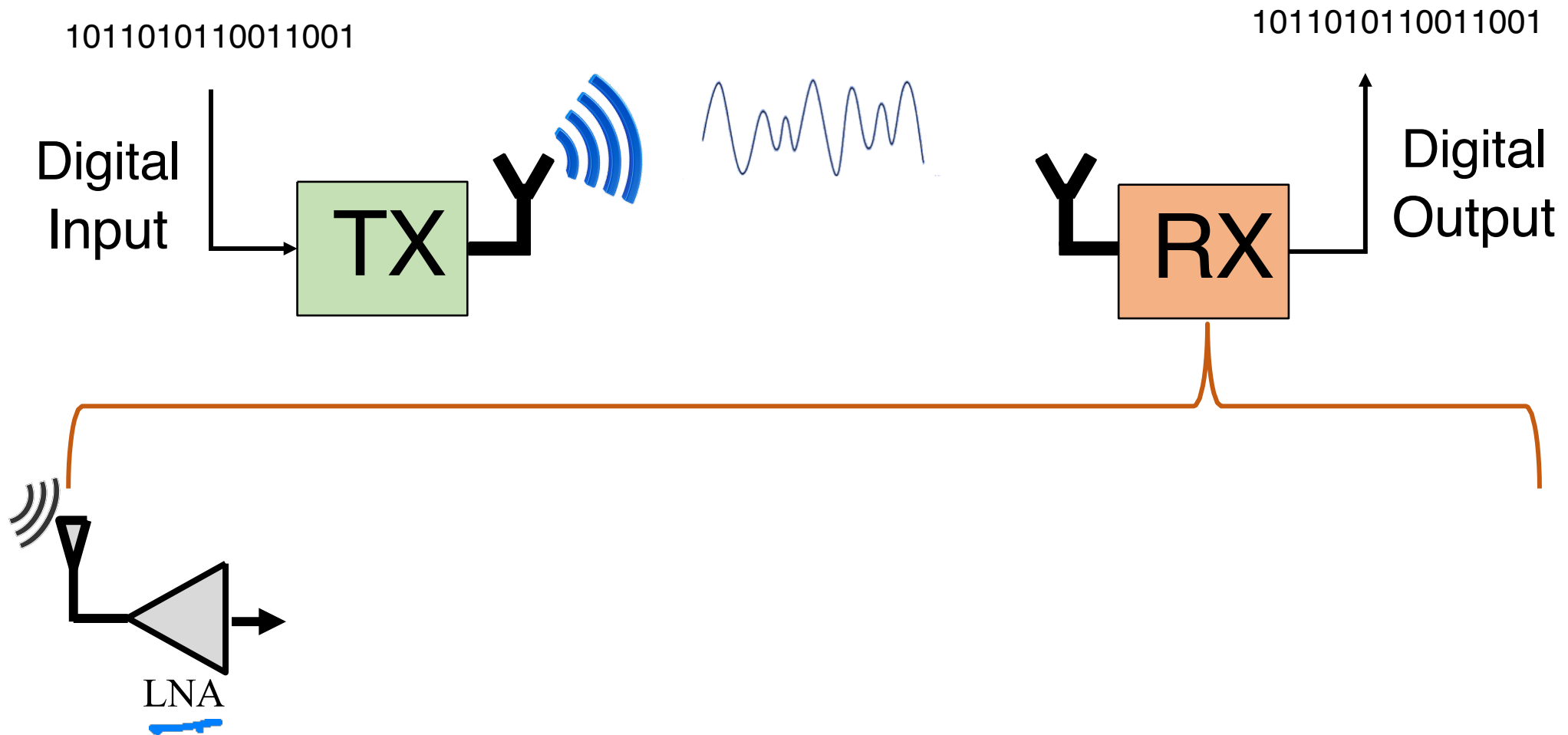
Wireless Communication System



Wireless Communication System



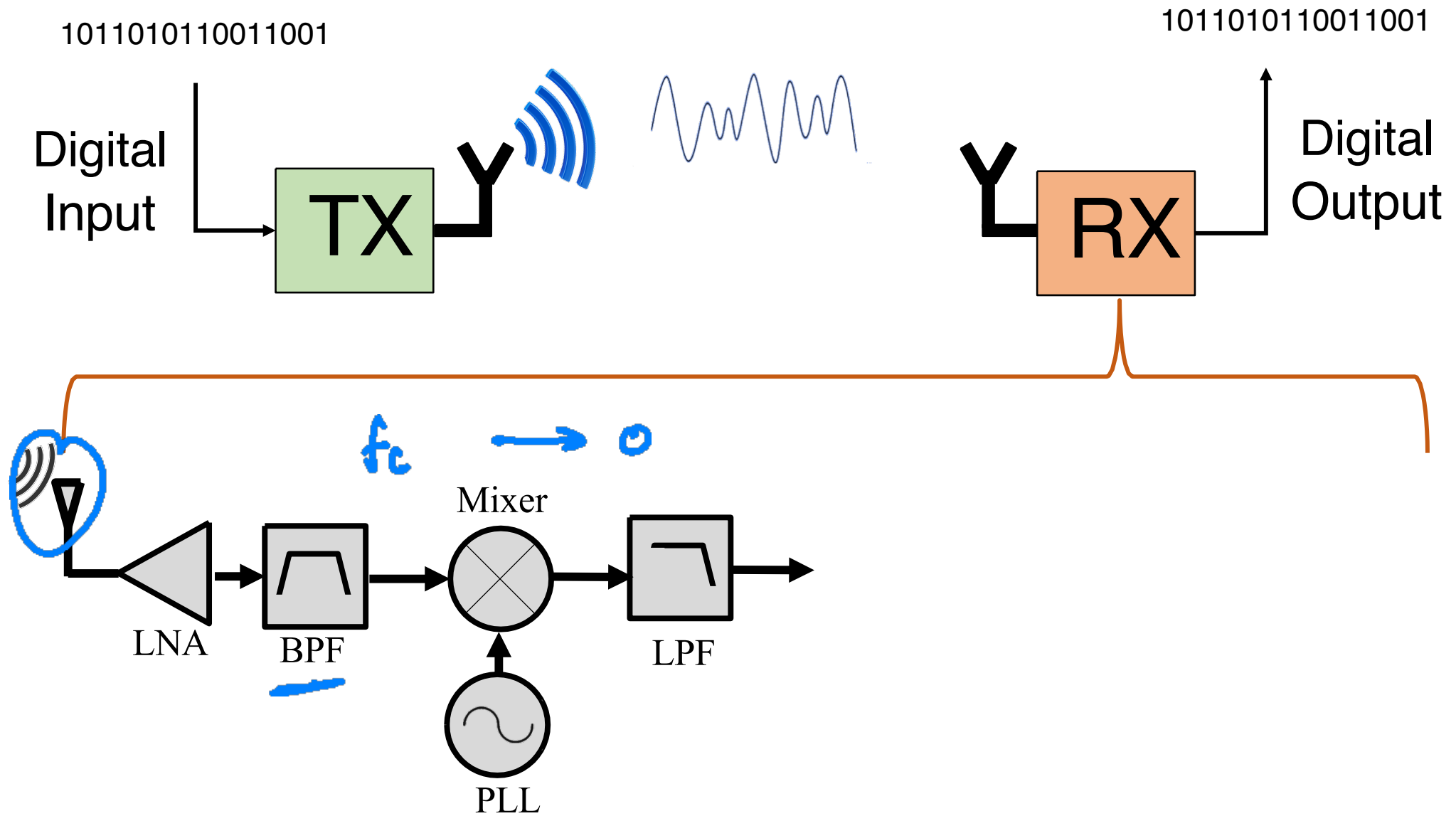
Wireless Communication System



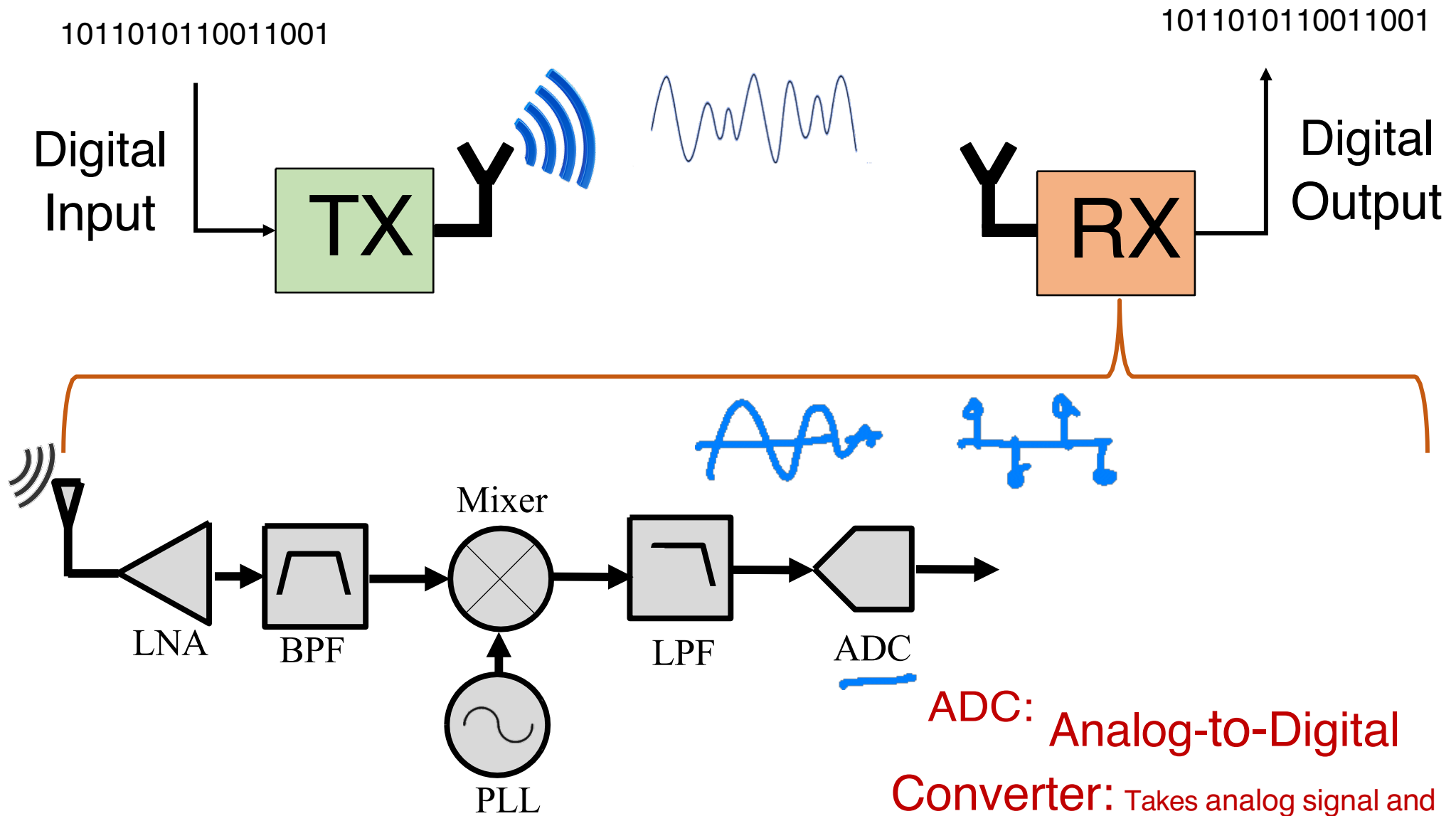
Low Noise Amplifier:

Amplify signal power with minimal
noise

Wireless Communication System

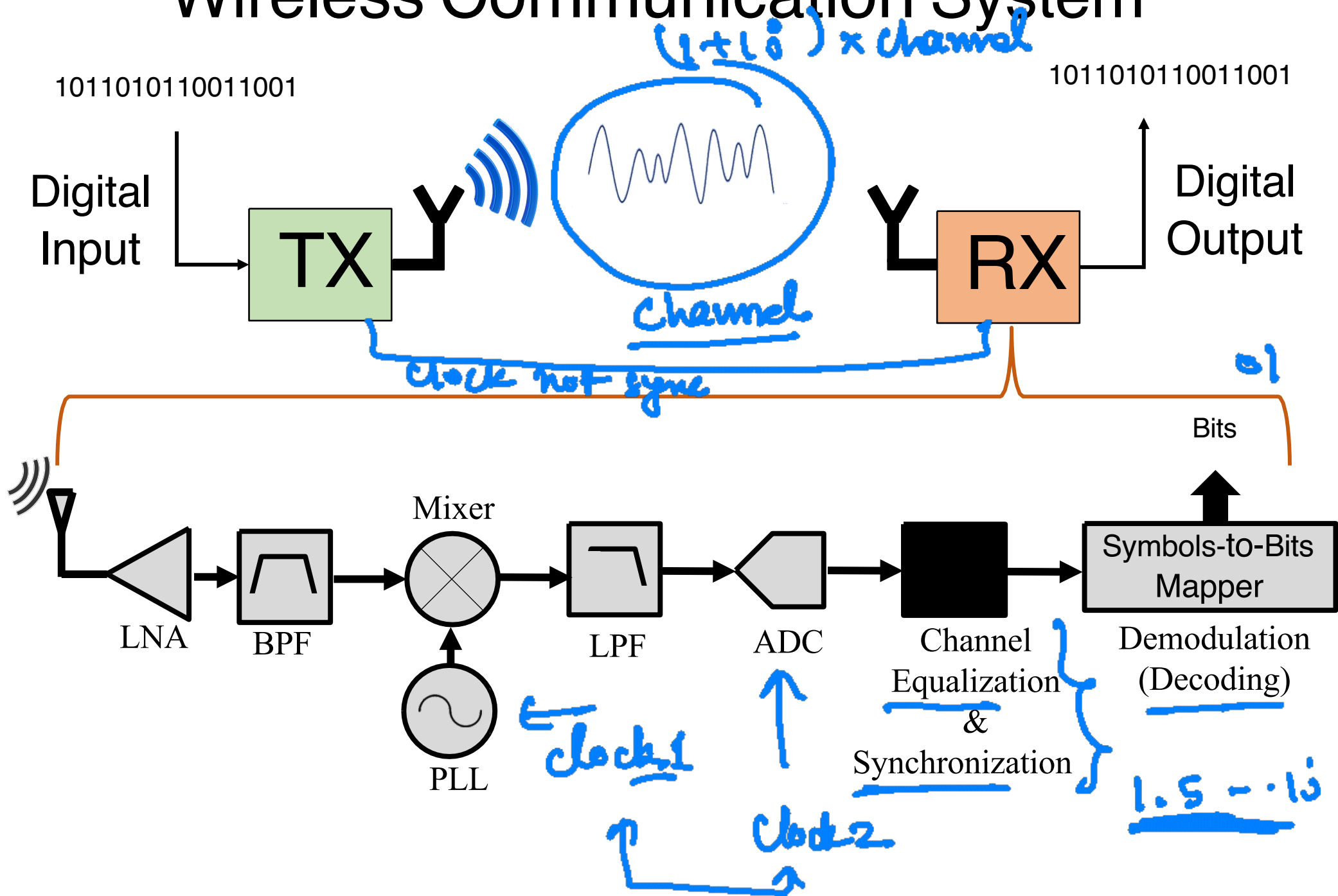


Wireless Communication System



ADC: Analog-to-Digital
Converter: Takes analog signal and digitizes it, i.e. samples and quantizes the signal values.

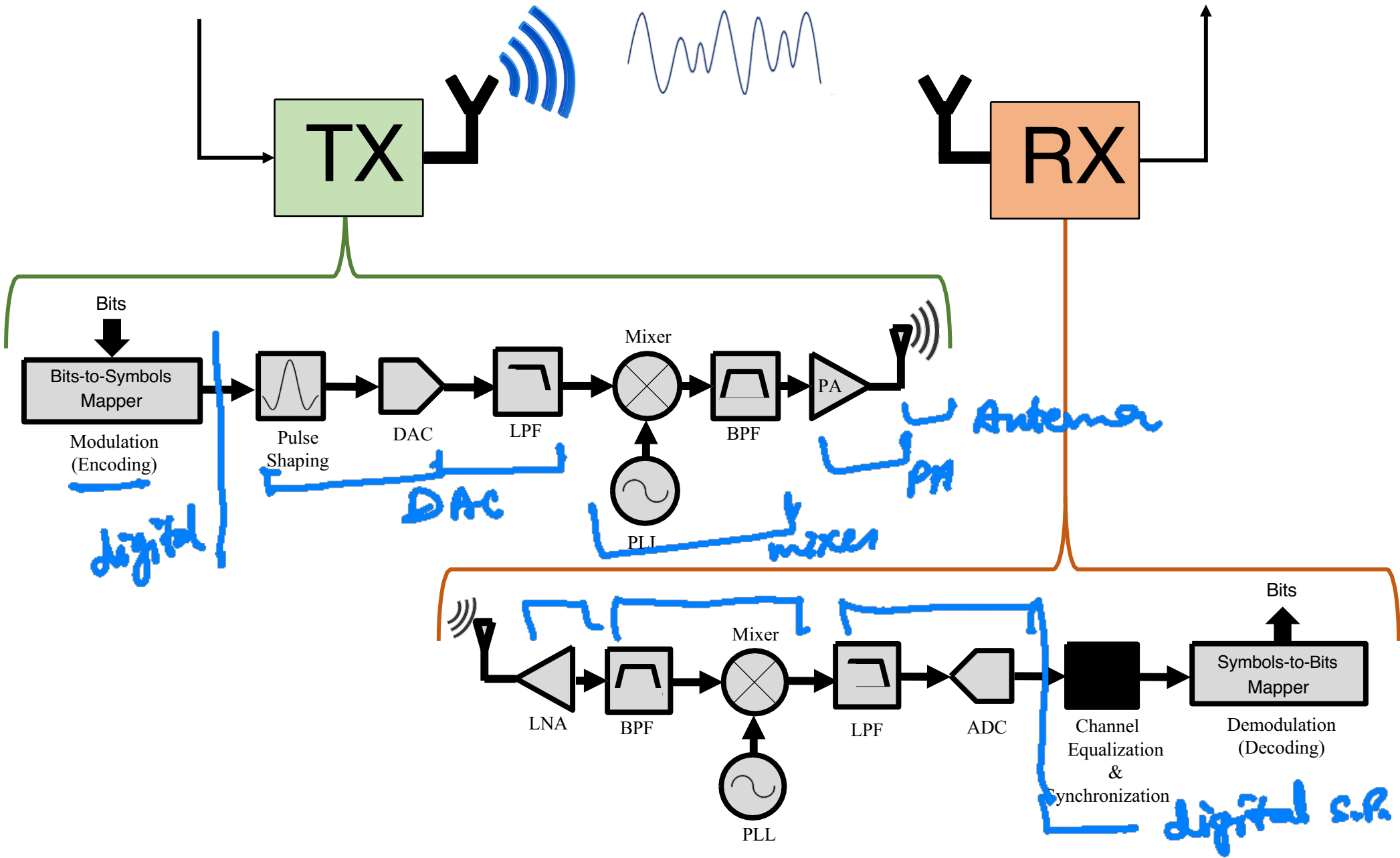
Wireless Communication System



Wireless Communication System

1011010110011001

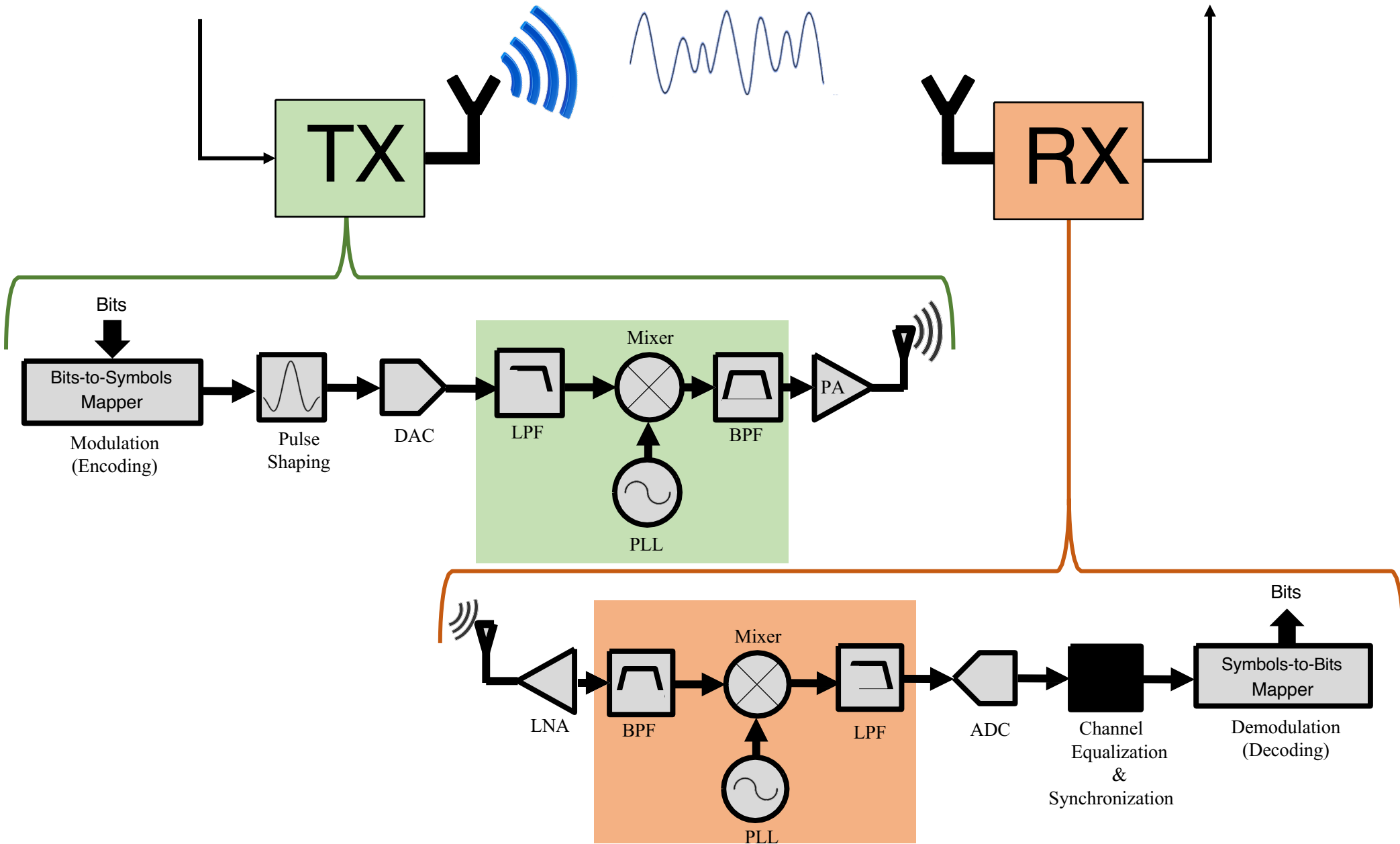
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Wireless Communication System

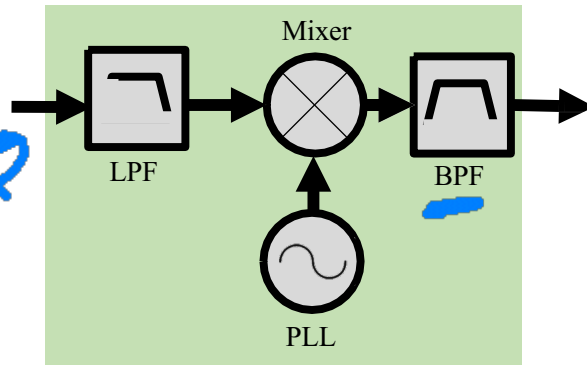
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1011010110011001

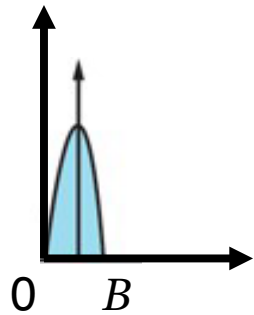


Up/Down Conversion

Transmitter
Up-conversion



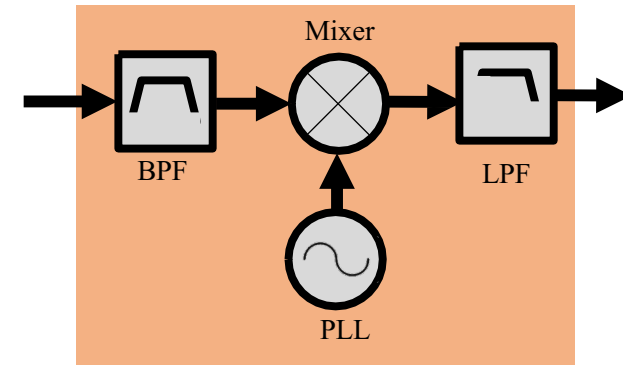
Baseband



Passband

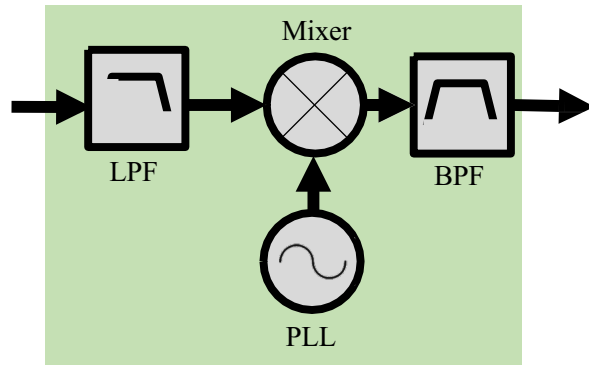


Receiver
Down-conversion

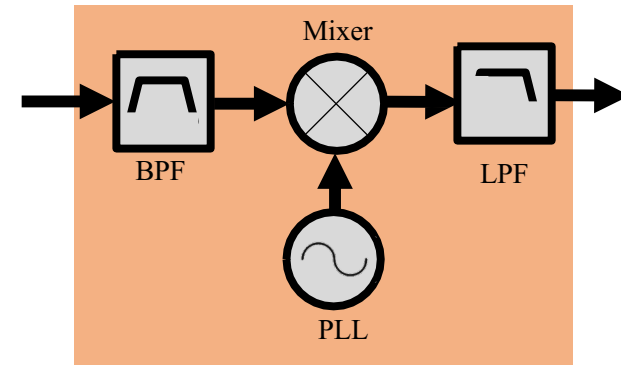


Up/Down Conversion

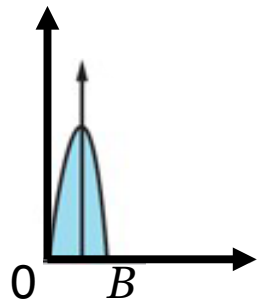
Transmitter
Up-conversion



Receiver
Down-conversion



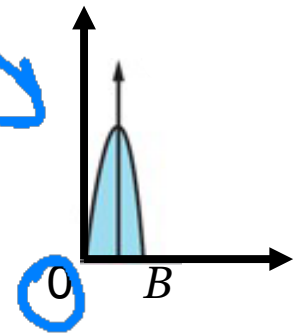
Baseband



Passband



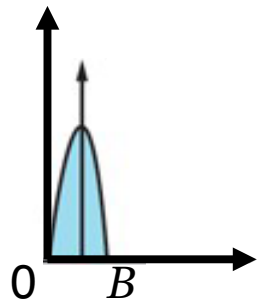
Baseband



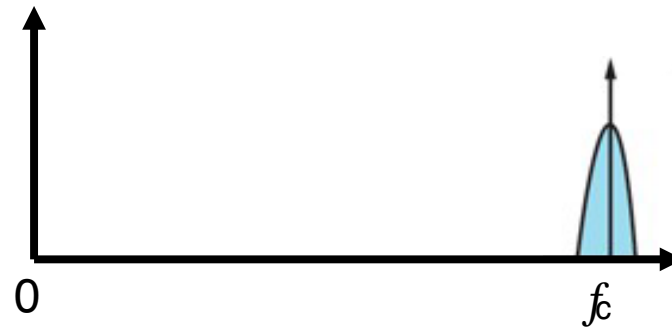
Why do we need to up/down conversion?

Up/Down Conversion

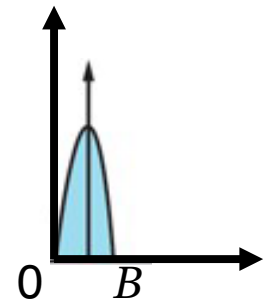
Baseband



Passband

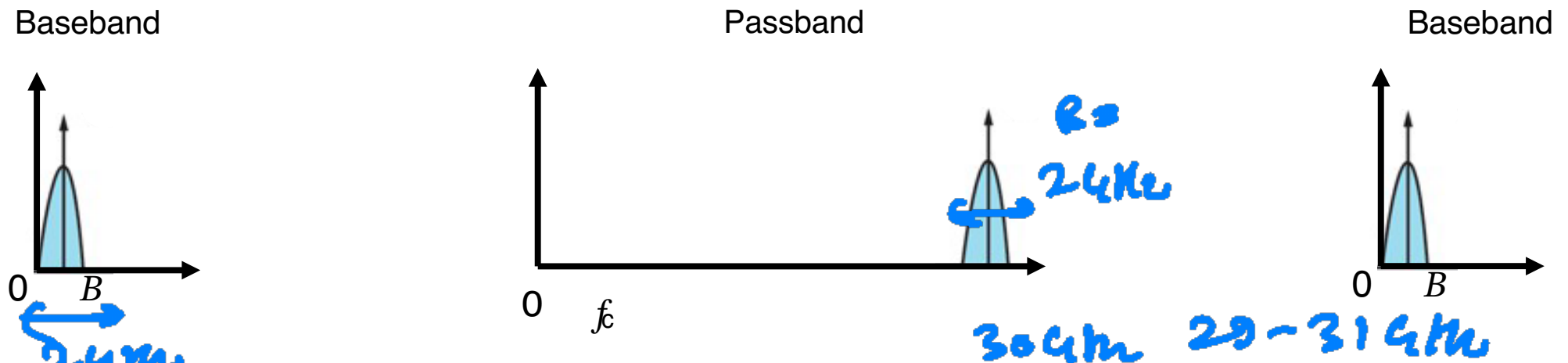


Baseband



Why do we need to up/down conversion?

Up/Down Conversion



Why do we need to up/down conversion?

Why not transmit everything at lower frequencies?

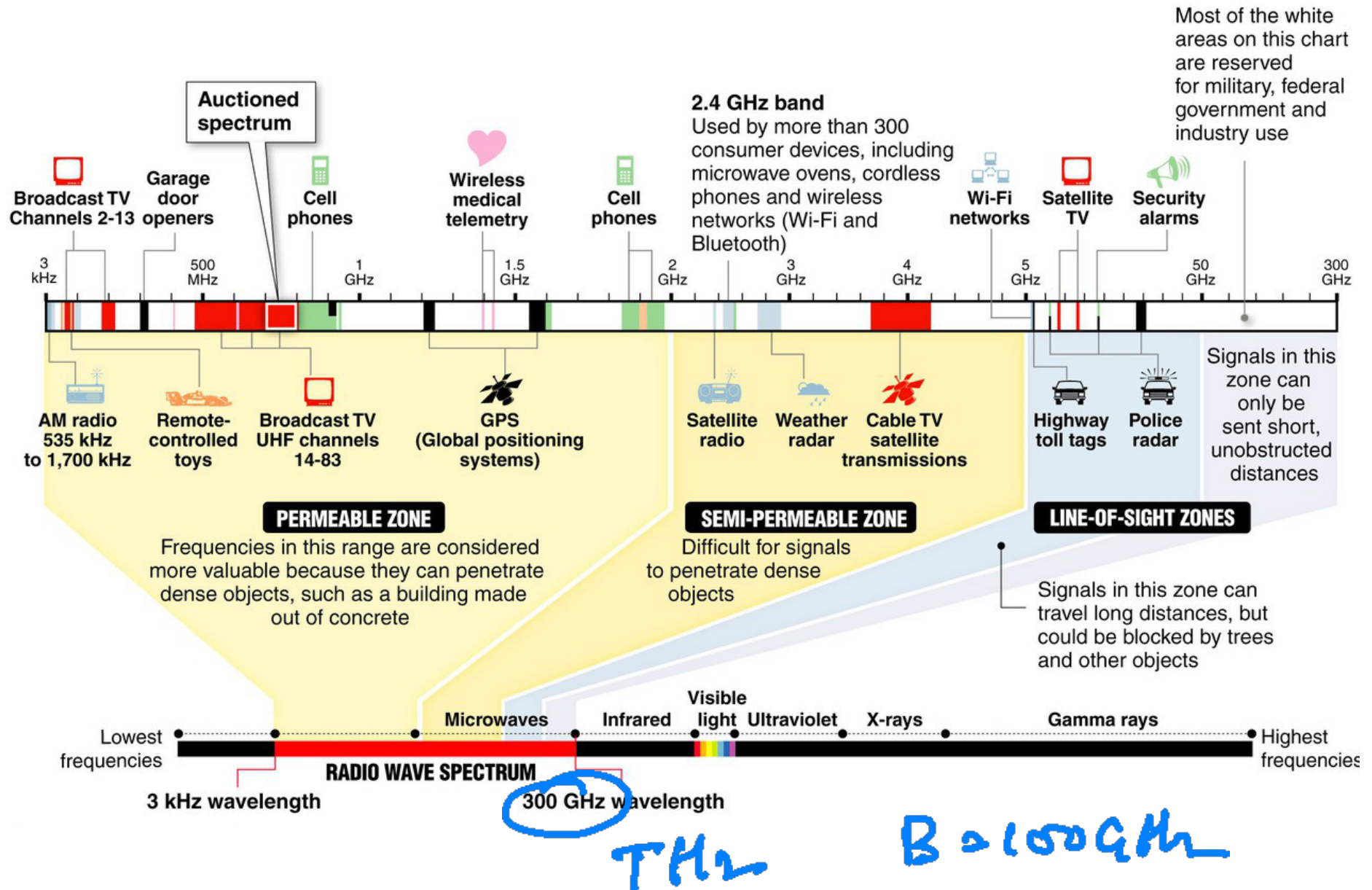
Not enough bandwidth:

Data rate $\propto$ bandwidth

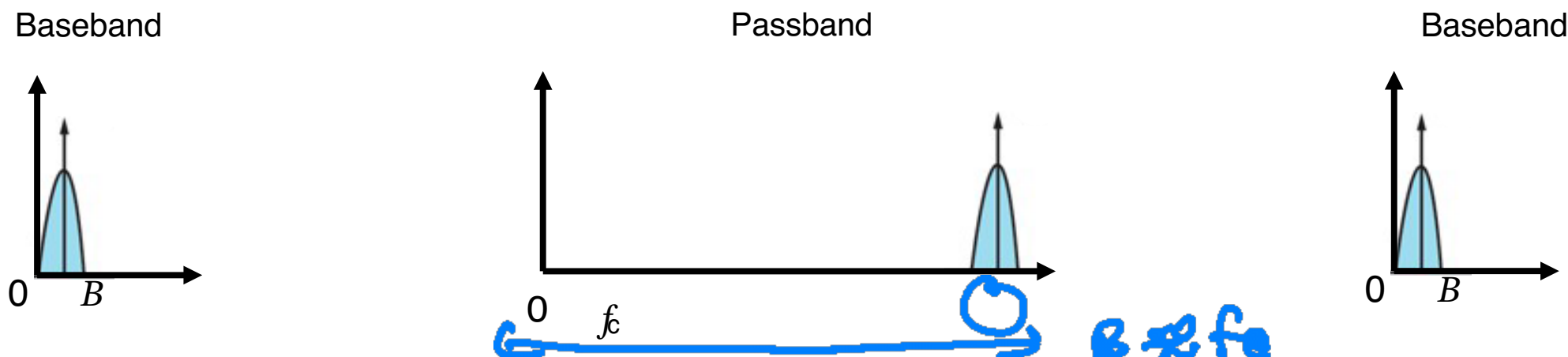
Different Technologies

Antenna size $\propto$ wavelength

The Wireless Spectrum



Up/Down Conversion



Why do we need to up/down conversion?

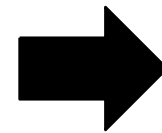
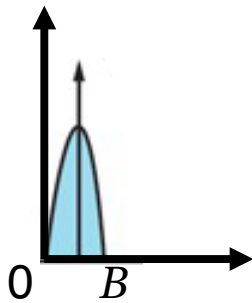
- Cannot transmit everything at low frequencies.
- Why not transmit & receive directly at high frequency?

Nyquist!

Nyquist Theorem

To recover the signal properly, we need to sample at twice the highest frequency, i.e. $2f_{\max}$

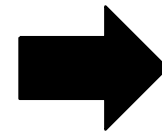
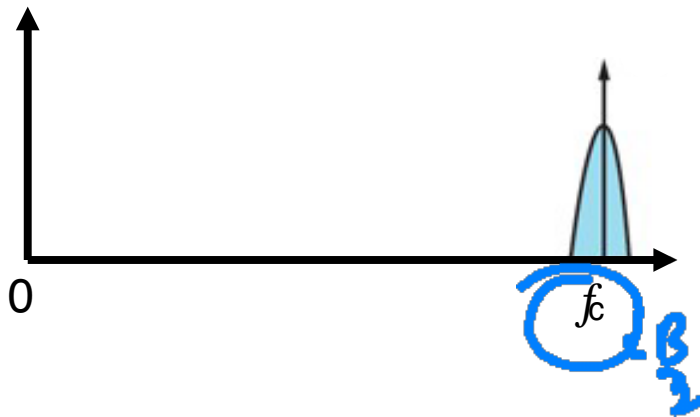
Baseband



Sample at $2B$

e.g. WiFi 802.11b ≈ 40 MS/s

Passband



Sample at $2f_c + B$

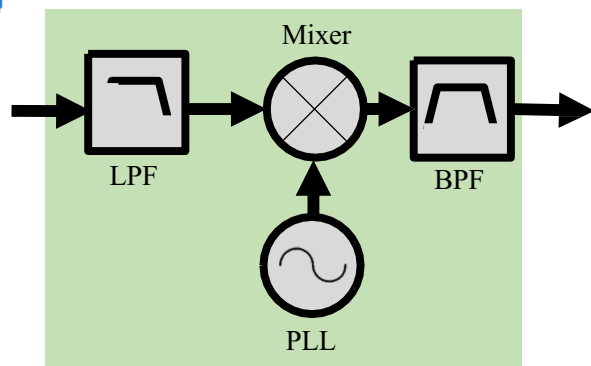
e.g. WiFi 802.11b ≈ 5 GS/s

Up/Down Conversion

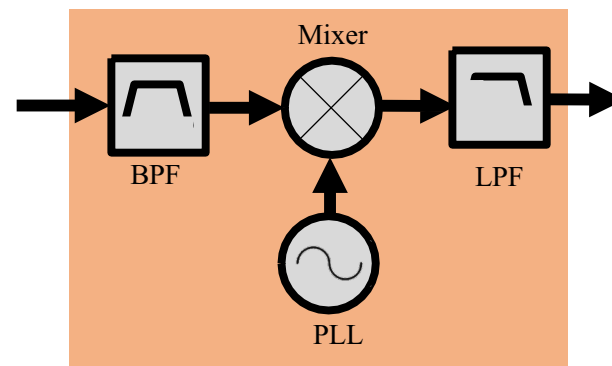
0101 $s(t)$ symbol

mod

Transmitter
Up-conversion



Receiver
Down-conversion



How to we do up/down conversion?

$$s(t) \xrightarrow{\times} s(t) \cos(2\pi f_c t) \quad x(t)$$

$$\cos(2\pi f_c t)$$

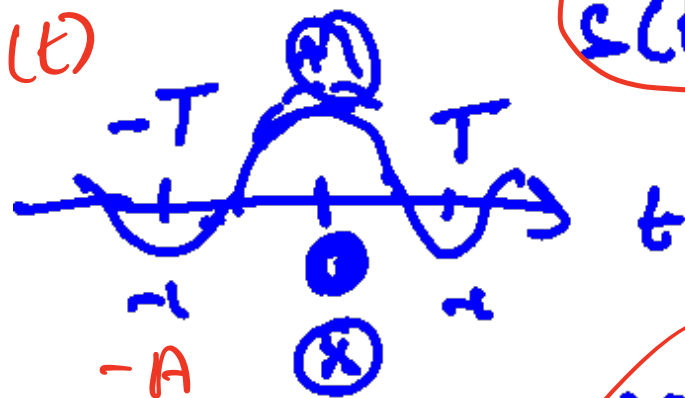
- PLL (Phased Lock Loop): create the cosine wave signal $\cos(2\pi f_c t)$



- Mixer: multiplies the two signals

BPSK + A

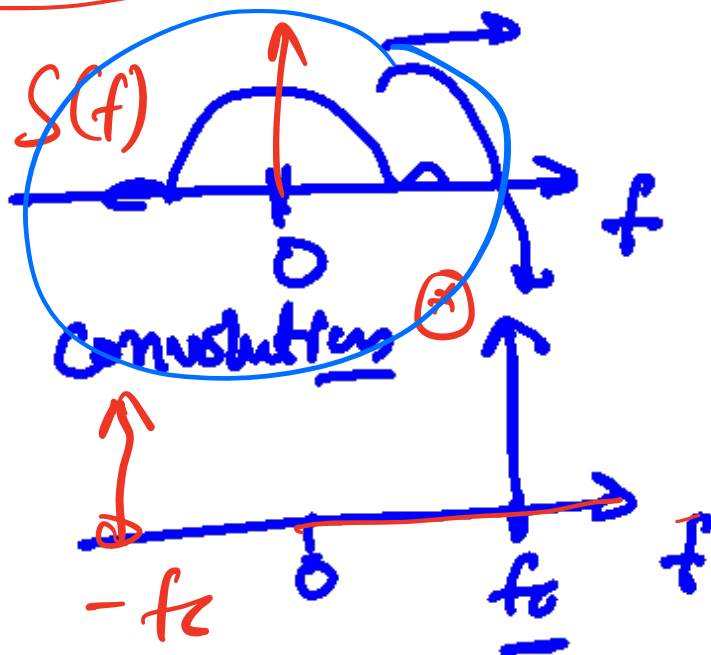
$s(t)$



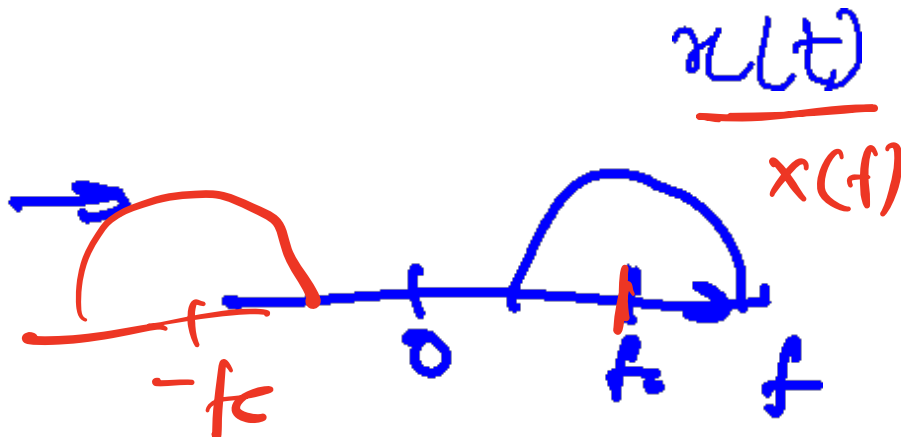
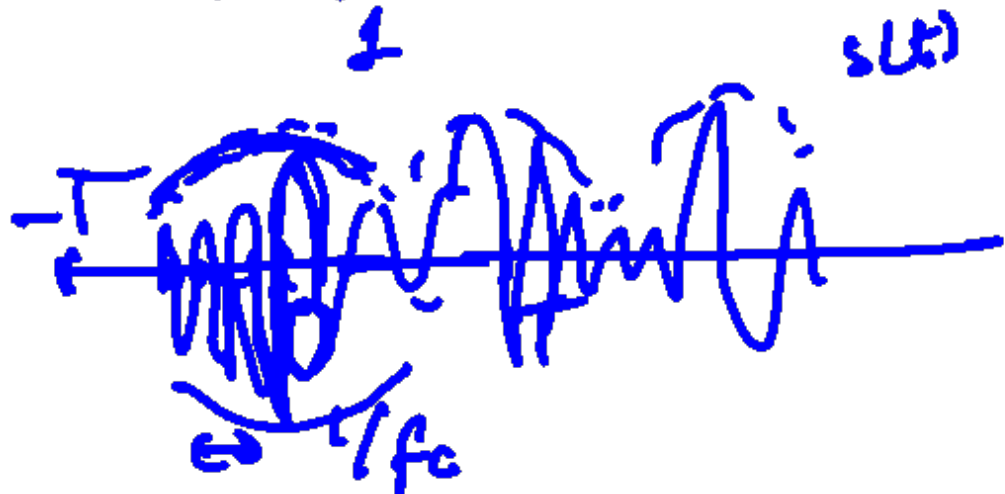
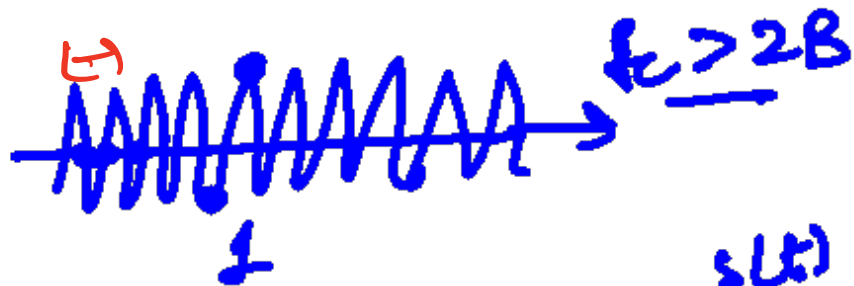
$s(t)$

$$\frac{1}{T} \int_0^T$$

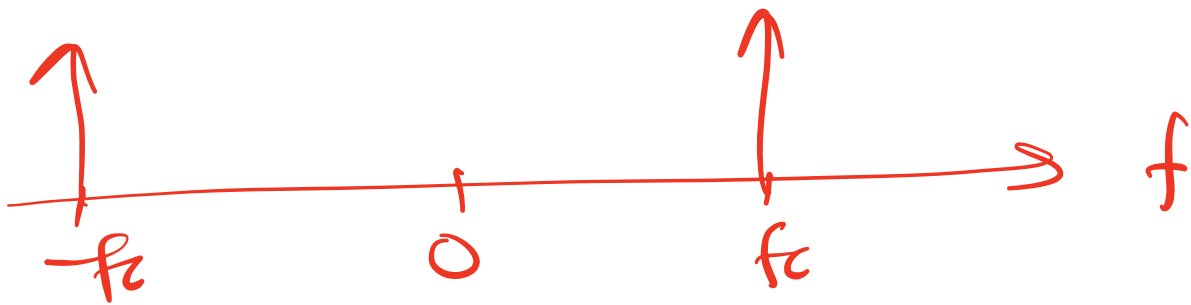
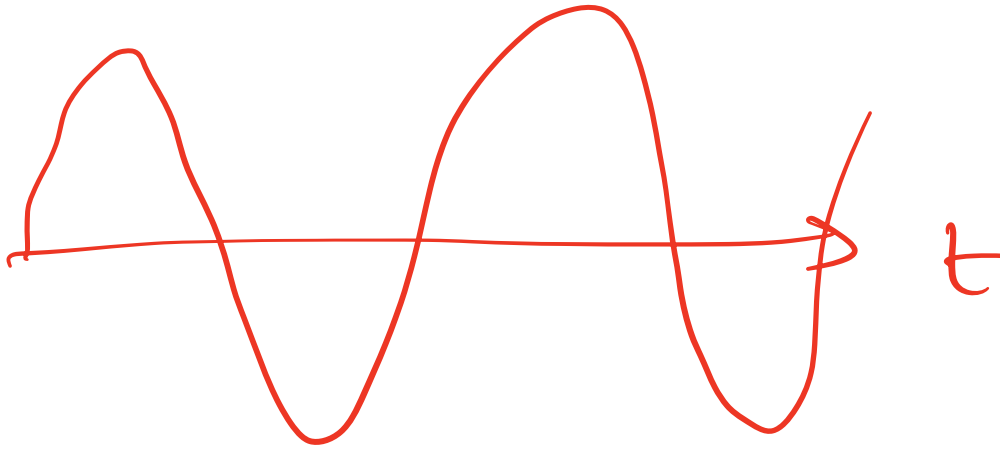
$$\cos 2\pi f_c t e^{-j2\pi f_c t} dt$$



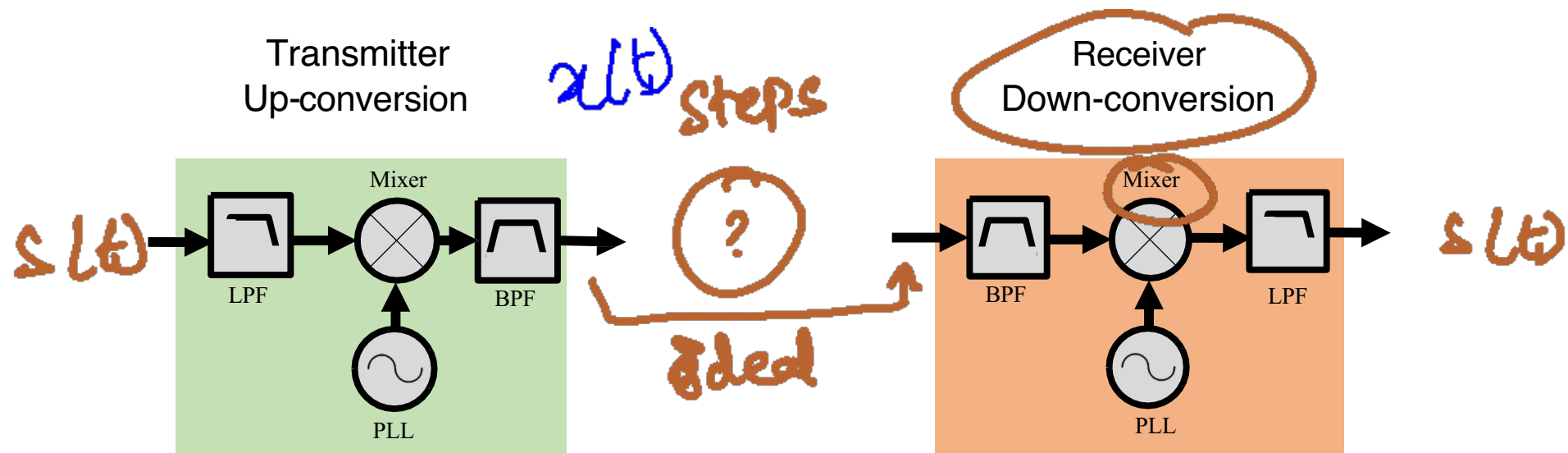
$\cos 2\pi f_c t$



$\cos 2\pi f_c t$ $\rightarrow \frac{1}{2} (e^{j2\pi f_c t} + e^{-j2\pi f_c t})$



Up/Down Conversion



How to we do up/down conversion?

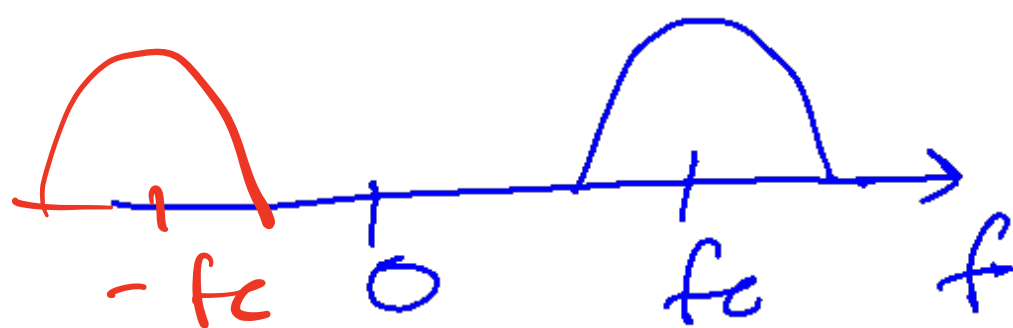
$$s(t) \xrightarrow{\cos(2\pi f_c t)} s(t) \cos(2\pi f_c t) \xrightarrow{\cos(2\pi f_c t)} s(t) \cos^2(2\pi f_c t)$$

Handwritten notes: $s(t)$ is circled, $\cos(2\pi f_c t)$ is circled, and $s(t) \cos(2\pi f_c t)$ is underlined. A box contains the identity: $2\cos^2\theta = 1 + \cos 2\theta$.

$$s(t) \cos^2(2\pi f_c t) = s(t) \left(\frac{1}{2} + \frac{1}{2} \cos(2\pi \cdot 2f_c t) \right)$$

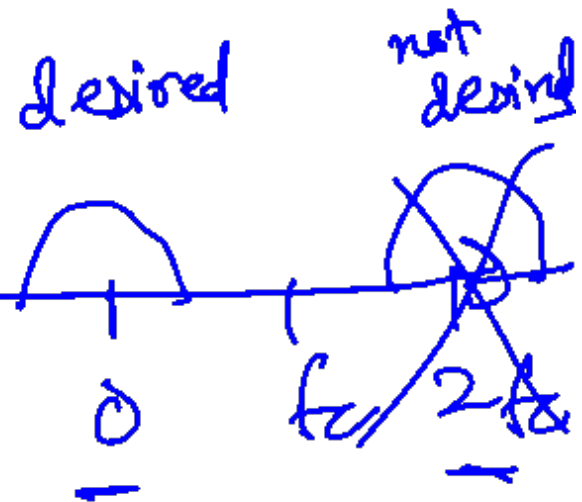
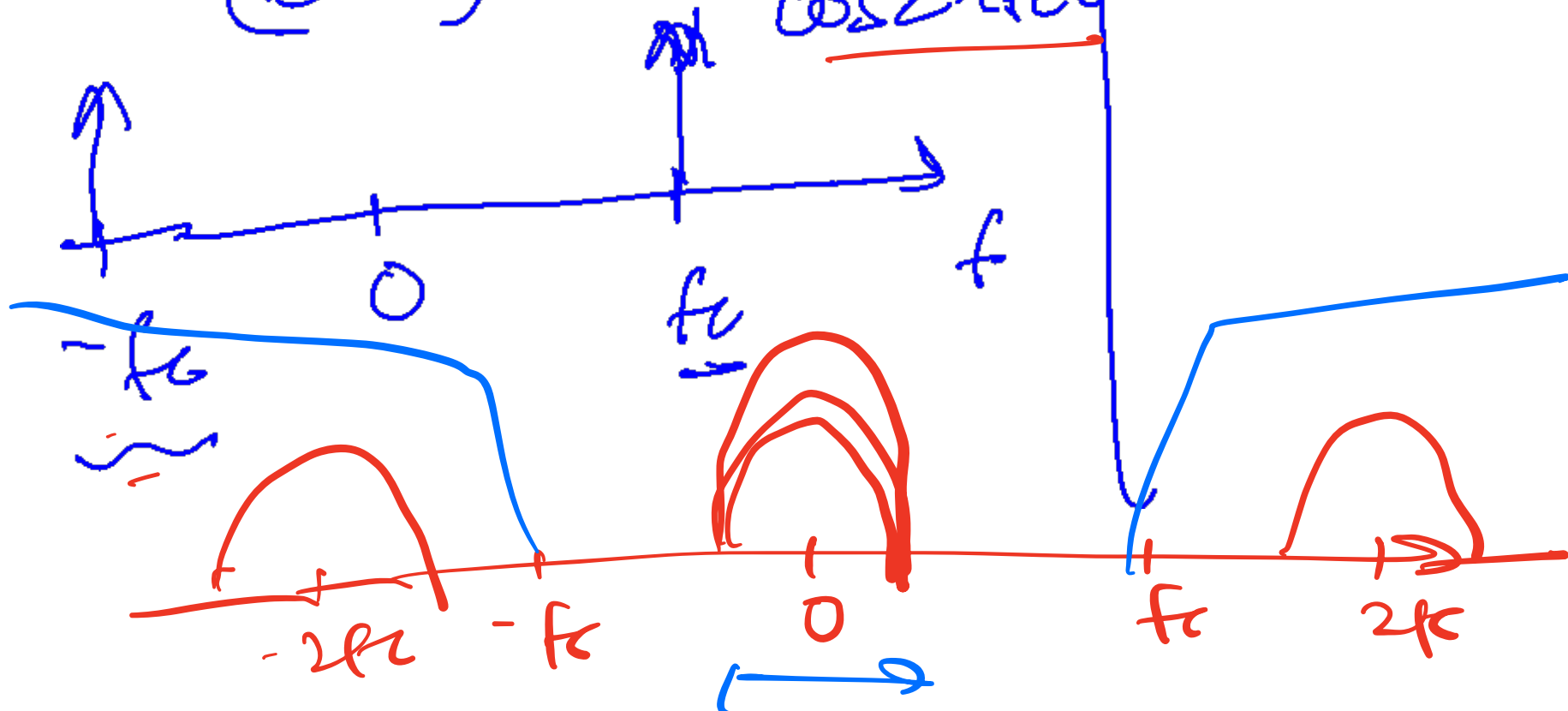
Handwritten notes: $s(t)$ is circled, $\frac{1}{2}$ is circled, and $2f_c$ is circled.

Rx chain $X(f)$



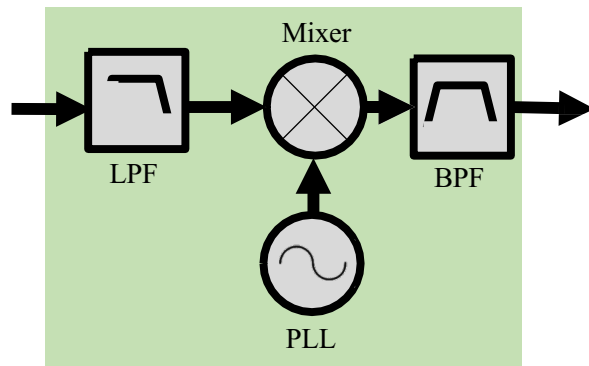
$\otimes$ (conv.)

$\cos 2\pi f c$

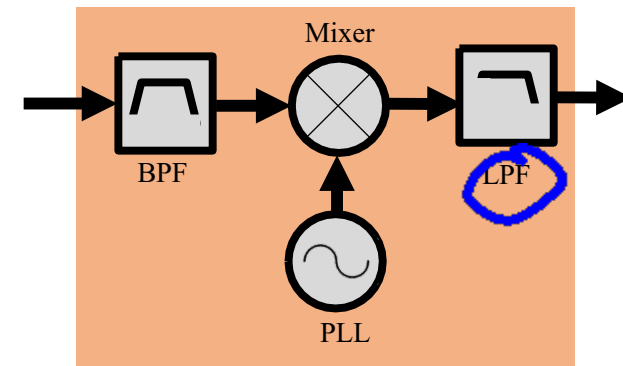


Up/Down Conversion

Transmitter
Up-conversion



Receiver
Down-conversion



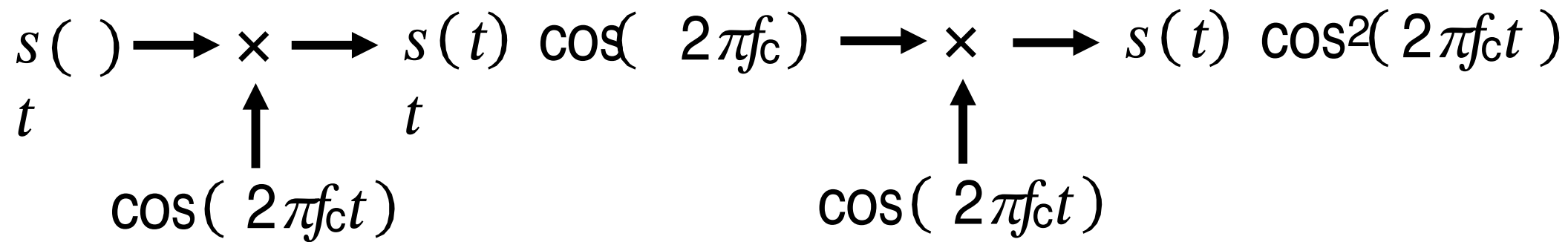
How to we do up/down conversion?

$$s(t) \xrightarrow{\times \cos(2\pi f_c t)} s(t) \cos(2\pi f_c t) \xrightarrow{\times \cos(2\pi f_c t)} s(t) \cos^2(2\pi f_c t)$$

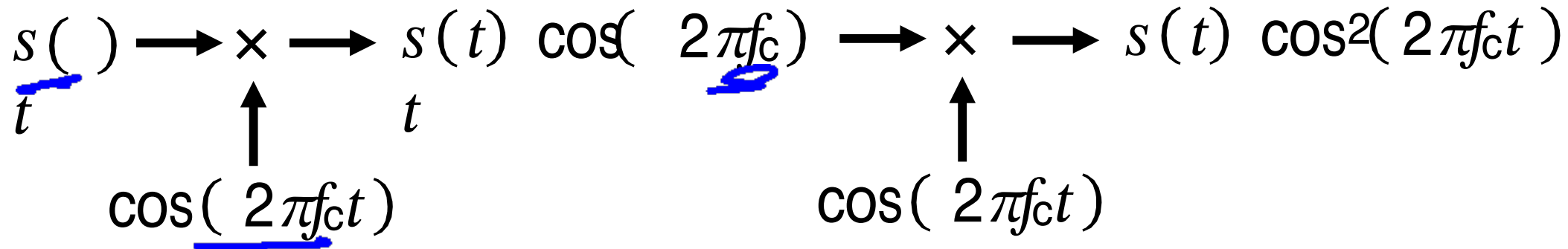
$$s(t) \cos^2(2\pi f_c t) = s(t) \left(\frac{1}{2} + \frac{1}{2} \cos(2\pi 2f_c t) \right) = \frac{1}{2} s(t)$$

Handwritten annotations: A red arrow points from the $\cos(2\pi 2f_c t)$ term to the word "Low" (part of "Low Pass Filter"). A red circle highlights the $\cos(2\pi 2f_c t)$ term with the word "LPF" written above it.

Up/Down Conversion



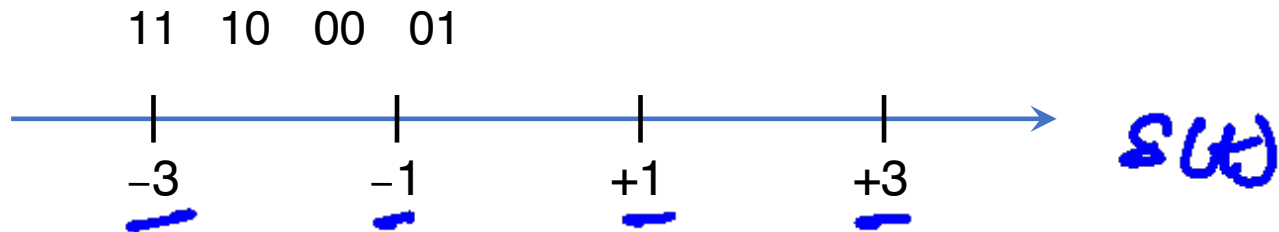
Up/Down Conversion



Consider using PAM: Pulse Amplitude Modulation

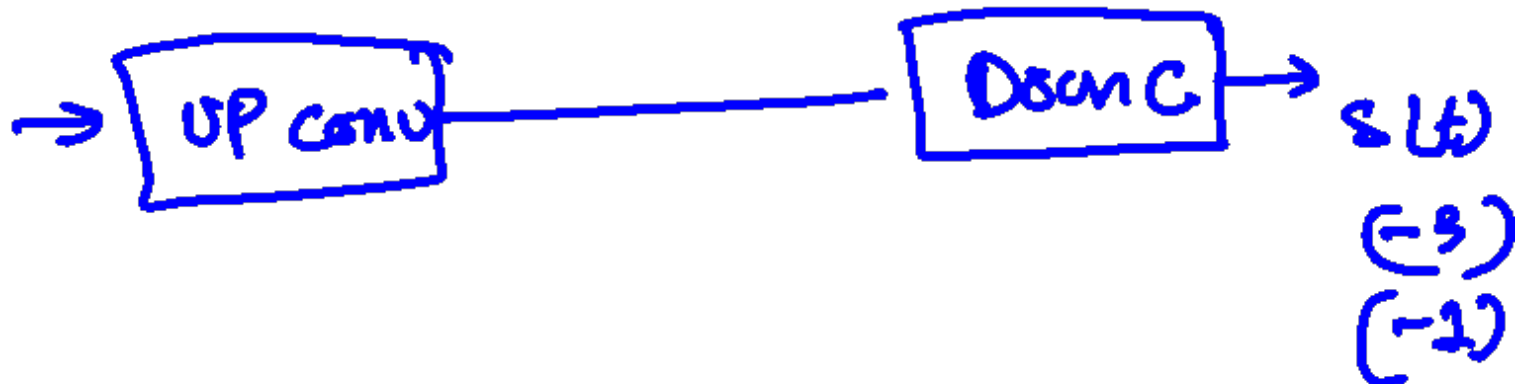
$2^2 = 4$

4 PAM:
2 bits/symbol

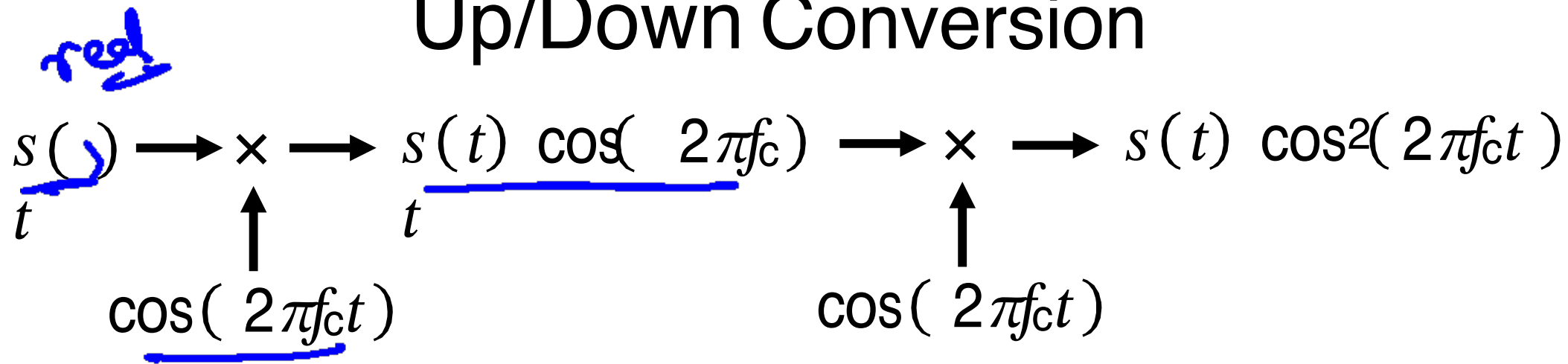


red

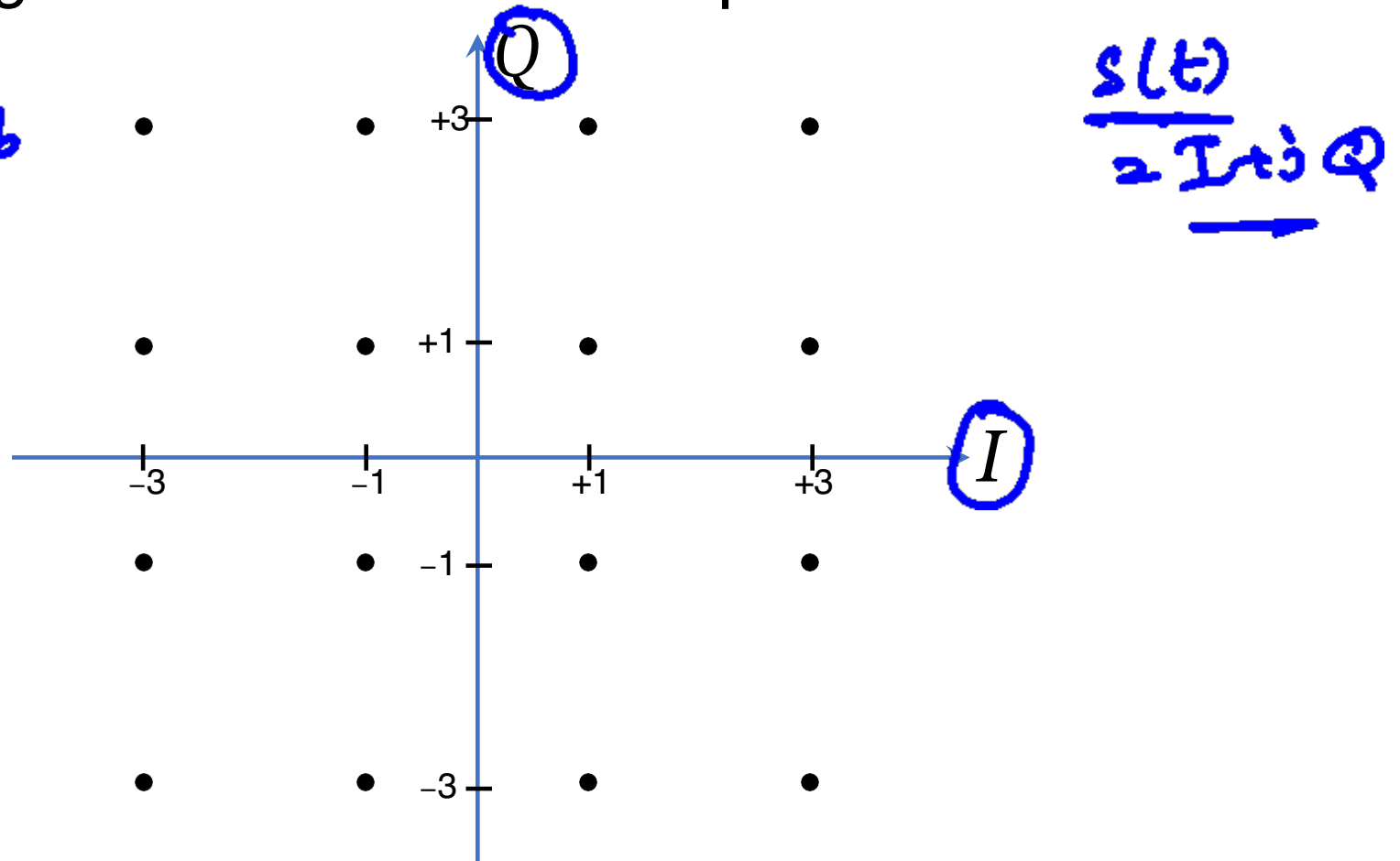
$s(t)$
 (-3)
 (-1)



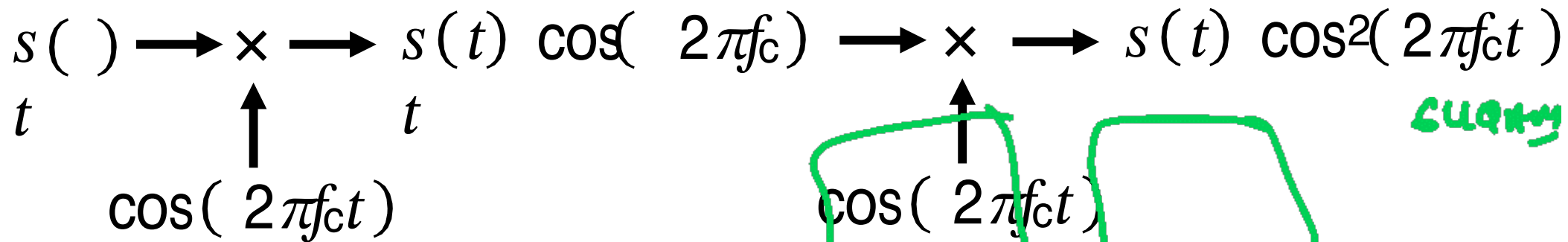
Up/Down Conversion



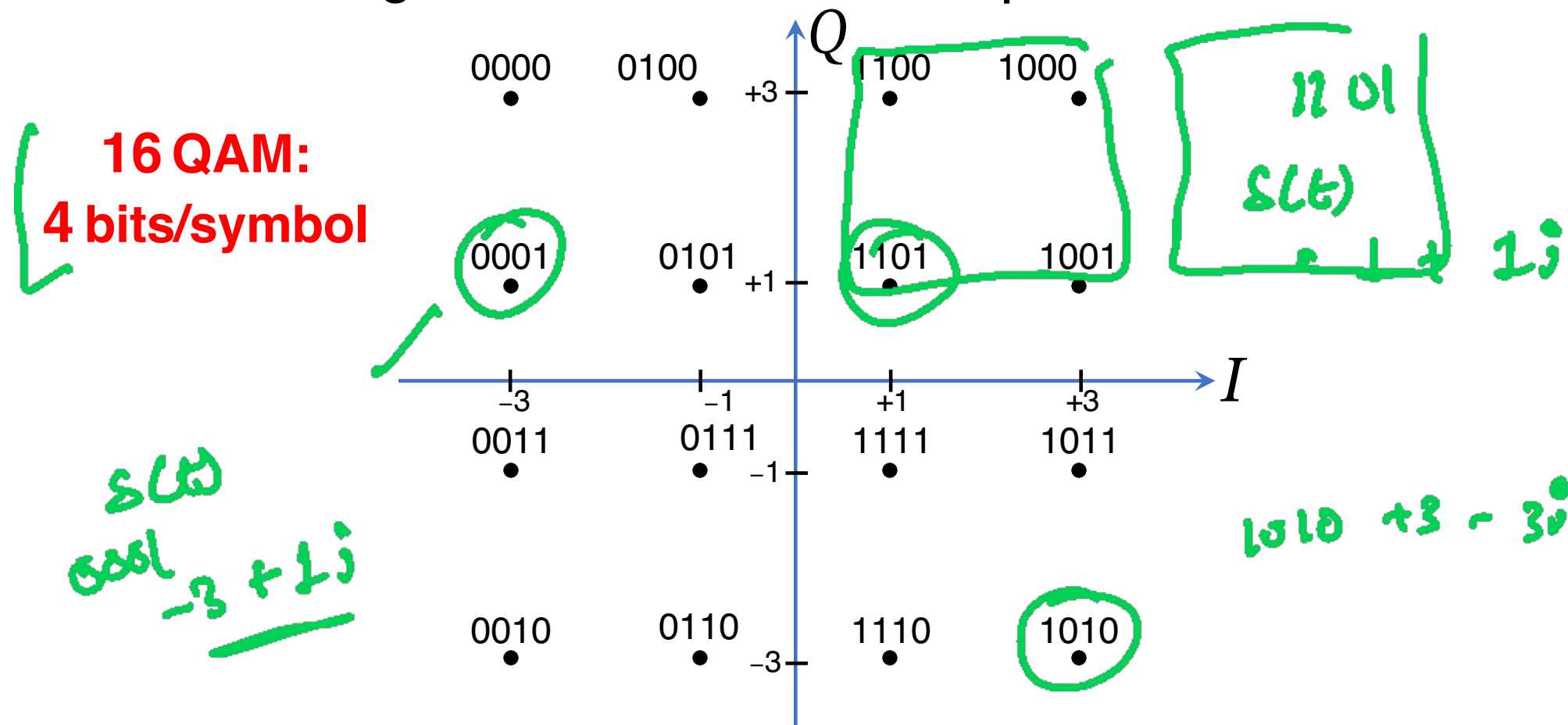
Consider using QAM: Quadrature Amplitude Modulation



Up/Down Conversion



Consider using QAM: Quadrature Amplitude Modulation



Up/Down Conversion

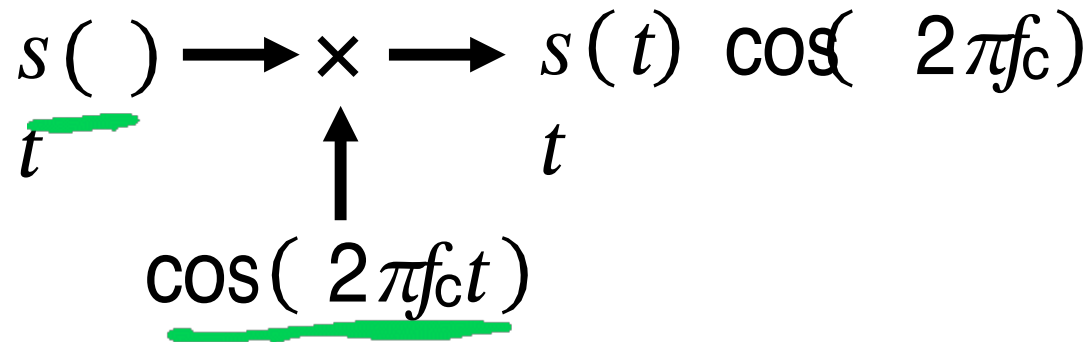
$$s(t) \xrightarrow{\cos(2\pi f_c t)} s(t) \cos(2\pi f_c t) \xrightarrow{\cos(2\pi f_c t)} s(t) \cos^2(2\pi f_c t)$$

The diagram illustrates the Up/Down Conversion process. It starts with a signal $s(t)$ on the left. An arrow points to a multiplication node (represented by $\times$) with a carrier wave $\cos(2\pi f_c t)$ entering from below. The result of the first multiplication is $s(t) \cos(2\pi f_c t)$. A green underline is placed under the $s(t)$ in this intermediate result. An arrow then points to a second multiplication node ($\times$) with another $\cos(2\pi f_c t)$ entering from below. The final result is $s(t) \cos^2(2\pi f_c t)$.

Consider using QAM: Quadrature Amplitude Modulation

$$s(t) \text{ is complex: } \underline{s(t) = I + jQ}$$

Up Conversion

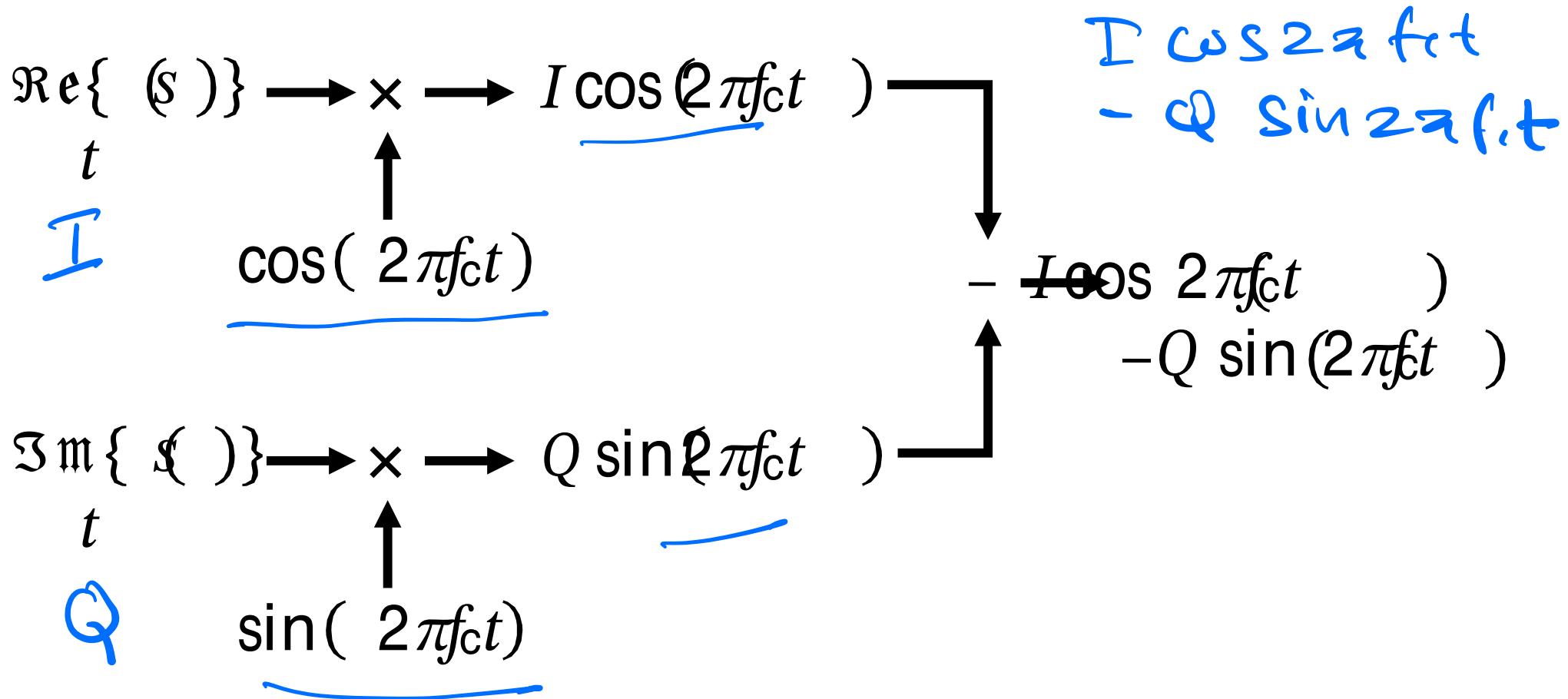


Traditional real chain

Consider using QAM: Quadrature Amplitude Modulation

$s(t)$ is complex: $s(t) = I + jQ$

Complex Up Conversion



Consider using QAM: Quadrature Amplitude Modulation

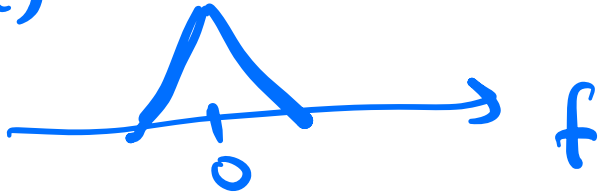
$$s(t) \text{ is complex: } s(t) = I + jQ$$

Complex Up Conversion

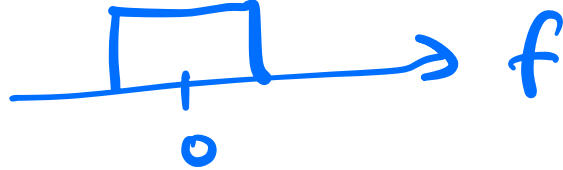
$$S(t) = I + jQ$$

$$I \cos 2\pi f_c t$$

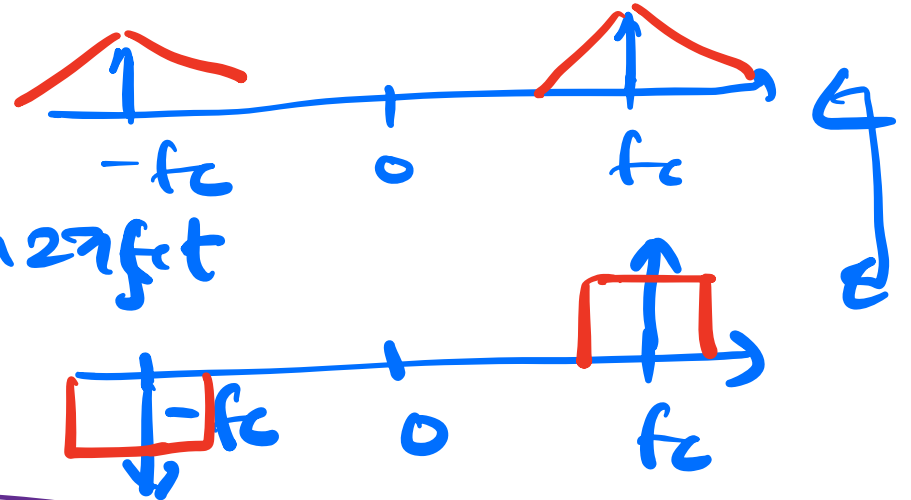
$I(t)$



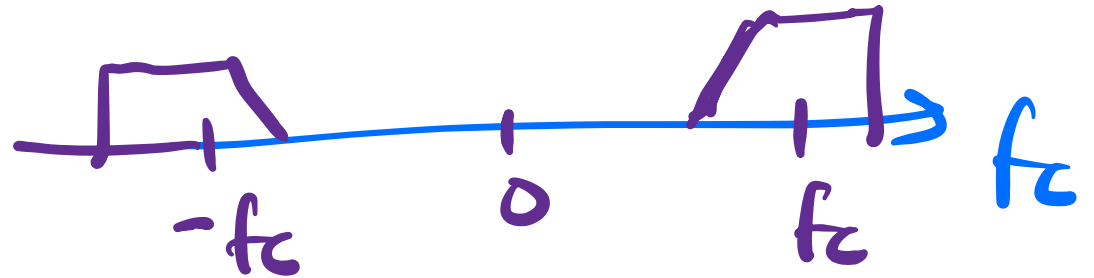
$Q(t)$



$$Q \sin 2\pi f_c t$$



$$I(t) \cos 2\pi f_c t - Q(t) \sin 2\pi f_c t$$



real signal

produce symmetric

magnitude freq response (ideally conj. symm.)

Sine and Cosine are Orthogonal

$$\langle \sin 2\pi f_c t, \cos 2\pi f_c t \rangle$$

✓

$$2\sin\theta\cos\theta = \sin 2\theta$$

$$= \frac{1}{T} \int_0^T \sin 2\pi f_c t \cos 2\pi f_c t dt$$

$$\underbrace{\left(\frac{1}{2T} \right)}_{\sin 2\pi f_c t} \int_0^T \sin(4\pi f_c t) dt = 0$$

$$\underline{T} \propto \frac{1}{2f_c}$$

$$T \rightarrow \infty$$

$$\text{norm} \rightarrow 0$$



Complex Down Conversion

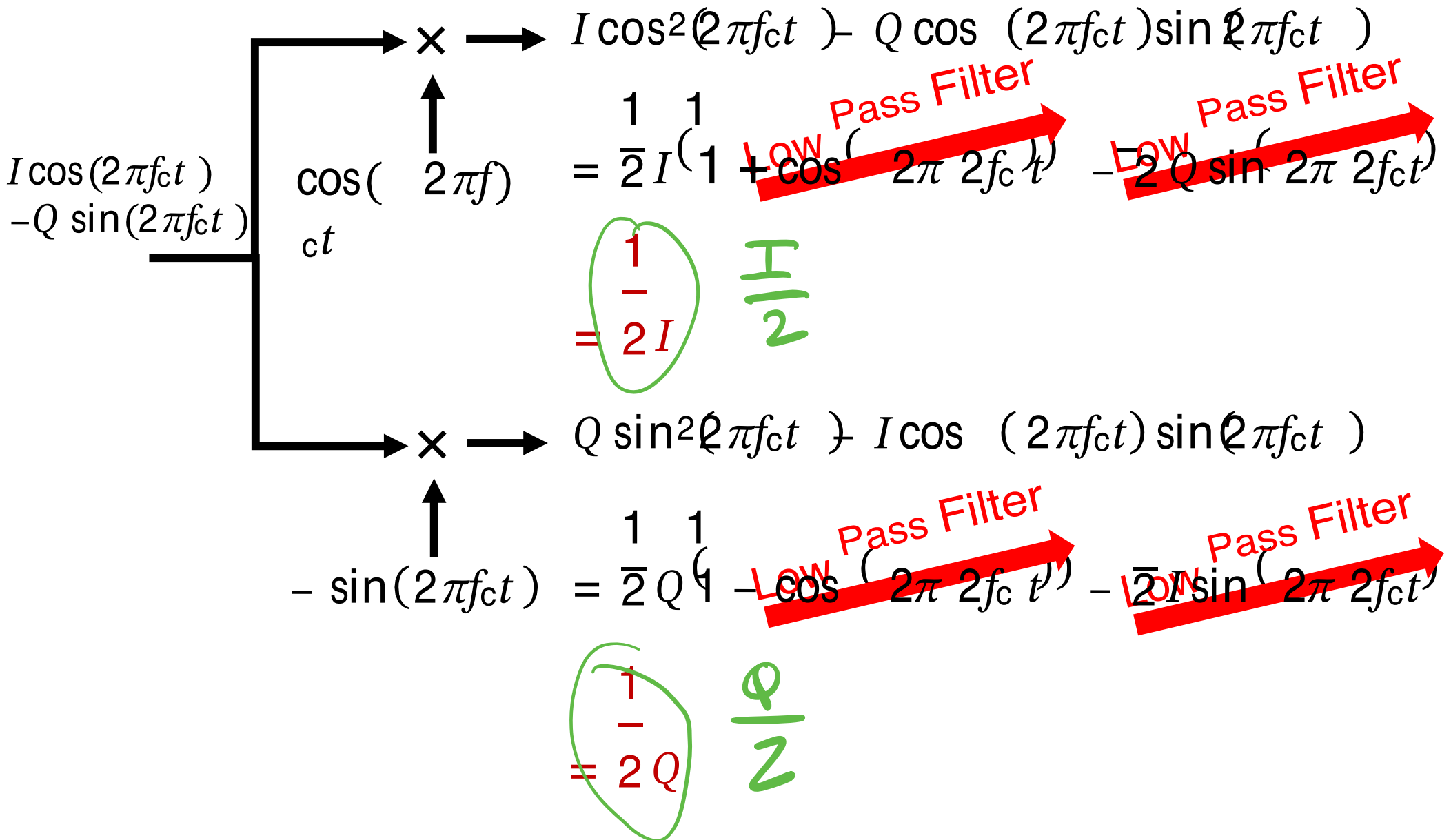
$$\begin{bmatrix} I \cos(2\pi f_c t) \\ -Q \sin(2\pi f_c t) \end{bmatrix} \times \begin{bmatrix} \cos(2\pi f_c t) \\ \sin(2\pi f_c t) \end{bmatrix}$$

$$\begin{aligned} & \rightarrow I \cos^2(2\pi f_c t) - Q \cos(2\pi f_c t) \sin(2\pi f_c t) \\ &= \frac{1}{2} I (1 + \cos(4\pi f_c t)) - \frac{1}{2} Q \sin(4\pi f_c t) \end{aligned}$$

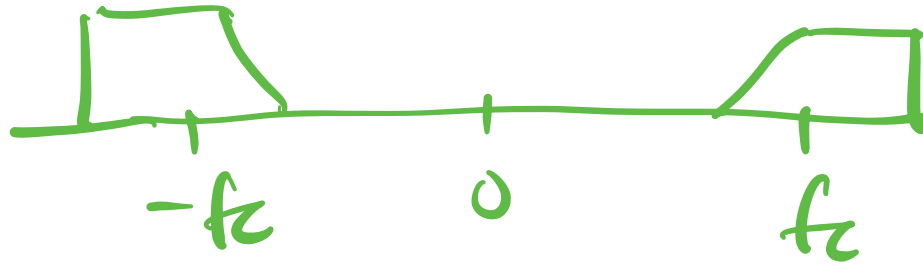
$$\begin{aligned} & \rightarrow Q \sin^2(2\pi f_c t) + I \cos(2\pi f_c t) \sin(2\pi f_c t) \\ &= \frac{1}{2} Q (1 - \cos(4\pi f_c t)) + \frac{1}{2} I \sin(4\pi f_c t) \end{aligned}$$

Handwritten notes:
 $2 \cos^2 \theta = 1 + \cos 2\theta$
 $4\pi f_c t$

Complex Down Conversion



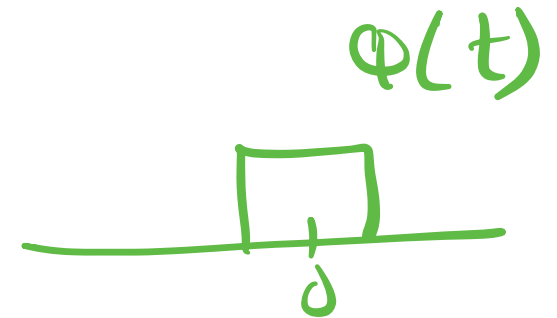
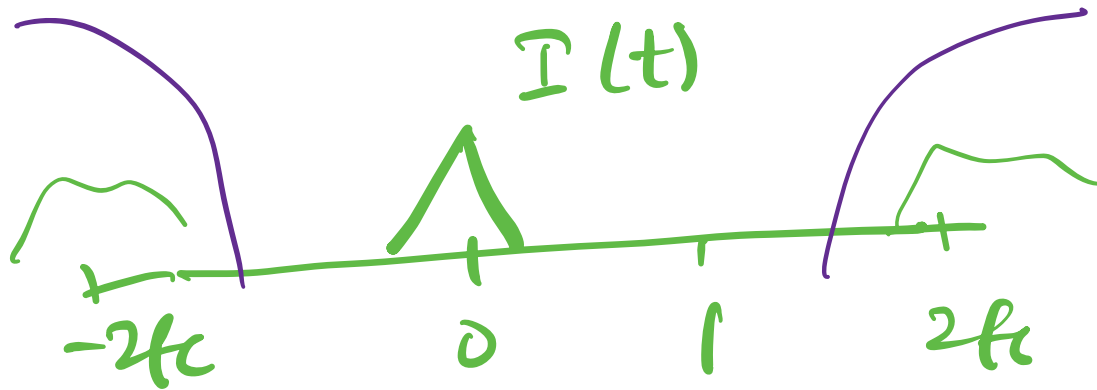
Complex Down Conversion



filter

$\cos 2\pi f_c t$

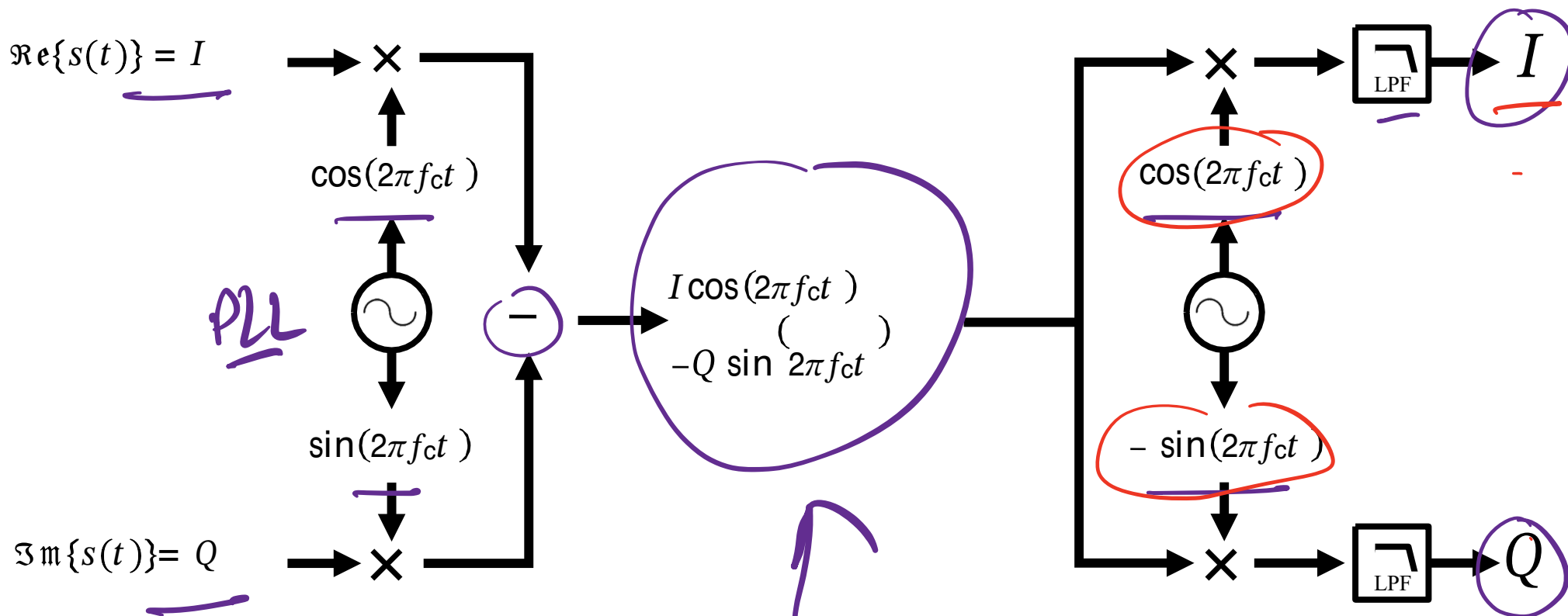
$-\sin 2\pi f_c t$



Up/Down Conversion

Transmitter

Receiver



$$s(t) \times e^{j2\pi f_c t} \rightarrow \Re\{s(t) e^{j2\pi f_c t}\} \rightarrow \times e^{-j2\pi f_c t} \rightarrow s(t)$$

Handwritten notes below the equation:

- $\Re\{s(t) e^{j2\pi f_c t}\}$ (circled in purple)
- $e^{-j2\pi f_c t}$ (underlined in purple)

Up/Down Conversion

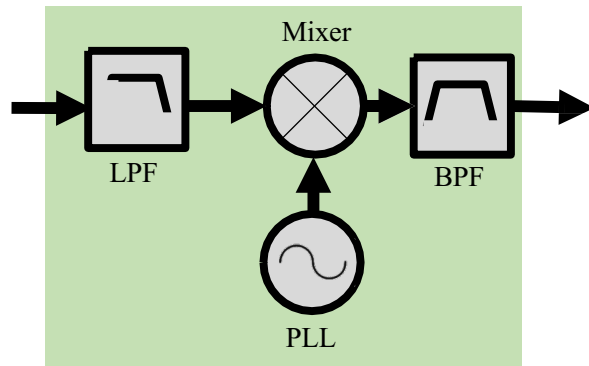
Transmitter

Receiver

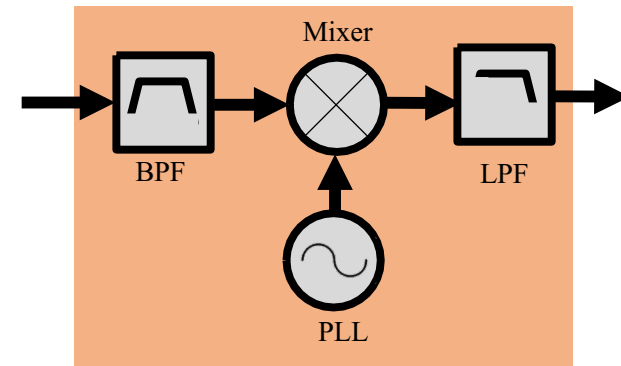
$$\begin{aligned}
 & s(t) \times e^{j2\pi f_c t} \rightarrow \Re \{ (s(t) e^{j2\pi f_c t}) \} \rightarrow \times \underline{e^{-j2\pi f_c t}} \rightarrow s(t) \\
 & \text{Receiver side: } \cos 2\pi f_c t - j \sin 2\pi f_c t \\
 & \Re(s(t) e^{j2\pi f_c t}) = (\hat{I} + j\hat{Q}) (\cos 2\pi f_c t + j \sin 2\pi f_c t) \\
 & = \underline{\hat{I} \cos 2\pi f_c t - \hat{Q} \sin 2\pi f_c t} + j(\text{some imag.})
 \end{aligned}$$

Up/Down Conversion

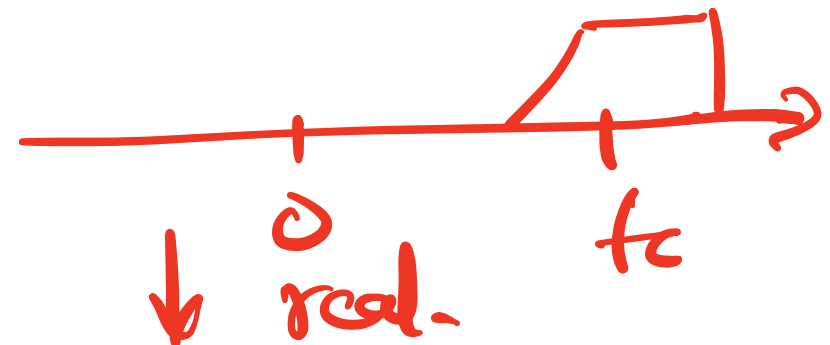
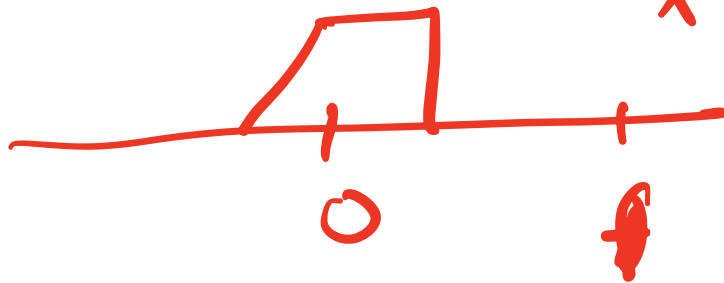
Transmitter
Up-conversion



Receiver
Down-conversion



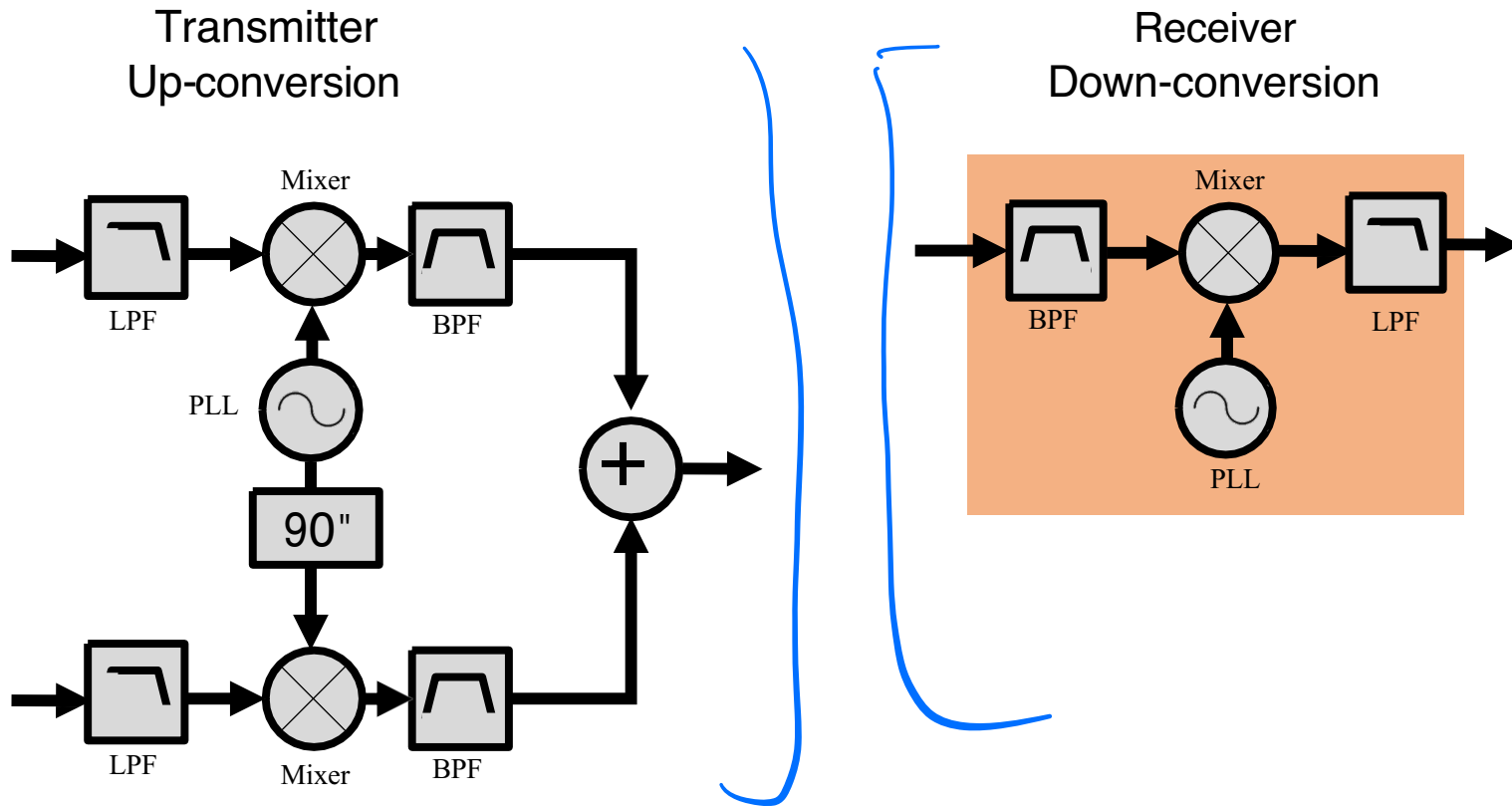
$S(t) \rightarrow S(f)$
complex
 $\times e^{j2\pi f_c t}$



$$\operatorname{Re}(x(t)) = \frac{1}{2} \left((a+ib)x(t) + \overline{(a+ib)}x(t) \right)$$

$$= \frac{1}{2} \left((a+ib)x(t) + (a-ib)x(t) \right)$$

Up/Down Conversion

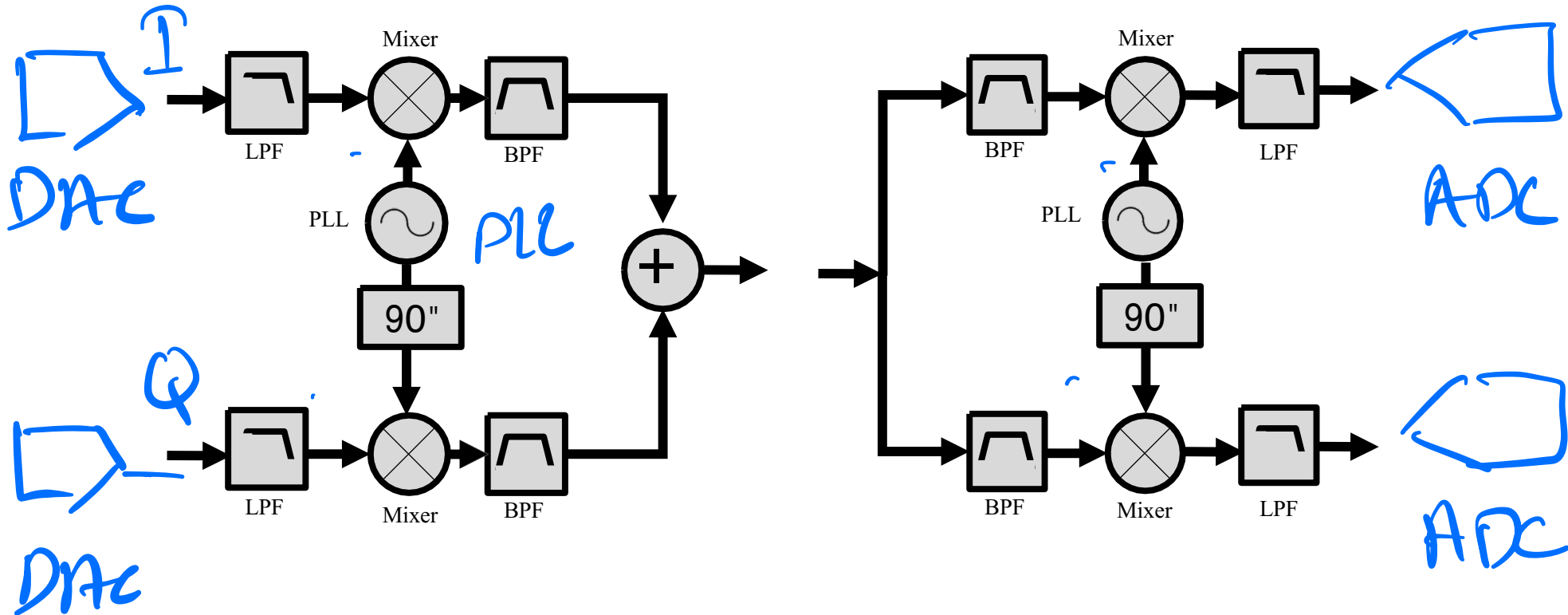


Up/Down Conversion

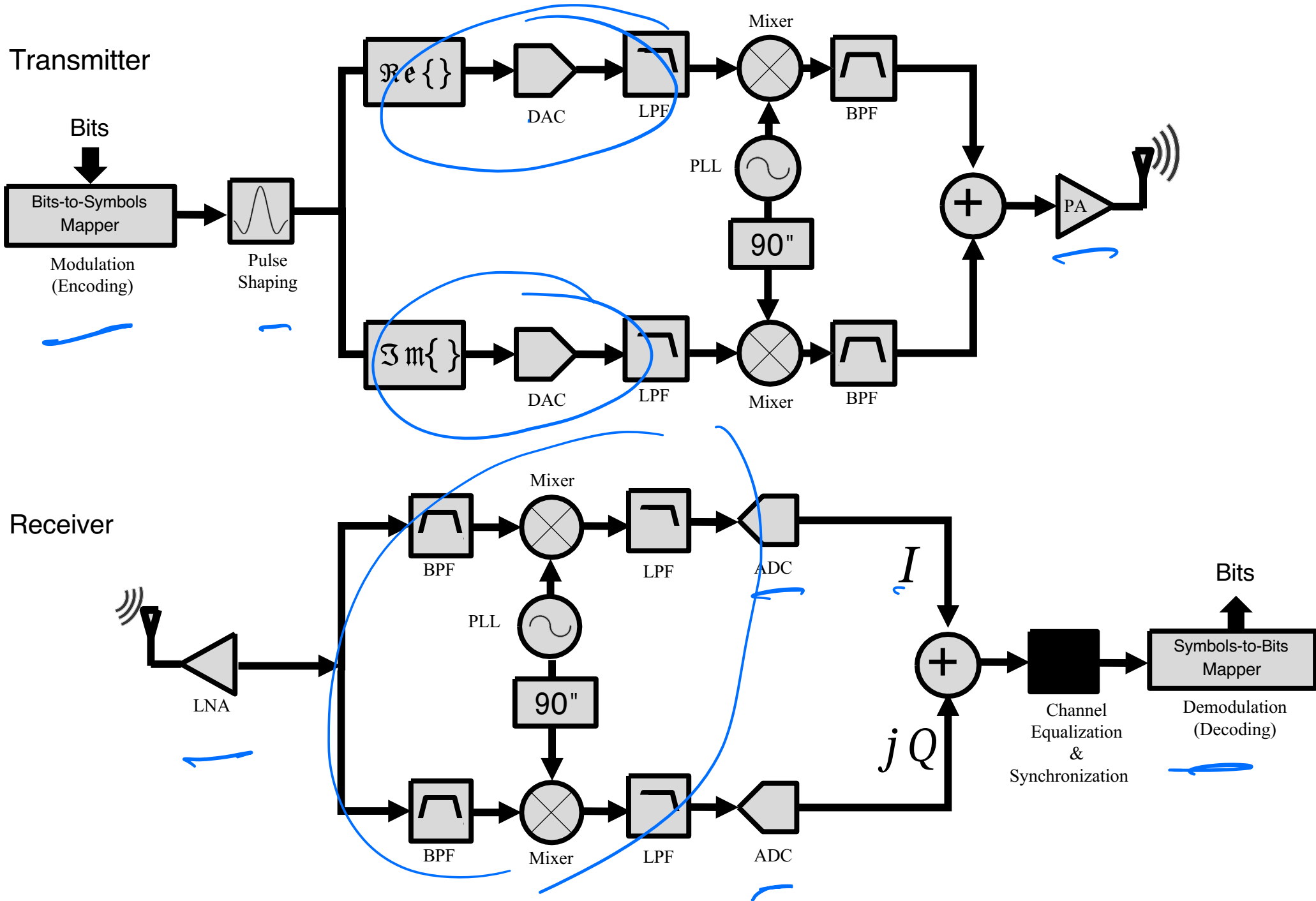
analog

Transmitter
Up-conversion

Receiver
Down-conversion

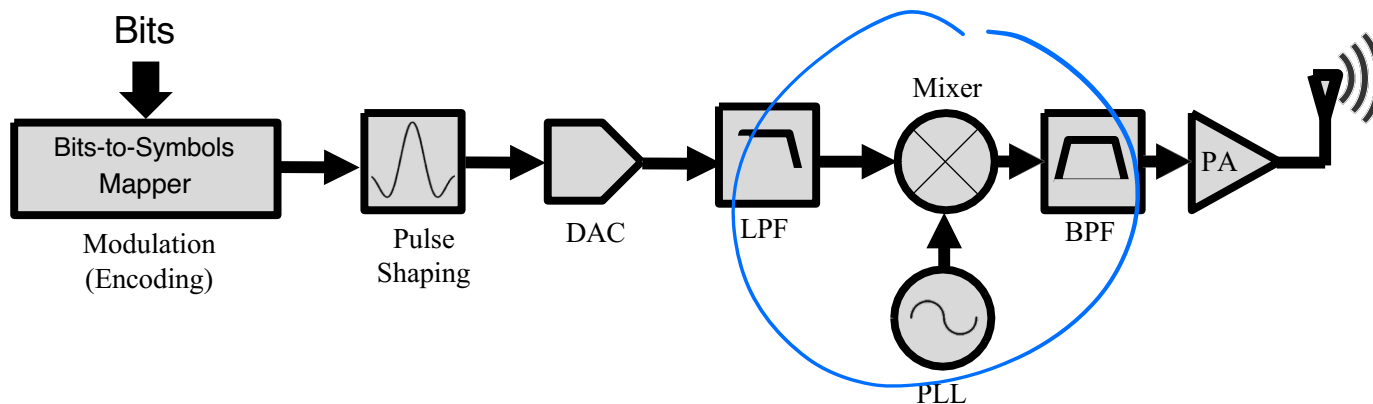


Wireless Communication System

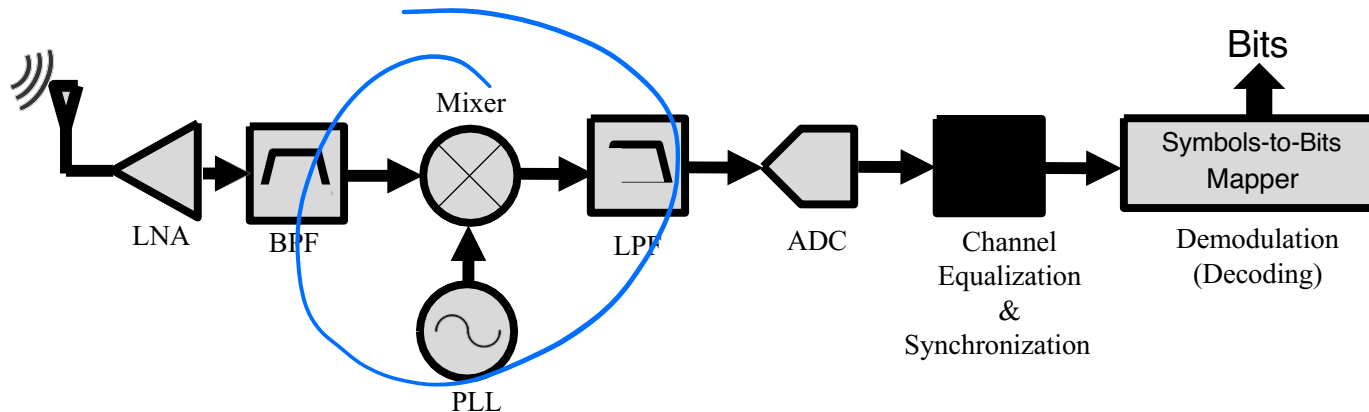


Wireless Communication System

Transmitter

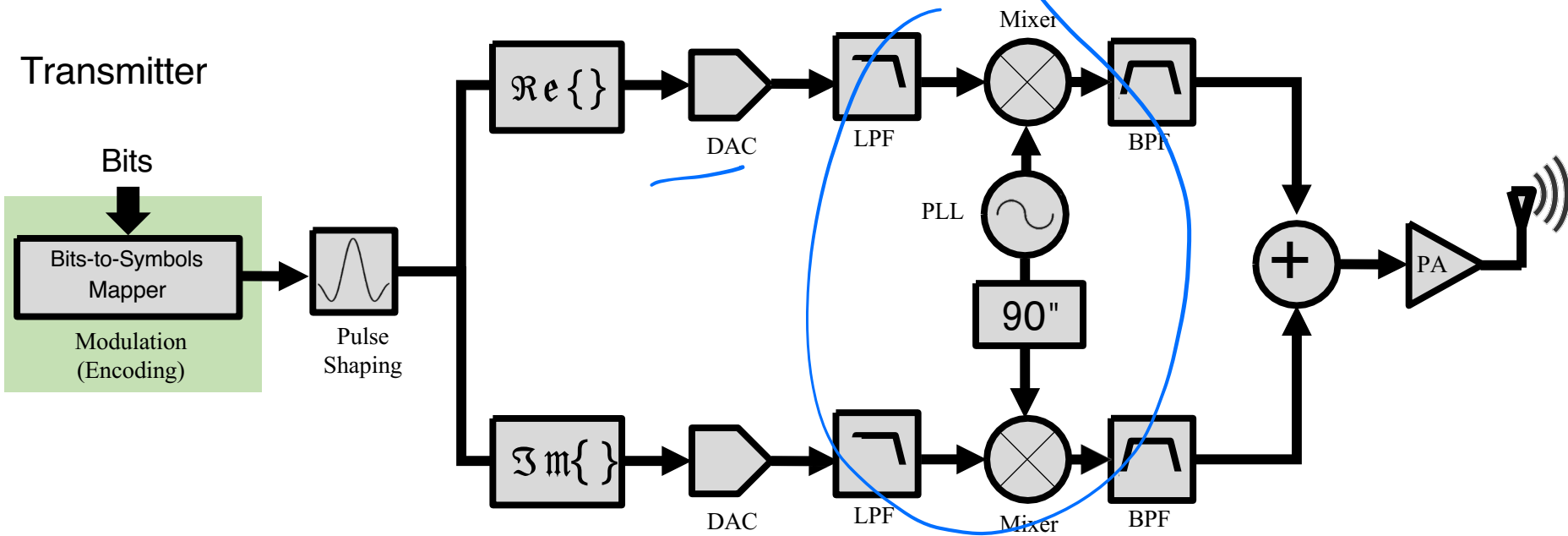


Receiver

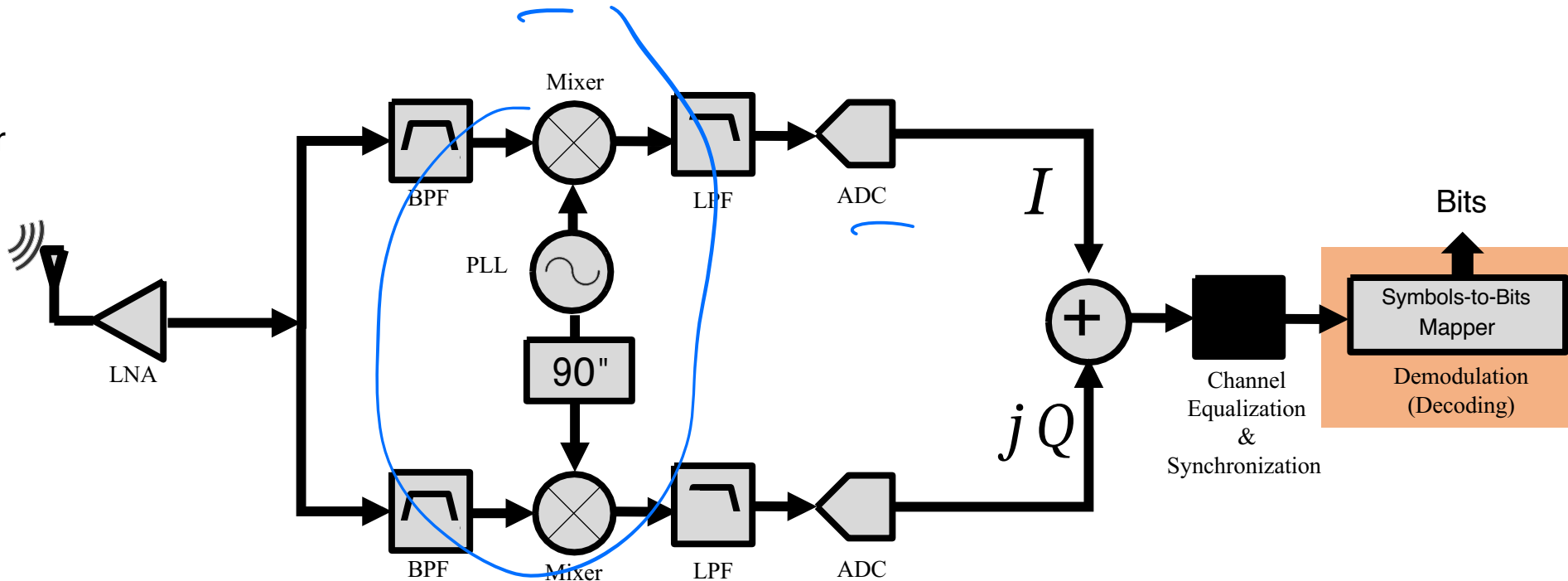


Wireless Communication System

Transmitter



Receiver

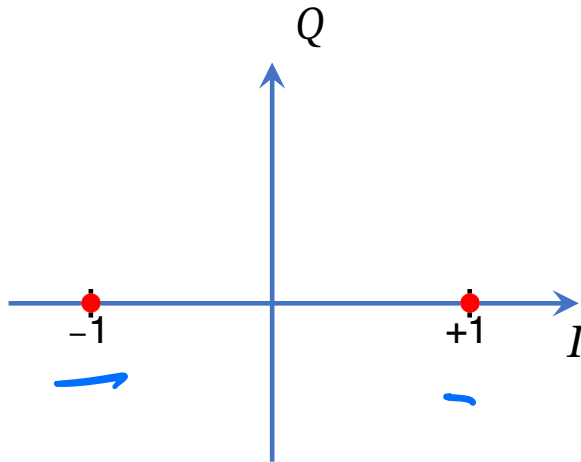


BPSK Modulation

noise

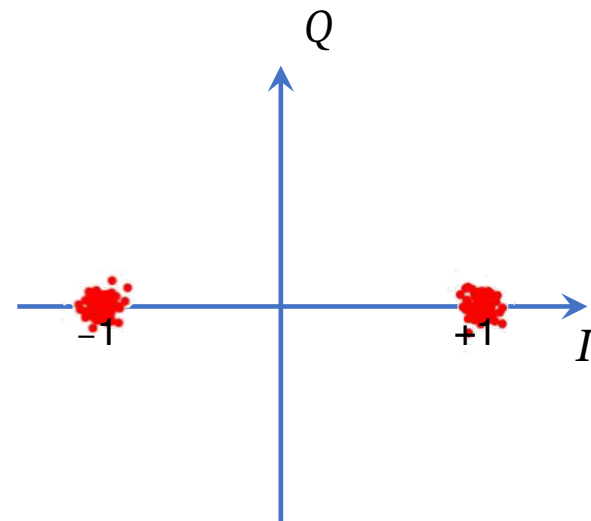
$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



$x(t)$

Transmitted Constellation



$$y(t) = x(t) + v(t)$$

Received Constellation

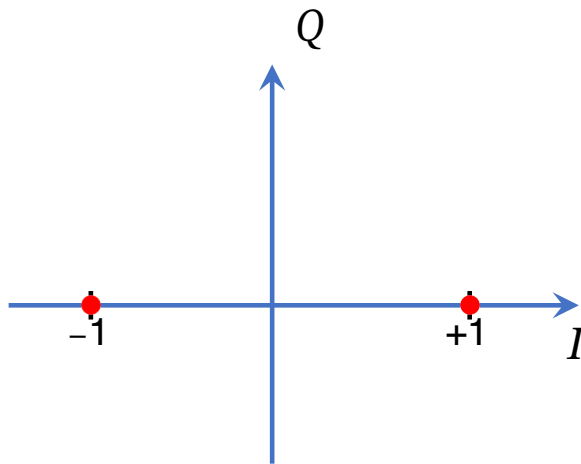
$$x(t) + v(t)$$

noise

BPSK Modulation

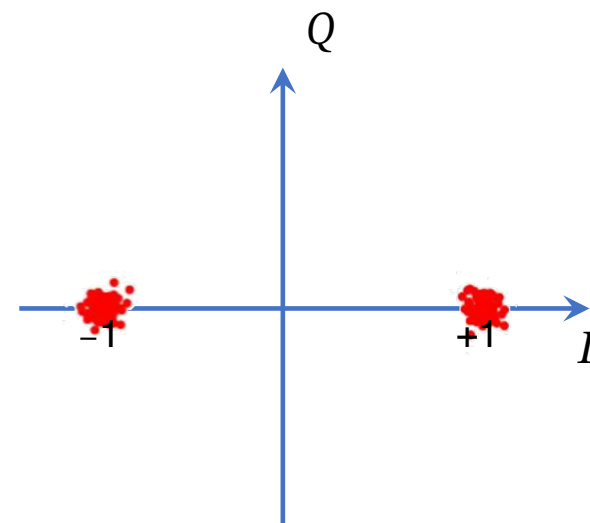
$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

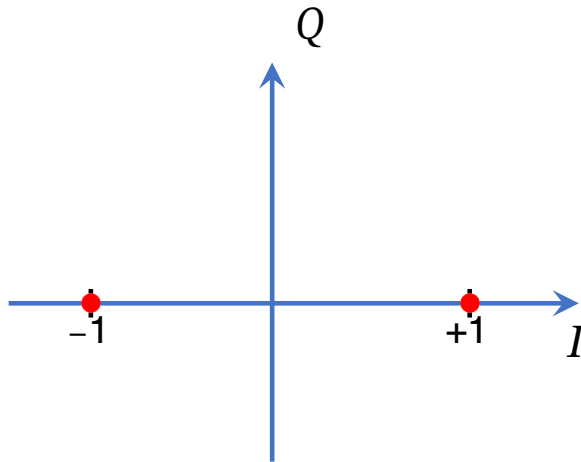
How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E[|x(t)|^2]}{E[|v(t)|^2]} = \frac{E_s}{N_0}$$

BPSK Modulation

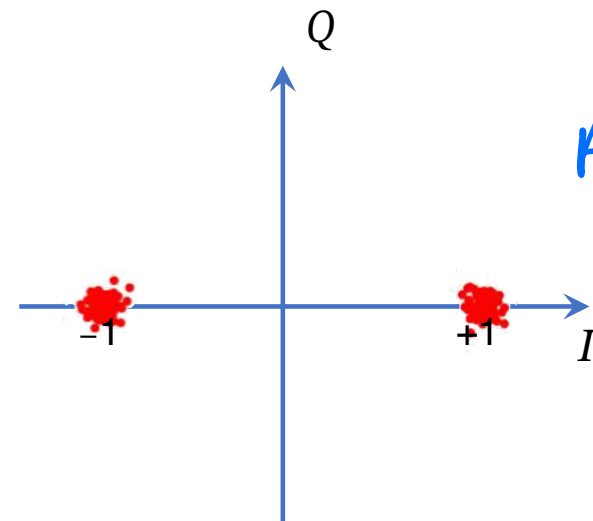
$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

Additive
white
Gaussian
noise

AWGN

$\sim \mathcal{N}(0, N_0)$

How well we can decode depends on SNR?

$$10 \log_{10}(\text{SNR}) = \text{SNR}_{\text{dB}}$$

$$\text{SNR} = \frac{\text{Signal Power}}{\text{Noise Power}}$$

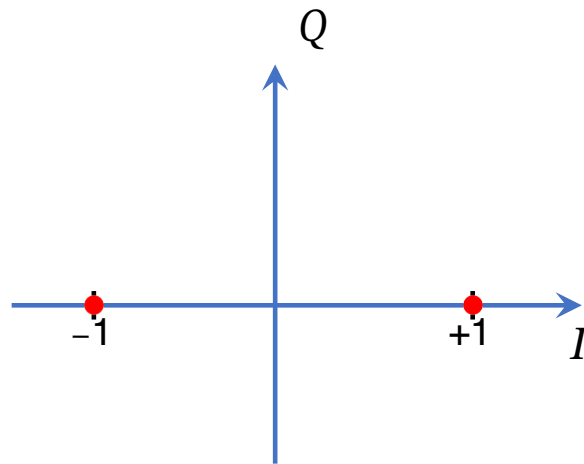
$$= \frac{E_s}{N_0} = 25 \text{ dB}$$

$$E[v(t)]^2 = N_0$$

BPSK Modulation

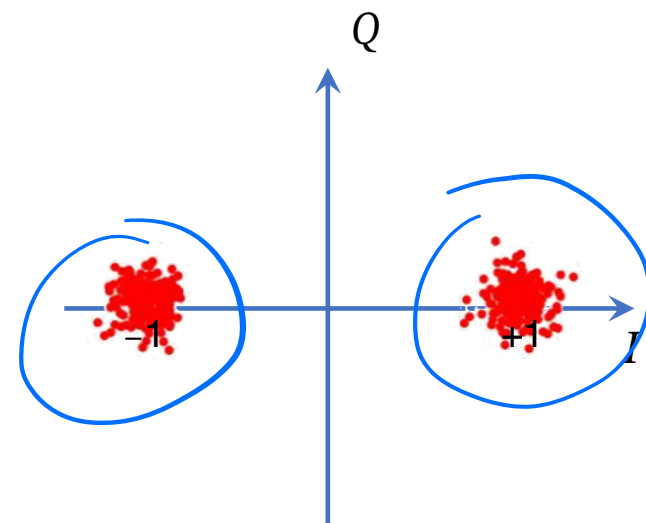
$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

AWGN

Received Constellation

How well we can decode depends on SNR?

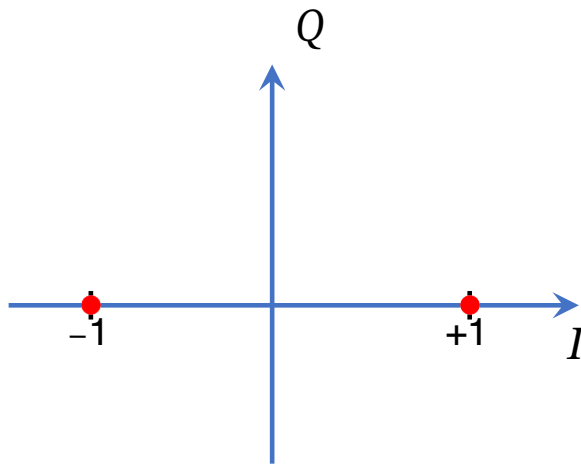
$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}}$$

$$= \frac{E_s}{N_0} = \underline{19 \text{ dB}}$$

BPSK Modulation

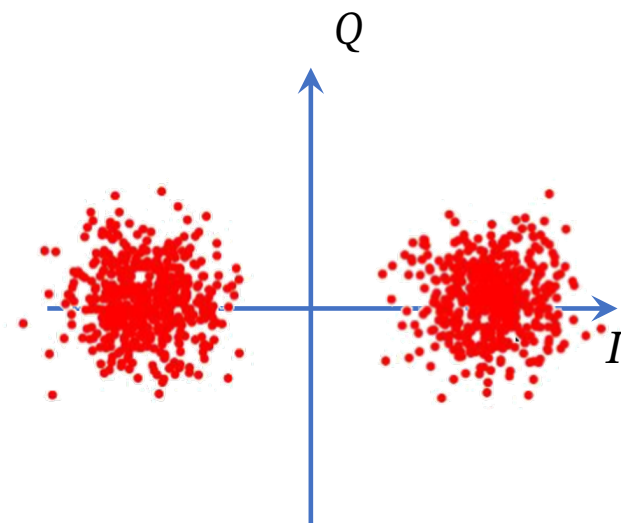
$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

AWGN

How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}}$$

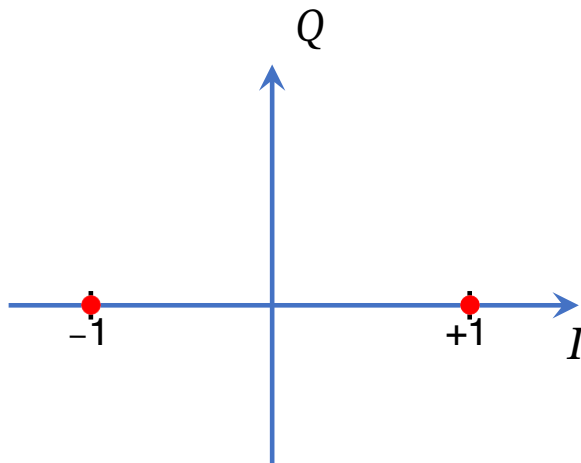
$$= \frac{E_s}{N_0} = \underline{13 \text{ dB}}$$

BPSK Modulation

bit errors

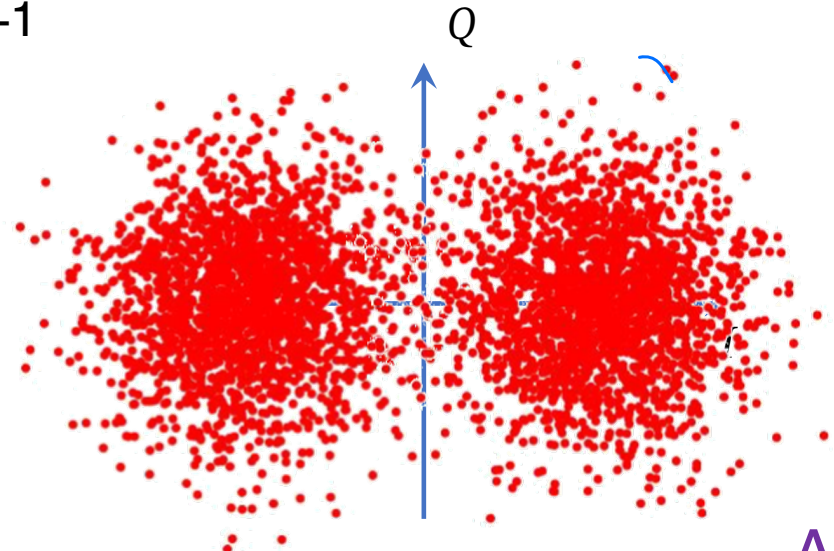
0 → -1

1 → +1



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

AWGN

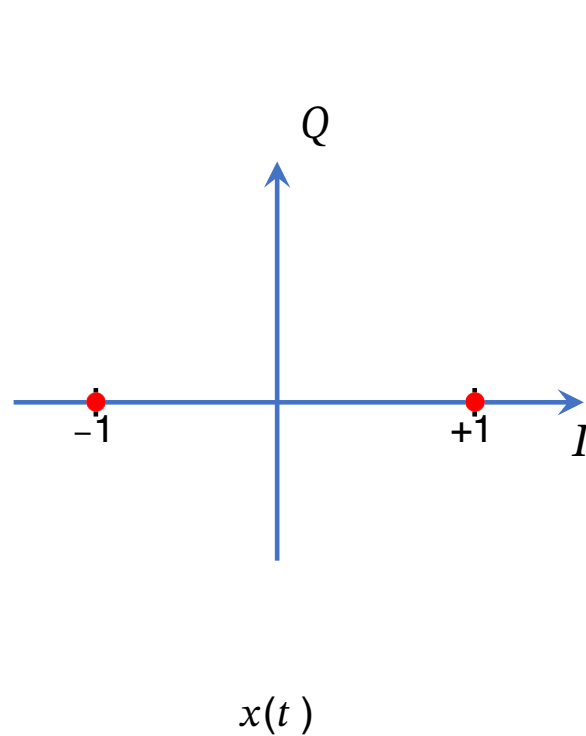
Received Constellation

How well we can decode depends on SNR?

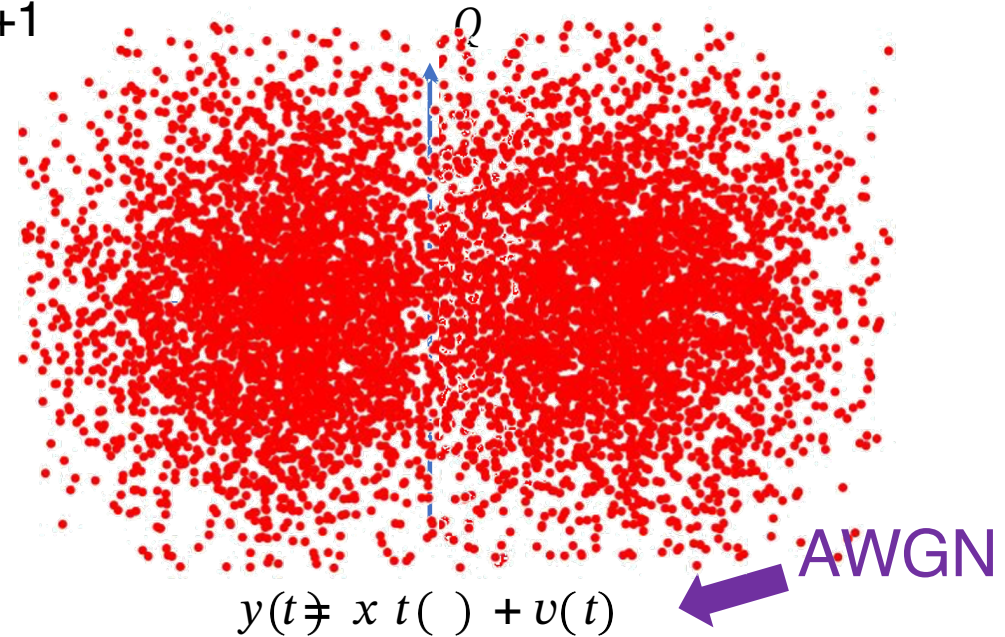
$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}}$$

$$= \frac{E_s}{N_0} = 7 \text{ dB}$$

BPSK Modulation



Transmitted Constellation



Received Constellation

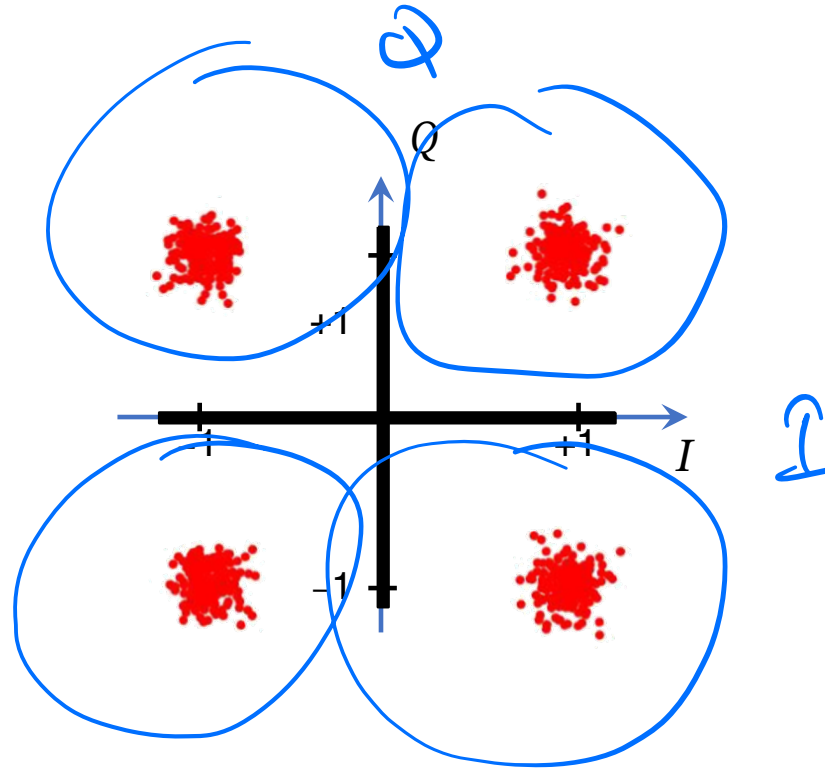
How well we can decode depends on SNR?

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}}$$

$$= \frac{E_s}{N_0} = \underline{3.5 \text{ dB}}$$

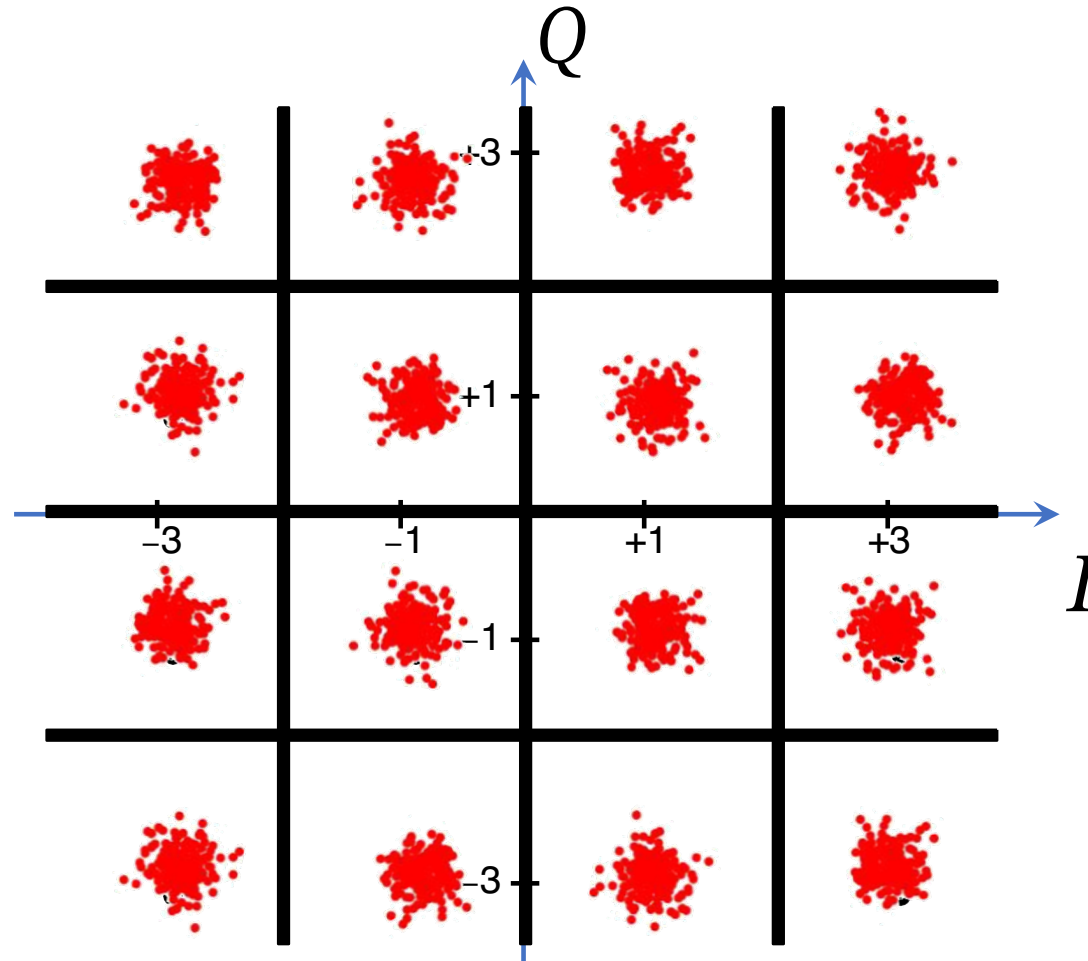


4-QAM



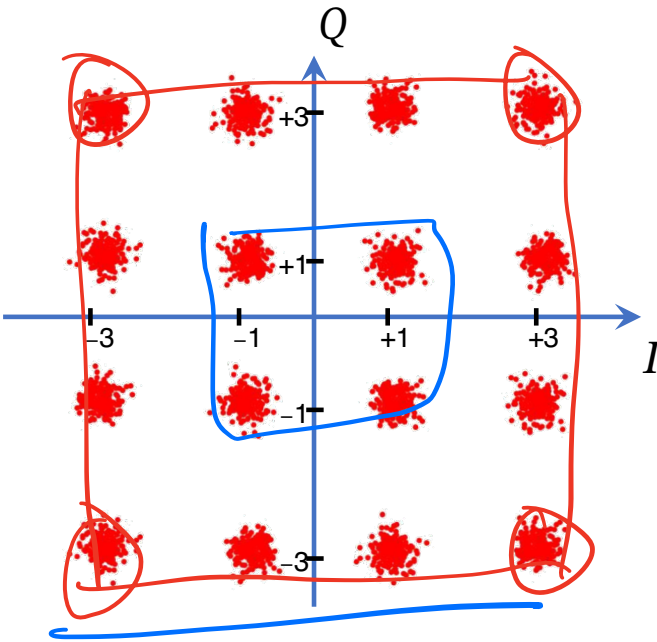
Encodes 2 bits per symbol resulting in Higher Data Rate

16-QAM

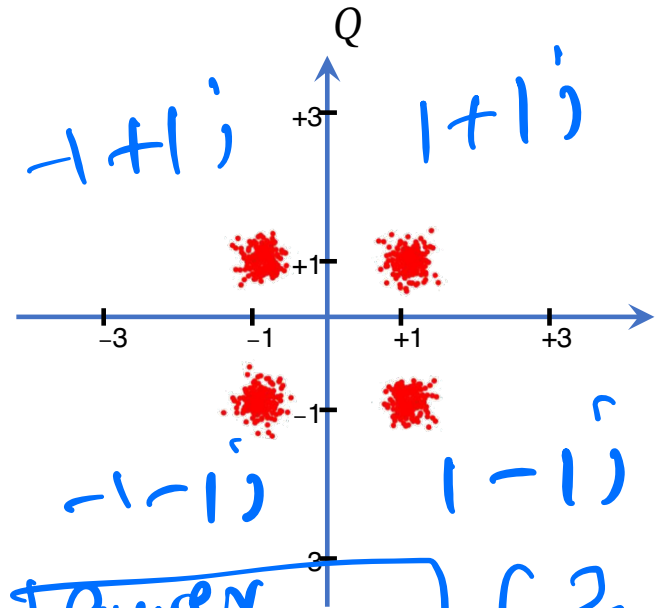


Encodes 4 bits per symbol resulting in Higher Data Rate

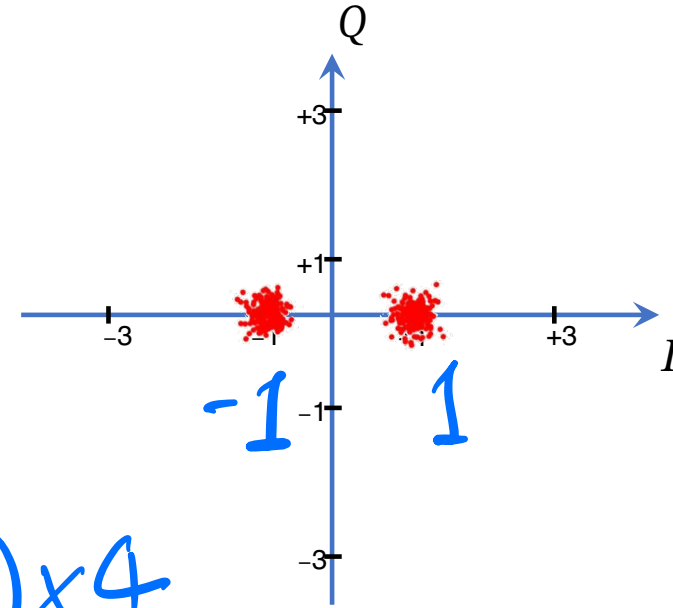
16-QAM



4-QAM



BPSK

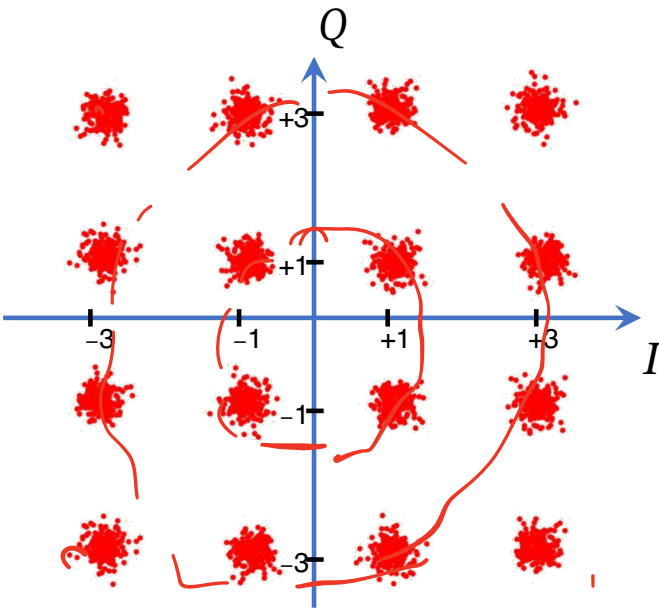


Is the SNR the same in these 3 constellations?

$$E_s = \frac{1 + 1}{2} = 1$$

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{E[|x(t)|^2]}{E[|v(t)|^2]} = \frac{E_s}{N_0}$$

16-QAM



$$SNR = \frac{E[|\mathbf{x}|^2]}{N_0}$$

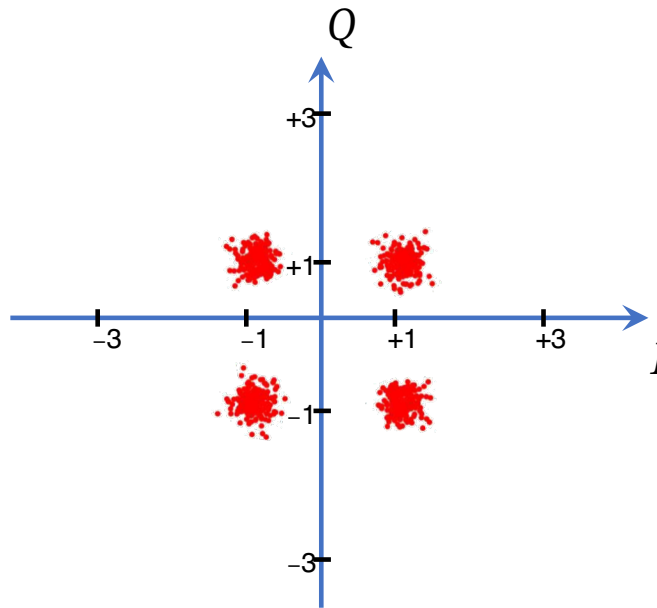
$$(4 \times 2 + 8 \times 10)$$

$$SNR = \frac{+4 \times 18}{N_0}$$

$$= \frac{10}{N_0}$$

>

4-QAM

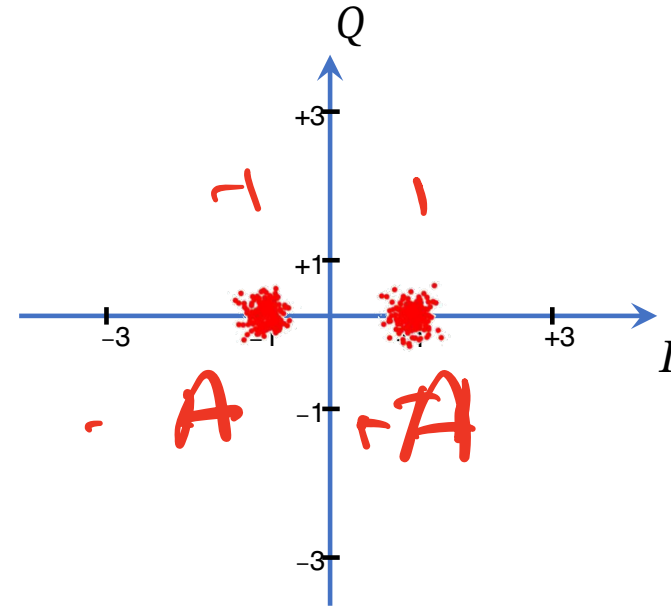


$$SNR = \frac{(2 + 2 + 2 + 2)}{N_0}$$

$$= \frac{2}{N_0}$$

>

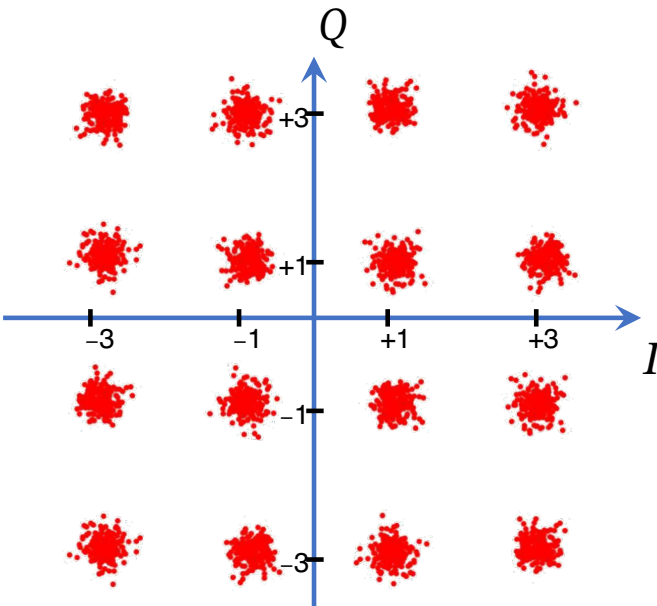
BPSK



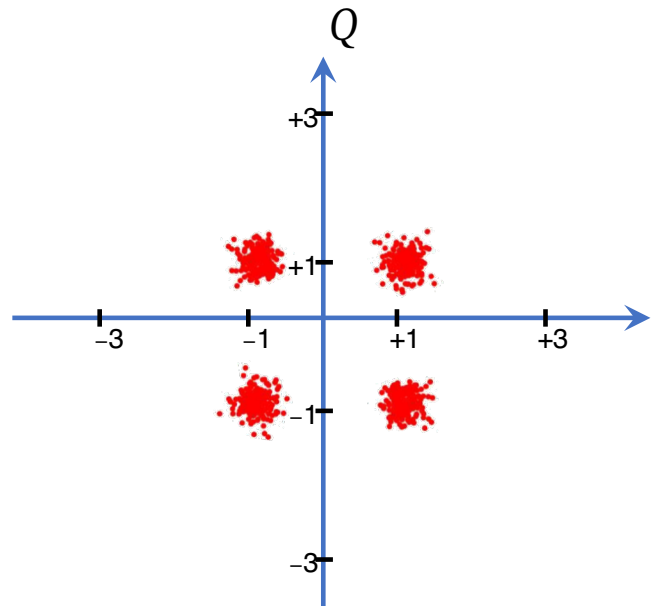
$$SNR = \frac{(1 + 1)}{N_0}$$

$$= \frac{1}{N_0}$$

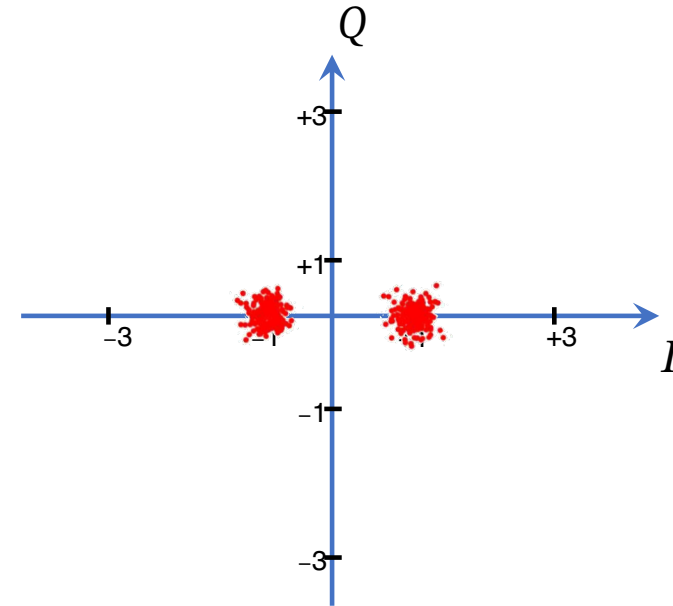
16-QAM



4-QAM



BPSK



Transmit power is not the same!

$$(4 \times 2 + 8 \times 10)$$

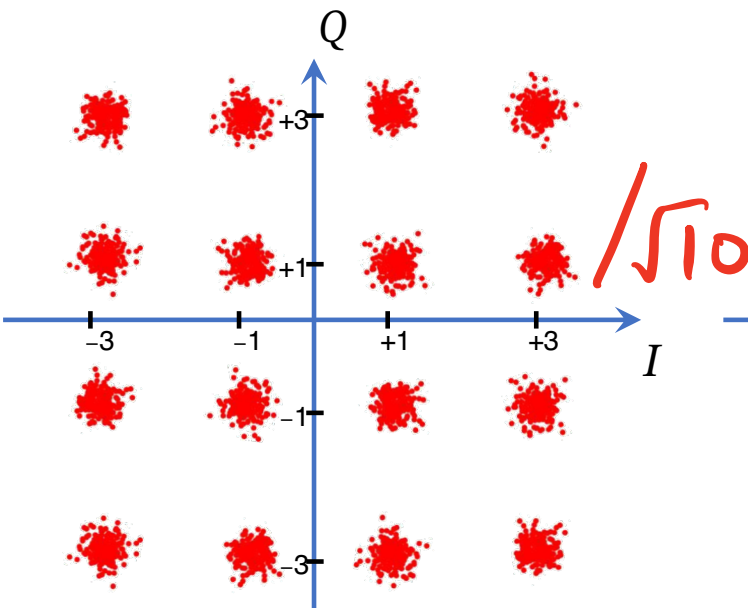
$$SNR = \frac{+4 \times 18}{N_0}$$

$$= \frac{10}{N_0}$$

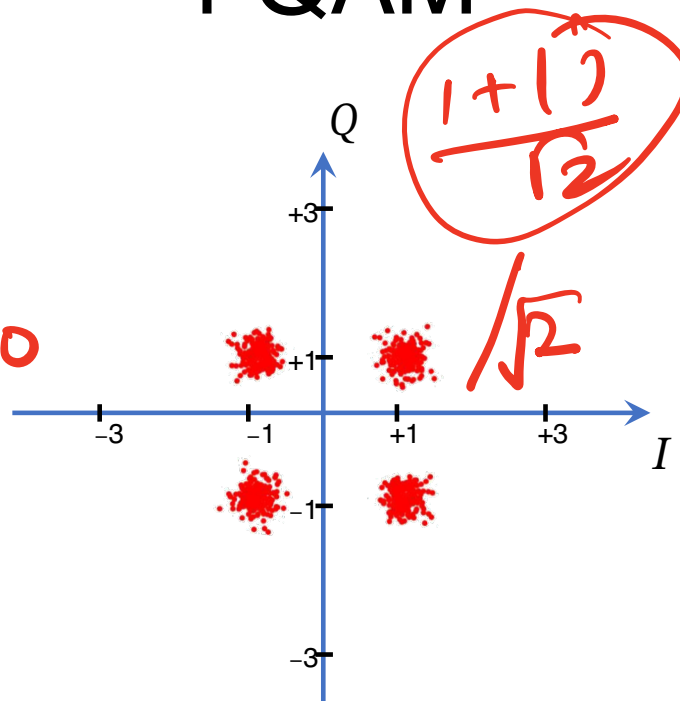
Must not increase transmit power to use higher order modulation!

$$\frac{1}{2} \sqrt{b}$$

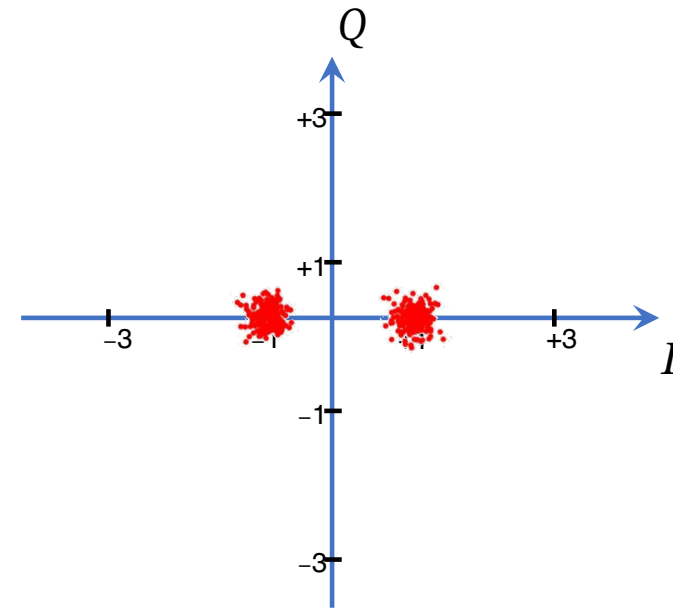
16-QAM



4-QAM



BPSK



Normalize to maintain constant transmit power.

$$SNR = \frac{(4 \times 2 + 8 \times 10 + 4 \times 18)/16}{N_0}$$

$$SNR = \frac{(2 + 2 + 2 + 2)/4}{N_0}$$

$$SNR = \frac{(1 + 1)/2}{N_0}$$

$$= \frac{10}{N_0}$$

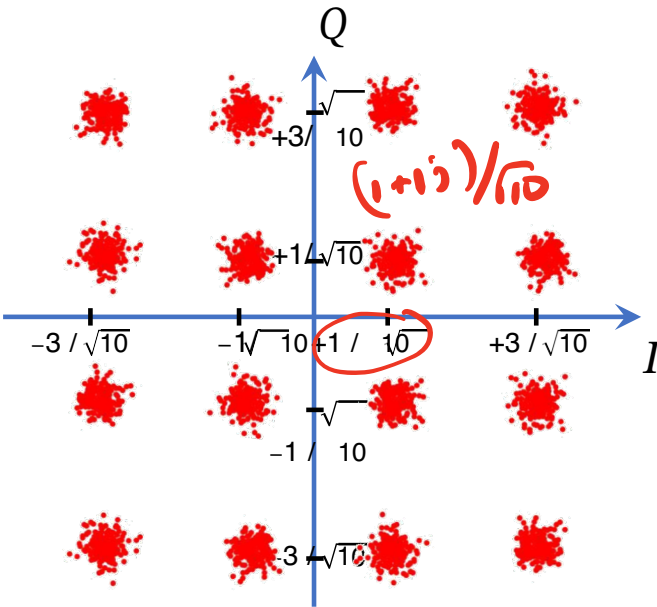
>

$$= \frac{2}{N_0}$$

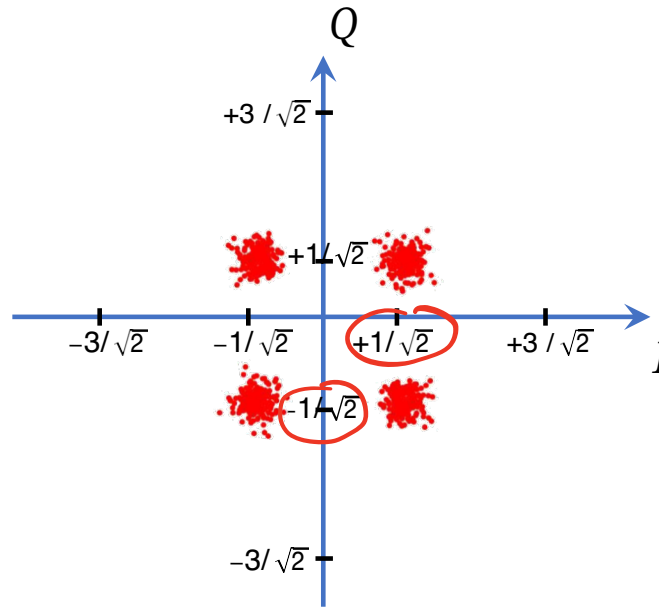
>

$$= \frac{1}{N_0}$$

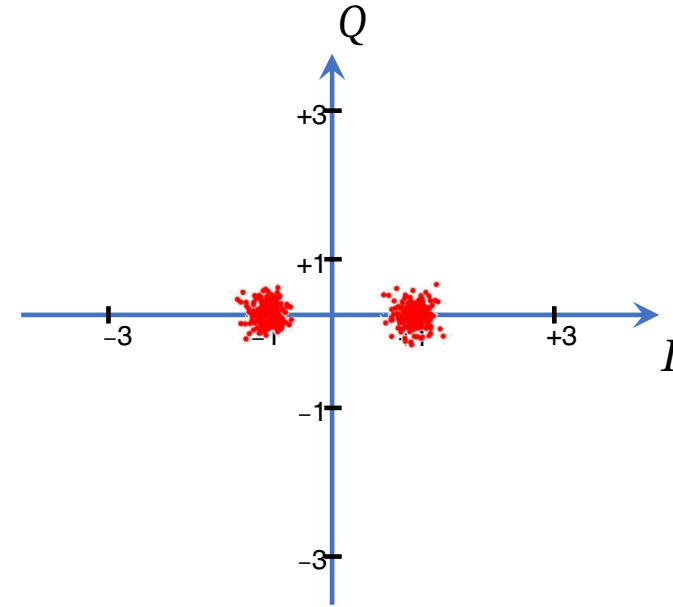
16-QAM



4-QAM



BPSK



Normalize to maintain constant transmit power.

$$SNR = \frac{(4 \times 2/10 + 8 \times 1 + 4 \times 18/10)/16}{N_0}$$

$$= \frac{1}{N_0}$$

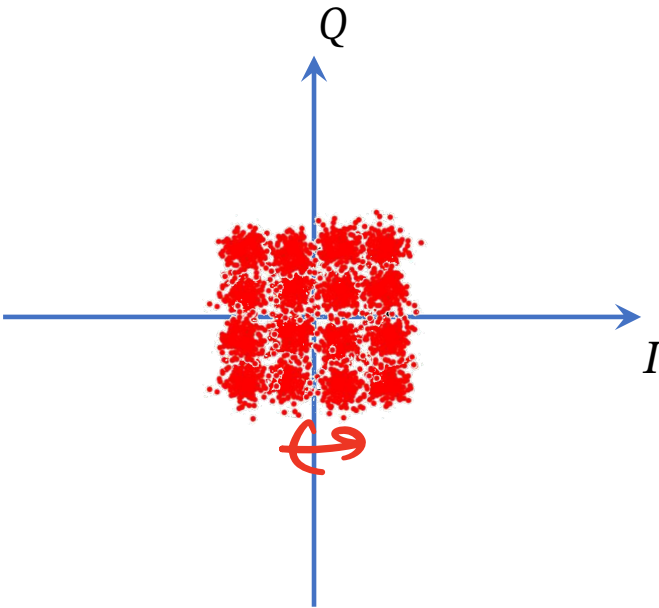
$$SNR = \frac{(1 + 1 + 1 + 1)/4}{N_0}$$

$$= \frac{1}{N_0}$$

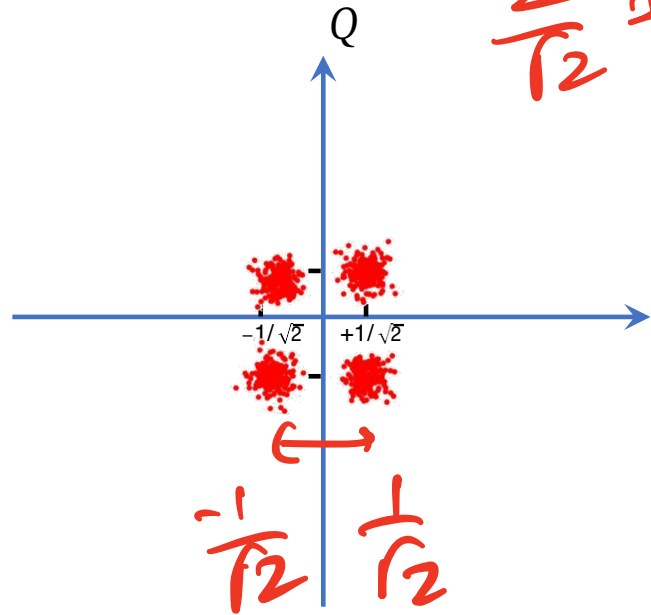
$$SNR = \frac{(1 + 1)/2}{N_0}$$

$$= \frac{1}{N_0}$$

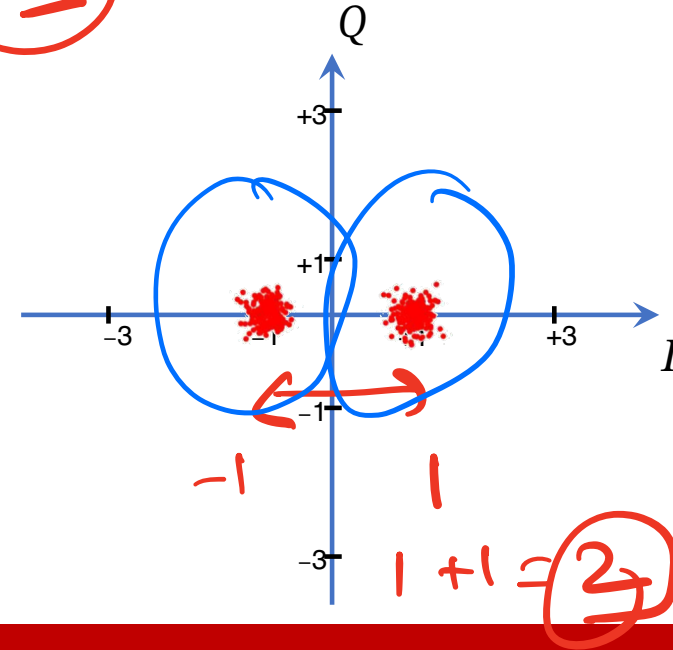
16-QAM



4-QAM



BPSK

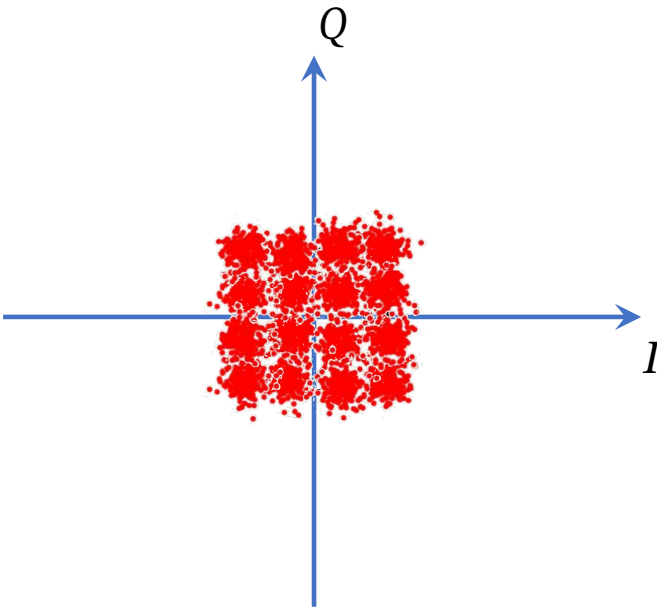


Normalize to maintain constant transmit power.

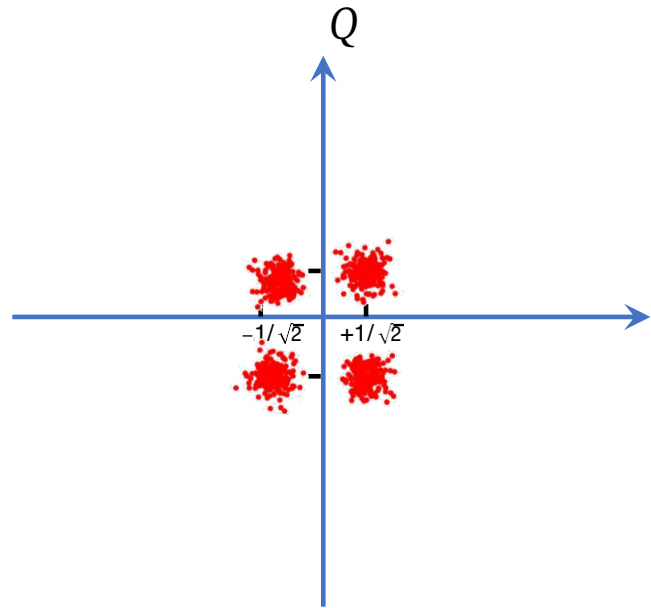
$$SNR = \frac{1}{N_0} \quad SNR = \frac{1}{N_0} \quad SNR = \frac{1}{N_0} \quad \text{---}$$

Need Higher SNR to Decode
Higher Order Modulation.

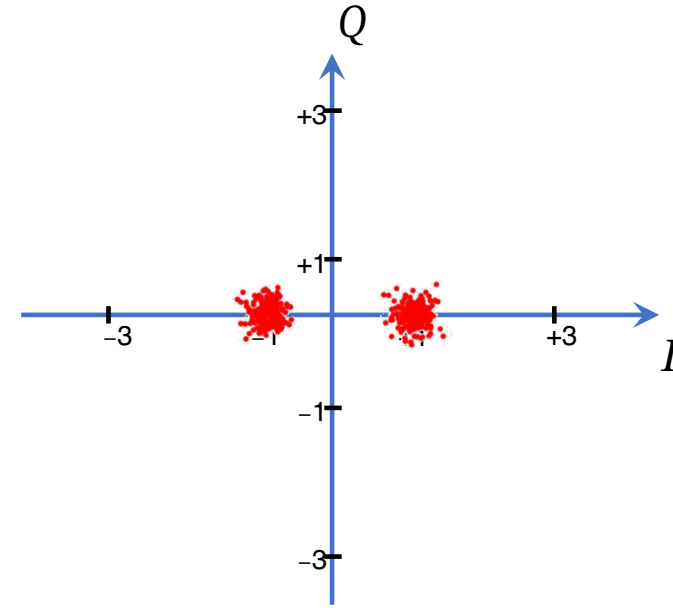
16-QAM



4-QAM



BPSK



4 bits/symbol 2 bits/symbol

1 bit/symbol

Higher Order Modulation:

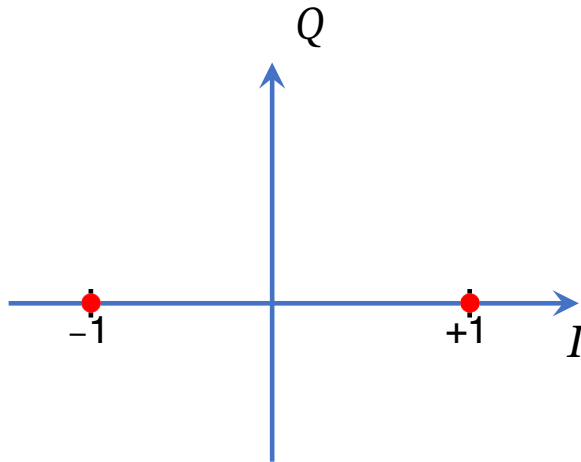
- Needs higher SNR to decode correctly.
- Achieves higher Bite Rate

Given an SNR, choose highest order modulation that guarantees minimal Bite Error Rate (BER)

Decoding under noise

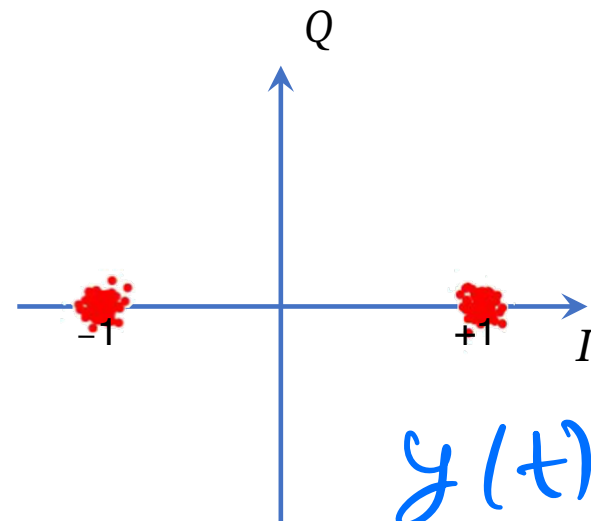
$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$

Transmitted Constellation



$y(t) = x(t) + v(t)$

Received Constellation

$$y(t) = x(t) + v(t)$$

How well we can decode depends on SNR?

$v(t)$

$\sim \mathcal{N}(0, N_0)$

$$SNR = \frac{\text{Signal Power}}{\text{Noise Power}}$$

$$= \frac{E[|x(t)|^2]}{E[|v(t)|^2]} = \frac{E_s}{N_0}$$

Decoding under noise

$$\underline{y(t)} \sim \mathcal{N}(\theta, \sigma^2) \quad \underline{\text{estimate } \theta}$$

Assume Data:

MLE (maximum likelihood): $\arg \max_{\theta} P(y | \theta)$
likelihood

Chooses the parameter that makes the observed data most likely.

MAP (maximum a posteriori): $\max_{\theta} P(\theta | y)$

Bayes

$$P(\theta | y) = P(y | \theta) \underline{P(\theta)} \leftarrow \text{prior}$$

likelihood

Maximum Likelihood Estimator

$$y \sim N(\theta, \sigma^2)$$

$$P(y|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(y-\theta)^2}{\sigma^2}}$$

$$\frac{\partial}{\partial \theta} P(y|\theta) = 0 = \frac{1}{\sqrt{2\pi\sigma^2}} \left(-\frac{1}{2} \frac{(y-\theta)^2}{\sigma^2} \right) (y-\theta)$$

$$\bar{y}_i = y_1, y_2, y_3, \dots$$

$$P(\bar{y}|\theta) = \prod_i P(y_i|\theta)$$

$$\max_{\theta} \prod_i P(y_i | \theta)$$

⋮

$$\theta = \frac{1}{N} \sum_{i=1}^N y_i$$



sample mean

Maximum A Posteriori Estimator

Decision Boundary

wireless

- Decide between BPSK symbols given .
- ML detector: $\rightarrow$ nearest-neighbor rule.
- MAP detector: $\rightarrow$ ML **plus** prior on symbol probabilities.
- Binary antipodal example y , :
Decide if

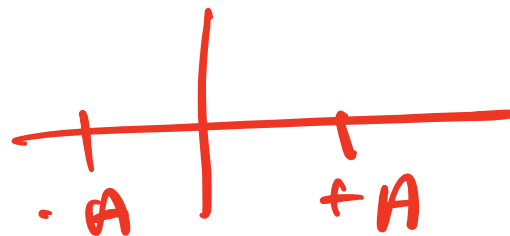
$$s \in \{+A, -A\}$$

$$y = s + n$$

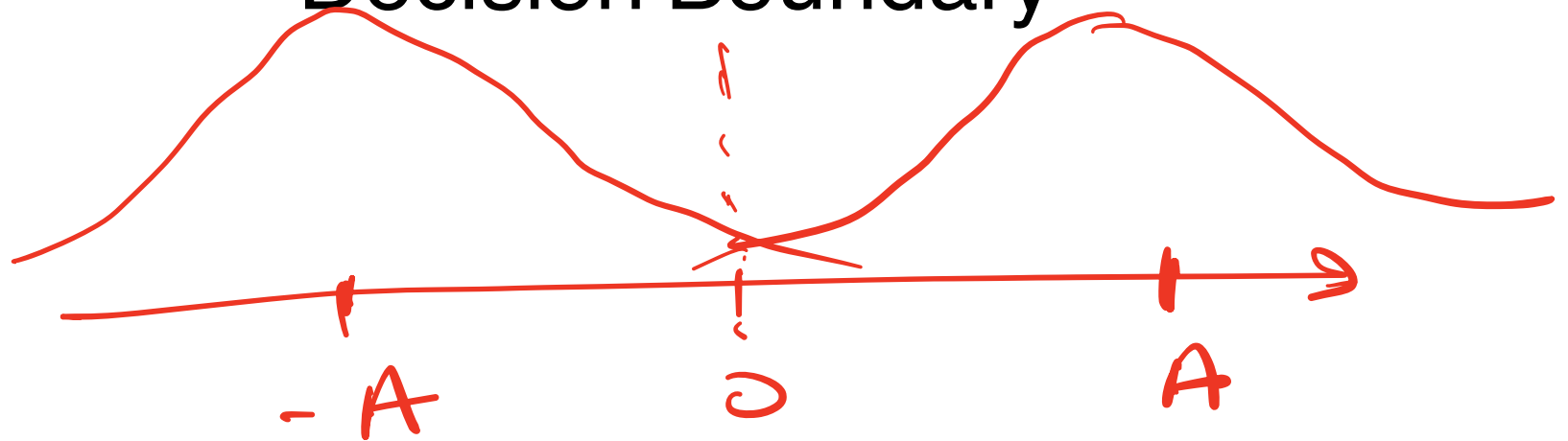
$$\uparrow \mathcal{N}(0, N_0)$$

$$\begin{aligned} &P(+A) \\ &P(s = -A) \\ &\underbrace{\hspace{10em}}_{\text{prior.}} \end{aligned}$$

- With equal priors the threshold is 0 $\rightarrow$ converges to ML.



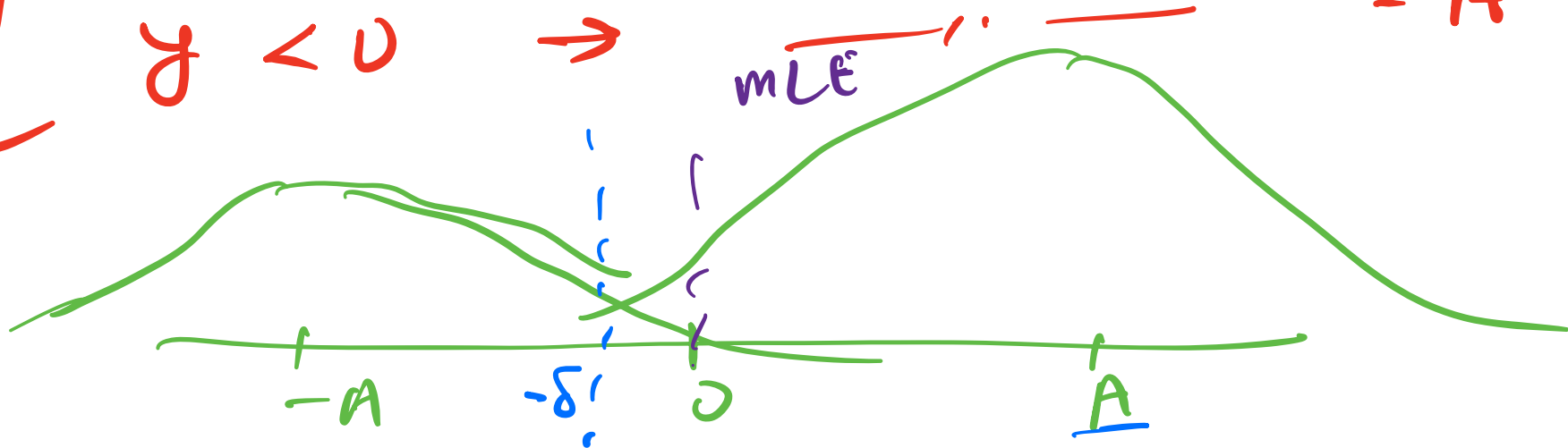
Decision Boundary



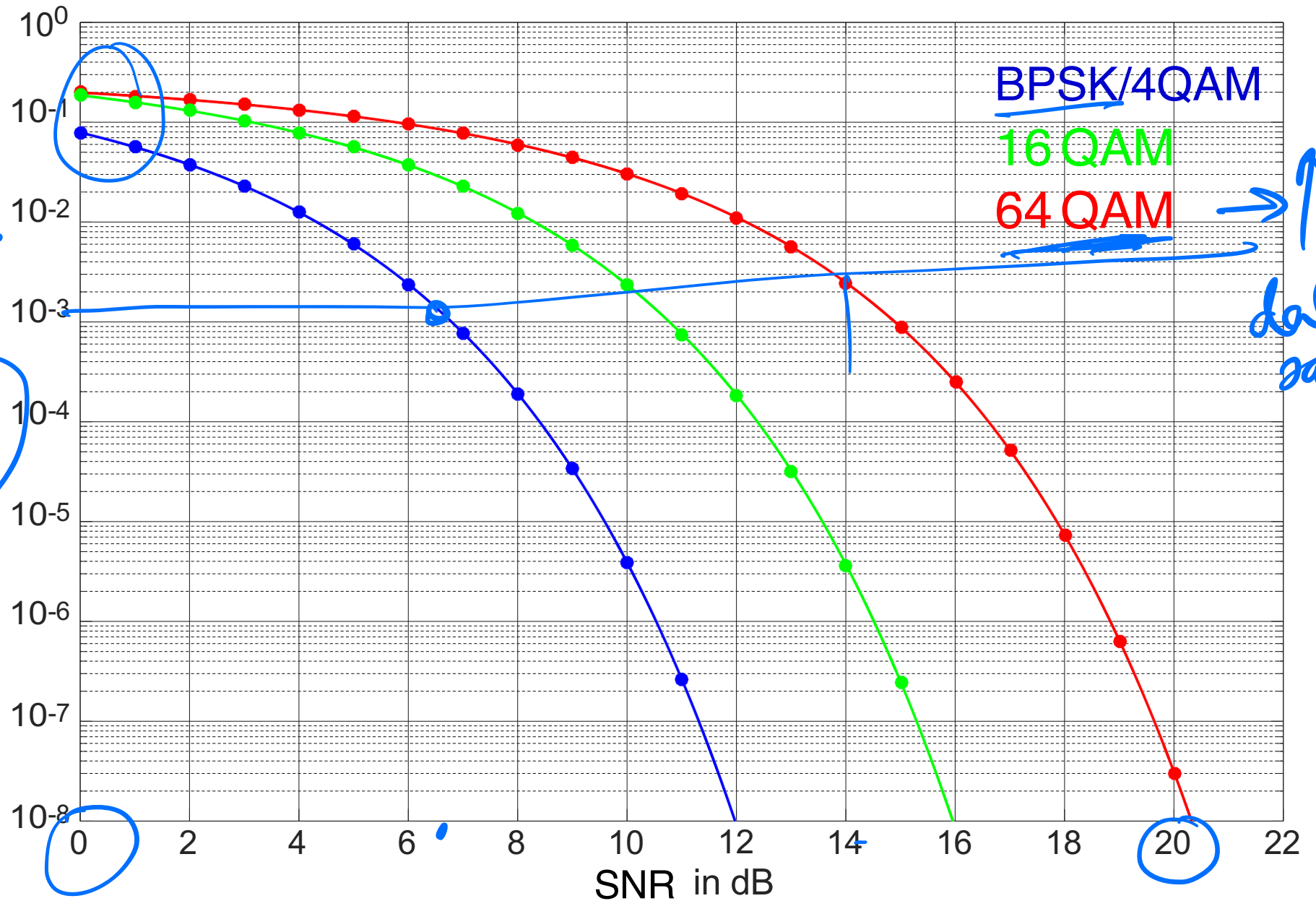
$S(t)$

MLE $\left\{ \begin{array}{l} y > 0 \rightarrow \text{you transmitted } +A \\ y < 0 \rightarrow \text{you transmitted } -A \end{array} \right.$

MAD

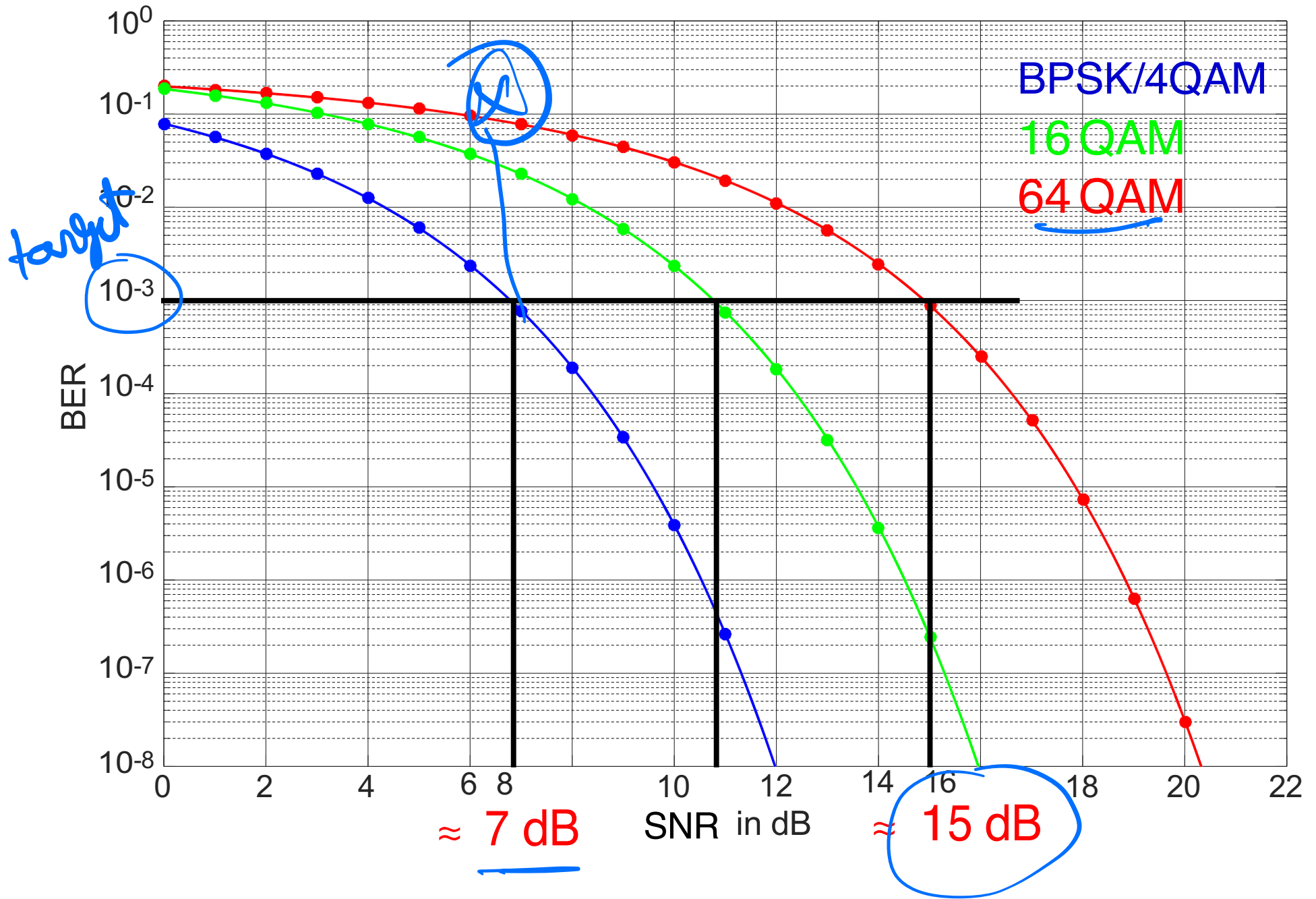


Bit Error Rate (BER)

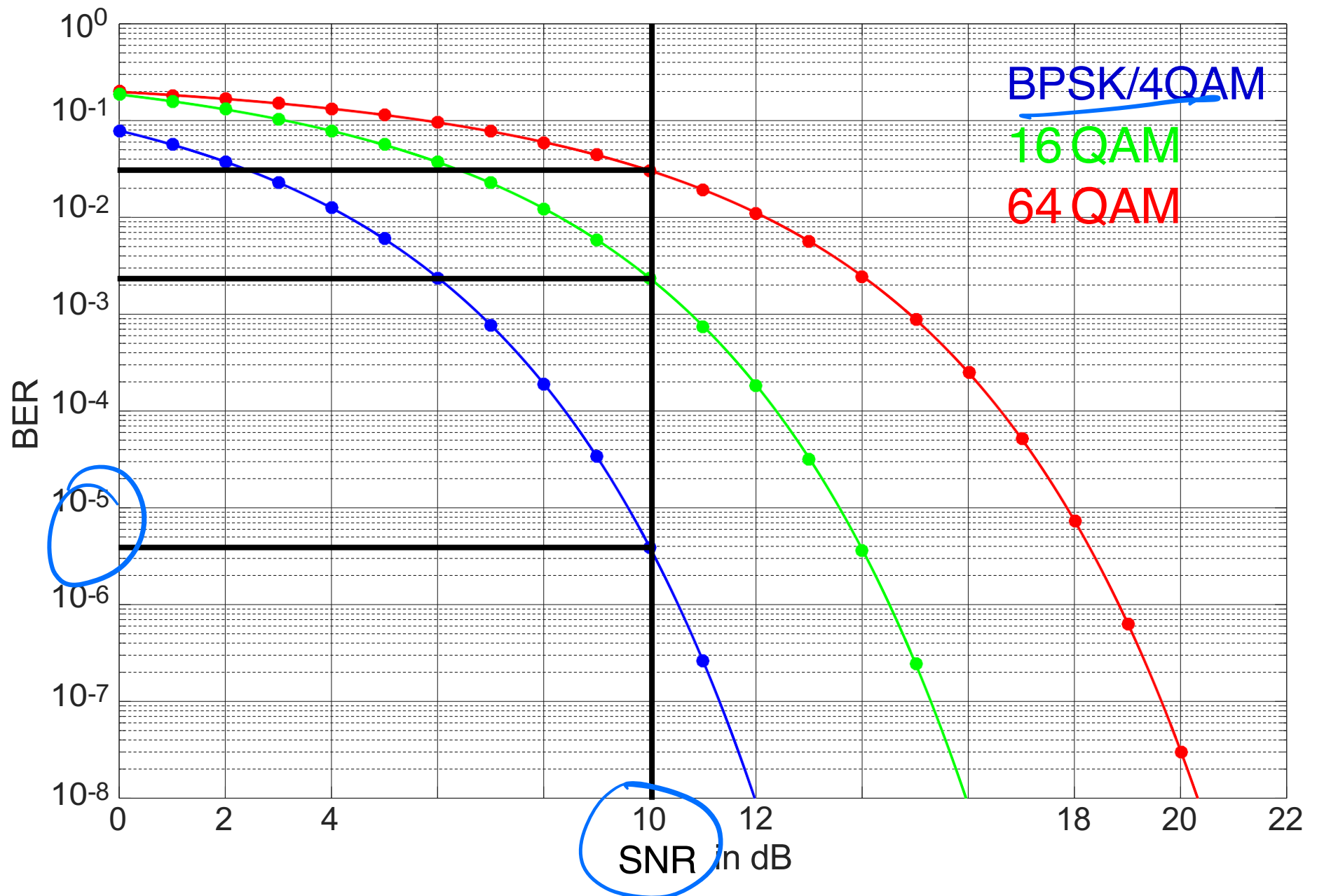


Bit Error Rate (BER)

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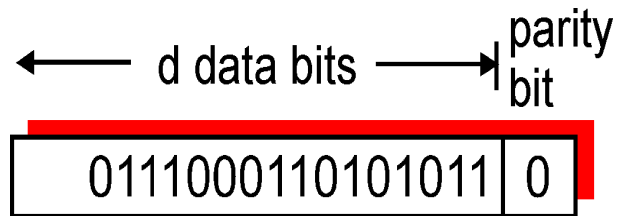


Bit Error Rate (BER)

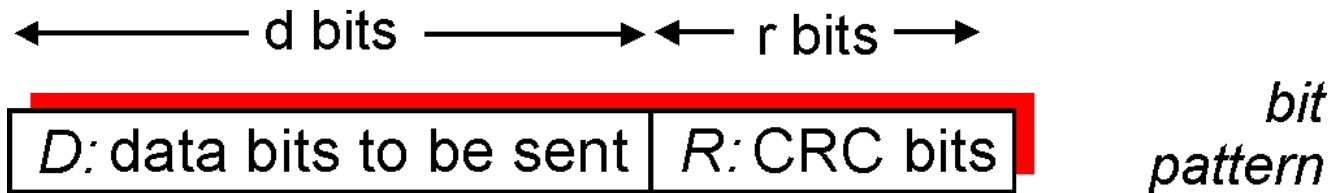


Error Detection

- Add Redundant bits in order to detect errors at receiver
- Checksums → Detect Errors
 - Parity Check



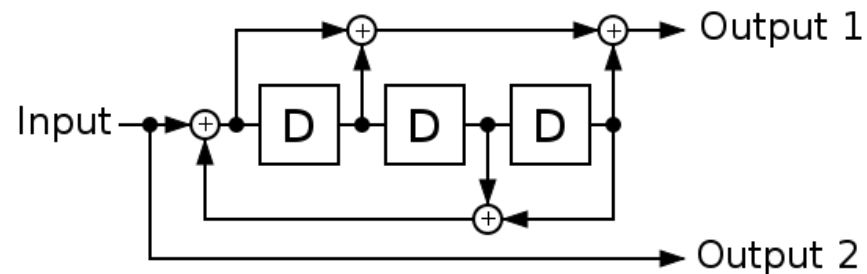
- CRC: Cyclic Redundancy Check



- If packet (frame, sequence of bits) fails the checksum test
 - Packet is dropped
 - Packet loss.

Coding: Forward Error Correction (FEC)

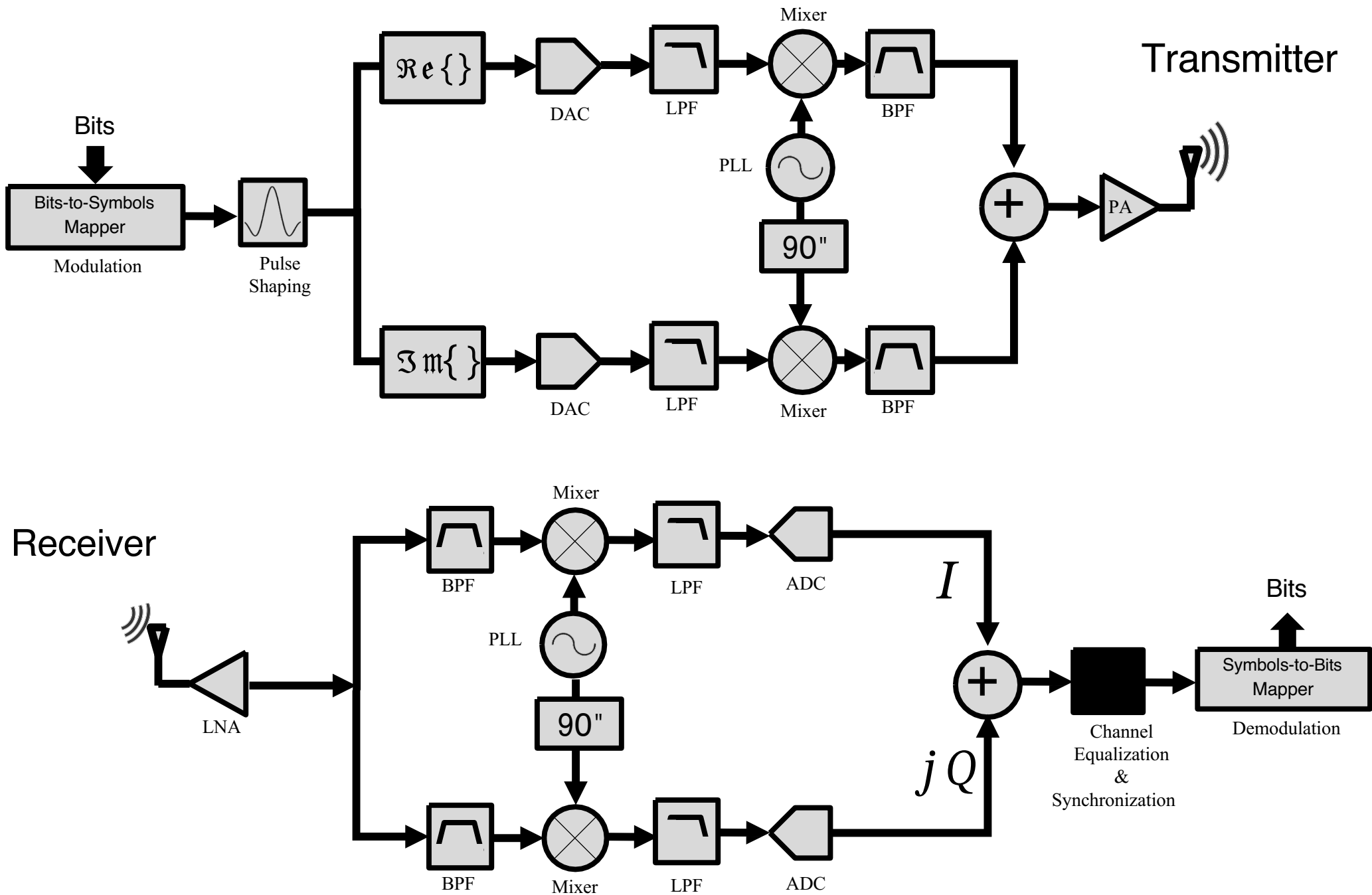
- Add Redundant bits in order to correct errors at receiver
- FEC: Forward Error Correction
 - Repetition Code
 - Convolutional codes
 - Reed Solomon codes
 - Turbo codes
 - Polar codes
 - LDPC codes



- Code Rate
$$\frac{\# \text{ of data bits (uncoded)}}{\# \text{ of transmitted bits (coded)}}$$

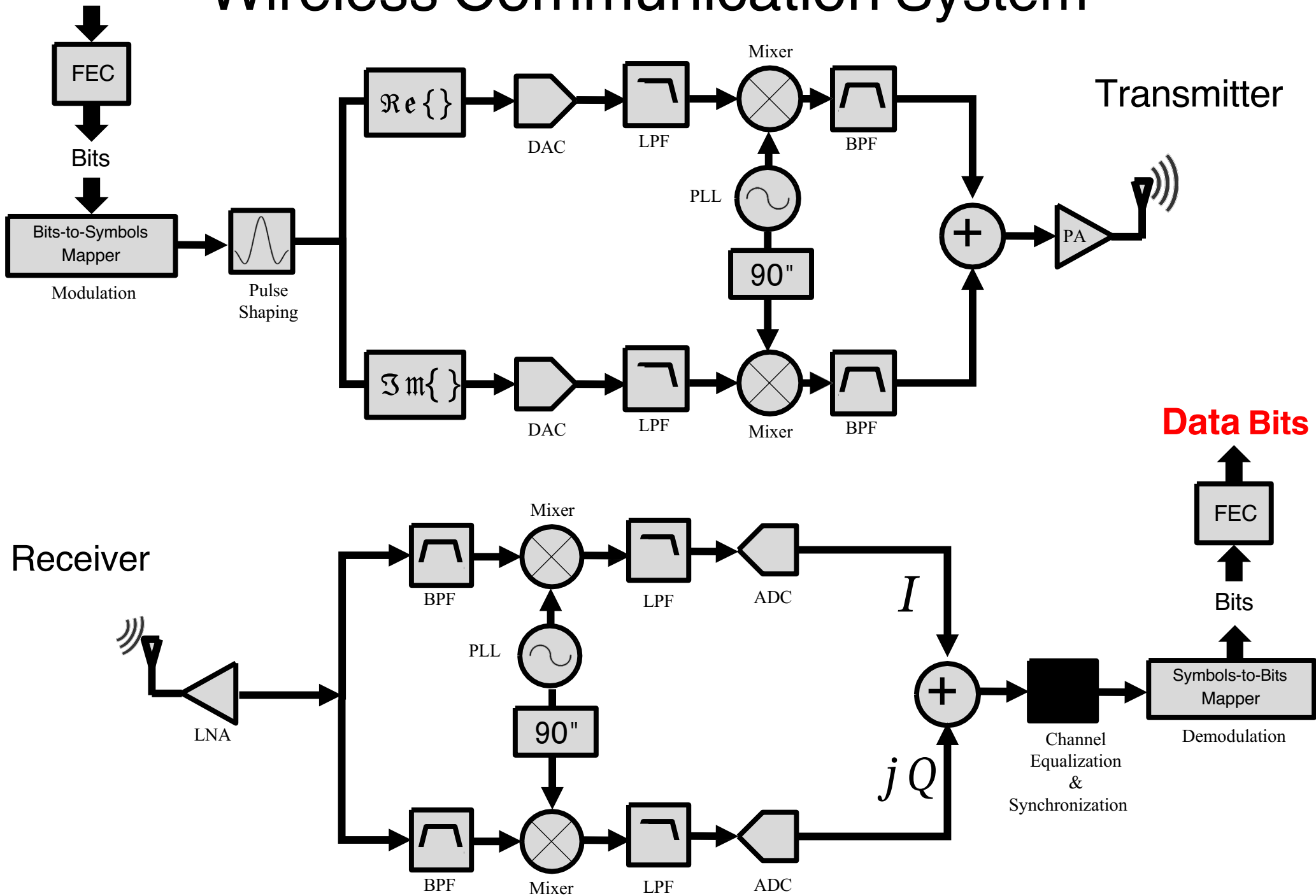
- E.g. coding rate 2/3, for every 2 data bits, I transmit 3 bits.

Wireless Communication System



Data Bits

Wireless Communication System



Wireless Data Rates

- *Data Rate*

- *Bandwidth: Samples/sec*
- *Modulation: Bits/sample*
- *Coding Rate: Data Bits/Coded Bits*

$$Data\ Rate = Bandwidth \times Bits/sample \times Code\ Rate$$

- *Capacity*

- *Maximum Achievable Data Rate*
- *Shannon Capacity Theorem:*

$$Capacity = Bandwidth \times \log_2 (1 + SNR)$$